

Essays in Urban and Political Economics

by
Vincent Rollet

Submitted to the Department of Economics
in partial fulfillment of the requirements for the degree of
DOCTOR OF PHILOSOPHY IN ECONOMICS
at the
MASSACHUSETTS INSTITUTE OF TECHNOLOGY
May 2026

© 2026 Vincent Rollet. All rights reserved.

The author hereby grants to MIT a nonexclusive, worldwide, irrevocable, royalty-free license to exercise any and all rights under copyright, including to reproduce, preserve, distribute and publicly display copies of the thesis, or release the thesis under an open-access license.

Authored by: Vincent Rollet
Department of Economics
May 15, 2026

Certified by: Dave Donaldson
Class of 1949 Professor of Economics, Thesis Supervisor

Certified by: David Atkin
Barton L. Weller (1940) Professor of Economics, Thesis Supervisor

Accepted by: Isaiah Andrews
Charles E. and Susan T. Harris Professor of Economics
Chairman, Departmental Committee on Graduate Studies

THESIS SUPERVISORS

Dave Donaldson

*Class of 1949 Professor of Economics
Department of Economics, Massachusetts Institute of Technology*

David Atkin

*Barton L. Weller (1940) Professor of Economics
Department of Economics, Massachusetts Institute of Technology*

Tobias Salz

*Castle Krob Career Development Associate Professor of Economics
Department of Economics, Massachusetts Institute of Technology*

Vincent Pons

*Byron Wien Professor of Business Administration
Harvard Business School*

THESIS READER

Nathaniel Hendren

*Ford Professor of Economics
Department of Economics, Massachusetts Institute of Technology*

Essays in Urban and Political Economics

by

Vincent Rollet

Submitted to the Department of Economics
on May 15, 2026 in partial fulfillment of the requirements for the degree of

DOCTOR OF PHILOSOPHY IN ECONOMICS

ABSTRACT

This dissertation consists of three essays in urban and political economics. The first studies how cities grow through redevelopment—demolishing old buildings to make way for new ones—and how this process is influenced by zoning, which regulates the size and uses of new buildings. Using New York City as a case study, I build the first parcel-level panel of a city’s buildings, zoning, and floorspace prices. This data allows me to estimate a dynamic spatial equilibrium model of floorspace supply and demand, validate the model using quasi-experimental variation from recent zoning reforms, and apply it to evaluate the effects of relaxing regulation on construction and affordability. While zoning strongly constrains city growth, the effects of relaxing regulation take decades to materialize and are limited in inexpensive or densely built areas. Large fixed costs of redevelopment, which rise sharply with the size of existing buildings, generate considerable persistence in city structure and substantially lower the expected gains from relaxing zoning. Furthermore, due to migration, the affordability benefits of zoning reform largely accrue to households outside the rezoned neighborhoods.

The second essay, coauthored with Benjamin Marx and Vincent Pons, studies electoral turnovers in national elections. In most national elections, voters face a key choice between continuity and change: electoral turnovers occur when the incumbent candidate or party fails to win reelection. To understand how turnovers affect national outcomes, we study presidential and parliamentary elections held globally between 1946 and 2018. We document the prevalence of turnovers over time and estimate their effects on economic performance, human development, and the quality of democracy. Using a close-elections regression discontinuity design across countries, we show that turnovers improve several measures of country performance. To explain these positive effects, we explore how electoral turnovers affect leader characteristics, shape policy decisions, reduce perceived corruption, and foster accountability.

The third essay, coauthored with Raphaël Descamps, Benjamin Marx, and Vincent Pons, explores the global incumbency advantage. We first show that existing subnational estimates of the incumbency advantage correlate positively with GDP per capita and democratic quality, and negatively with corruption across countries. Building on this meta-analysis, we then consider presidential and parliamentary elections held since 1945 to estimate how a national electoral victory affects the probability that winning parties and candidates retain power beyond the term for which they were elected. On average, national election winners benefit from an incumbency advantage, but this effect is short-lived and differs markedly

across contexts: it is large in Africa, North America, and Western Europe, but muted or even reversed in Latin America and in Asia and Oceania. We explore how standard incumbency effects and electoral manipulation contribute to these results. In established democracies, the national incumbency advantage reflects gains in the subsequent electoral performance of election winners. In less democratic regimes, it mainly stems from manipulation of the fairness and timing of elections.

Thesis supervisor: Dave Donaldson

Title: Class of 1949 Professor of Economics

Thesis supervisor: David Atkin

Title: Barton L. Weller (1940) Professor of Economics

Thesis supervisor: Tobias Salz

Title: Castle Krob Career Development Associate Professor of Economics

Thesis supervisor: Vincent Pons

Title: Byron Wien Professor of Business Administration

Acknowledgments

I have been immensely fortunate to grow as an economist at MIT under the guidance of Dave Donaldson, David Atkin, and Tobias Salz. I could not have hoped for better teachers, advisors, or role models. With patience and constant encouragement, they taught me how to pursue research that is useful, rigorous, and ambitious. They always pushed me to ask important questions, and I deeply admire their intellectual breadth and clarity. They helped me understand what it means to be an economist, and to find my place in academia. My PhD has been extraordinarily rewarding and fun, largely because I was lucky enough to be guided by such kind and generous advisors. I am profoundly grateful to them, and I know that the example they set will continue to guide me for many years to come.

I would likely not have embarked on this academic journey without the impulse of Pierre Boyer, who invited me to regularly spend time at Polytechnique's Economics Department and introduced me to academia. He and Benoit Schmutz-Bloch patiently showed me what research was about and illuminated a path that I have been very happy to follow.

Before starting my PhD, I was also particularly lucky to meet Ben Marx and Vincent Pons. They taught me how to conduct empirical research with the rigor, perseverance, and attention to detail that characterize good academic work. They allowed me to understand early on what this work involved, which helped me begin the PhD in the best possible conditions. Working with them over the past seven years has been incredibly rewarding and fun.

In Cambridge, I benefited from the advice and feedback of many faculty members. I am especially grateful to Daron Acemoglu, Nikhil Agarwal, Charles Angelucci, David Autor, Arnaud Costinot, Ed Glaeser, Adam Guren, Ben Vatter, and Bill Wheaton for conversations that helped shape my thinking and my work. I have also been lucky to learn from an extraordinary group of coauthors: Pierre Boyer, Thomas Delemotte, Raphaël Descamps, Dave Donaldson, Germain Gauthier, Olivier Gossner, Federico Huneus, Raphaël Lafrogne-Joussier, Jean Lavallée, Yves Le Yaouanq, Ben Marx, Jean-Baptiste Michau, Vincent Pons, Benoit Schmutz-Bloch, and Laura Weiwu. Thinking about economics with them has been one of the great joys of my PhD.

One of the most rewarding parts of doing a PhD in Cambridge has been belonging to a dense international community full of interesting, active, and kind people. My friends and roommates made me feel at home while I was far from my roots. I am grateful to Agnes, Amory, Angie, Arthur, Ben, Brad, Camille, Carole, Charlotte, Claire, Danny, Deivy, Drew, Evan, Fra, Isa, Ishaana, Isaiah, Jett, Jules, Laura, Léa, Lena, Leo, Lia, Lindsey, Manon, Martina, Mathilde, Moïse, Namrata, Nicolas, Olivia, Quentin, Romain, Salo, Sankalp, Theo, Tishara, Victor, and Vishan for the many conversations, dinners, walks, and trips we shared.

over the years.

I owe deep gratitude to my parents and brothers for their continuous support and care. They have been a source of stability, love, and perspective throughout this journey. Their confidence in me has helped me find my own.

I dedicate this dissertation to my grandparents, who nurtured me with boundless love from the very beginning and shaped so much of who I am. I owe them more than I could ever express. They are in my heart every day, wherever I am.

Contents

<i>Acknowledgments</i>	7
<i>List of Figures</i>	11
<i>List of Tables</i>	13
1 Zoning and the Dynamics of Urban Redevelopment	15
1.1 Introduction	15
1.2 Data and Context	20
1.3 Motivating Facts	23
1.4 A Model of Urban Evolution	27
1.4.1 Supply	27
1.4.2 Demand	30
1.4.3 Equilibrium	33
1.5 Model Estimation	33
1.5.1 Supply	33
1.5.2 Demand	37
1.6 Measuring the Effects of Relaxing Zoning Regulations	41
1.6.1 What are the aggregate effects of relaxing zoning?	42
1.6.2 Where is zoning a binding constraint?	44
1.6.3 Who benefits from relaxing zoning?	46
1.6.4 How efficient are policies favoring construction and affordability?	48
1.6.5 How do zoning and historical persistence shape city structure?	49
1.6.6 How does zoning affect floorspace supply elasticities?	50
1.7 Why did NYC’s planners impose such costly regulations?	52
1.8 Conclusion	53
References	54
2 Electoral Turnovers	59
2.1 Introduction	59
2.2 Key Data Sources	64
2.2.1 Data on Elections, Leaders, and Institutions	64
2.2.2 Data on National Outcomes	66
2.3 Empirical Framework	67
2.3.1 Defining Electoral Turnovers	67
2.3.2 Sample Description	68
2.3.3 Outcome Variables	69

2.3.4	Regression Discontinuity Estimation	70
2.3.5	Turnovers in the Executive Branch	71
2.4	Main Results	75
2.4.1	Identification Checks	75
2.4.2	Main Results: Effects of Electoral and Executive Turnovers	77
2.4.3	Robustness	85
2.4.4	Comparing Effects in Close Elections and Other Elections	86
2.4.5	The Role of Leader Characteristics	89
2.5	Mechanisms	90
2.5.1	Policy Changes	90
2.5.2	Political Accountability	91
2.6	Conclusion	94
	References	95
3	The Global Incumbency Advantage	101
3.1	Introduction	101
3.2	Meta-analysis	105
3.2.1	Analysis sample	105
3.2.2	Findings	106
3.3	Data	111
3.4	Empirical Framework	112
3.4.1	Power trajectories	112
3.4.2	Next-election incumbency advantage	115
3.5	Results	117
3.5.1	Testing the identification assumption	117
3.5.2	Estimates of the national incumbency advantage	118
3.5.3	Heterogeneity and robustness	120
3.6	Mechanisms	123
3.6.1	Next-election incumbency advantage	124
3.6.2	Fairness and timing of elections	126
3.6.3	Decomposing the incumbency advantage	127
3.6.4	Democratic quality and the national-level incumbency advantage	129
3.7	Conclusion	130
	References	132

List of Figures

1.1	NYC's zoning map	22
1.2	Redevelopment happens when it can substantially increase a parcel's floorspace	24
1.3	Effects of upzoning on built FAR	25
1.4	Redevelopment reallocates land to its most profitable use	26
1.5	Model-implied vs. quasi-experimental effects of upzoning on built FAR	38
1.6	Effects of new construction on floorspace quantities and rents	40
1.7	Effects of relaxing zoning	43
1.8	Heterogeneous effects of removing zoning on additional built FAR by 2060 .	45
1.9	Distributional effects of removing zoning	47
1.10	Effects of alternative policies on floorspace growth	48
1.11	Counterfactual FAR profiles of NYC	50
1.12	Floorspace supply elasticities	51
2.1	Share of elections with a turnover	61
2.2	Effect of an electoral turnover on the probability of turnover in the executive branch	73
2.3	Density tests	76
2.4	Effects of electoral turnovers on country performance	78
2.5	Effects of turnovers on country performance across subsamples	80
2.6	Dynamic effects of electoral turnovers on country performance	83
2.7	Comparing RD with OLS estimates	87
3.1	Distribution of subnational incumbency advantage estimates across countries	106
3.2	Average incumbency advantage at the subnational level	107
3.3	Correlates of subnational incumbency advantage effects	109
3.4	Placebo tests: Effect of election victory on being in power before the election	118
3.5	Incumbency advantage at the national level	119
3.6	Heterogeneity in the post-term incumbency advantage	121
3.7	Incumbency advantage before and after 1990	123
3.8	Next-election incumbency advantage, party-level estimates	124
3.9	Decomposition of the incumbency advantage	128
3.10	Incumbency advantage measures by level of democracy	130

List of Tables

1.1	Cost parameter estimates	36
1.2	Spillover parameter estimates	41
2.1	Effects of electoral turnovers on country performance	77
2.2	Heterogeneity analysis	81
2.3	Effects of turnovers in the executive branch on country performance	84
3.1	Power trajectories for the 2008 elections in the United States	114

Chapter 1

Zoning and the Dynamics of Urban Redevelopment

1.1 Introduction

Urban neighborhoods constantly evolve to accommodate growing populations and new economic activities. This continuous transformation is essential for a city’s success, yet in many places, it appears to be faltering. In particular, the world’s largest cities face acute housing shortages and soaring rents. Insufficient construction there is holding back aggregate economic growth, as these cities are the most productive and innovative parts of the modern economy.

Zoning regulations, which limit the size and permitted uses of buildings, are widely seen as a major contributor to these issues. In response, many cities have begun to relax these rules or are considering deregulation to spur construction and improve affordability. However, whether these reforms will achieve their aims remains unclear. Indeed, regulation is not the only constraint preventing large cities from growing. In city cores, vacant land is scarce and most new construction comes through redevelopment—demolishing existing buildings to replace them with new ones. While this process has recently become the dominant mode of urban expansion ([Frolking et al., 2024](#)), it is much costlier than building on undeveloped land.

Using New York City as a case study, this paper examines how zoning and the existing building stock influence a city’s evolution and evaluates the effects of relaxing zoning on construction and equilibrium prices. Such an analysis has long been hindered by two challenges. First, it requires data about developers’ behavior: which buildings they demolish and when, what they build in their place, and the zoning constraints they face. This information is not readily available and must be assembled from numerous sources. Second, modeling the evolution of a large city is technically challenging: developers make dynamic, durable investment choices on hundreds of thousands of land parcels, each with distinct buildings depreciating slowly over time and facing diverse zoning constraints. These development decisions depend on local floorspace prices now and far into the future, which in turn depend on development decisions across the entire city. Indeed, construction in one

neighborhood influences other areas via migration and commuting spillovers.

To overcome these challenges, I build the first parcel-level panel dataset tracking a city’s buildings, zoning, and floorspace prices. Using this data, I first provide new facts about redevelopment and evaluate the impact of recent zoning reforms. I then estimate a new model of urban evolution that integrates a dynamic model of developers’ redevelopment decisions into a spatial equilibrium model of households and firms. Finally, I leverage this framework to evaluate the effect of relaxing zoning on construction, prices, and the city’s structure.

The data enabling my analysis is presented in Section 1.2. To track land uses in NYC, I first link cadastral maps issued in different years using a custom algorithm and manual reconciliation. This allows me to partition the city into about 833,000 parcels and follow the buildings on each parcel for each year between 2004 and 2022. I extract building characteristics from property tax records (obtained through FOIA requests and agency portals) and from scraped online listings. Over the study period, about 22,000 parcels were redeveloped (i.e., a new building was constructed there). For these parcels, I track the redevelopment process using building permits and certificates of occupancy, hundreds of thousands of which had to be digitized. I match this data with the universe of real estate transactions in NYC since 2003, rent data from scraped building-level tax notices, and zoning maps for each year. This panel allows me to measure individual development decisions and the constraints developers face: zoning and the demolition of existing buildings.

In Section 1.3, I provide new stylized facts about redevelopment and reduced-form evidence on the effects of relaxing zoning. Tracking individual parcels over time, I show that redevelopment is associated with densification: 96% of demolished buildings are replaced with larger ones, with new structures on average 3.4 times larger than the ones they replace. Conversely, redevelopment seldom happens when it cannot increase the amount of floorspace on a parcel, such as when the parcel is already at the maximum density allowed by zoning. By substantially increasing the amount of floorspace on a parcel, developers can overcome the fixed costs associated with demolishing an old building. I show that, in addition to increasing the aggregate supply of floorspace in a city, redevelopment enables adjustments in the ratio of residential to commercial floorspace within each neighborhood. Developers tend to build the most valuable type of floorspace, reallocating land to its most profitable use.

I leverage the quasi-exogenous timing of recent zoning changes to evaluate the effect of relaxing regulatory constraints (upzoning) in a difference-in-differences design at the parcel level. While upzoning does increase construction, its effects materialize slowly over time: ten years after the policy change, only 9% of the newly allowed floorspace has been built. The effects of upzoning are concentrated on parcels where redevelopment is the most profitable—those with high floorspace prices and low levels of existing development.

To extrapolate beyond observed rezonings (i.e., to evaluate more ambitious policies, apply them to neighborhoods not recently targeted by planners, and assess their long-term effects) and to measure their impact on local and distant prices, I develop a dynamic spatial equilibrium model of floorspace supply and demand, described in Section 1.4. To model floorspace supply, I consider that each parcel is associated with a potential developer who periodically decides whether to redevelop it or not. When redeveloping a parcel, developers choose what to build subject to zoning limits on use and size. Redevelopment is costly.

Some costs are variable and increase with the size of the new building. Others, such as those associated with permitting, eviction, and demolition, are fixed and vary from one structure to the next. For instance, demolition will be costlier for tall buildings in dense neighborhoods.

I integrate this model of developers' behavior into a spatial equilibrium model of workers and firms. Workers choose where to live and work in the city, consume residential floorspace, vary in their productivity, and have non-homothetic preferences. Firms consume commercial floorspace. Over time, redevelopment changes the supply of floorspace in each neighborhood, leading to changes in local amenities and productivity levels due to agglomeration spillovers both within and across residential and commercial land uses. Buildings deteriorate over time, reducing their value and increasing the profitability of redevelopment. Each period, floorspace rents adjust so that markets clear. Developers, aware of these dynamics, make forward-looking decisions with rational expectations.

Section 1.5 describes how I estimate the model. For the supply side, which models the behavior of developers, I rely on methods developed in the empirical industrial organization literature and their application to urban economics (Rust, 1987; Murphy, 2018; Almagro and Domínguez-Iino, 2025; Hsiao, 2024). Specifically, I proceed in three steps. First, I use real estate sales data to estimate the value of buildings as a function of their characteristics. This allows me to estimate the value of properties in NYC and assess how redevelopment could increase their worth. Then, I estimate variable construction costs using a revealed-preferences approach. When redevelopment takes place, I observe the choices made by developers as well as the floorspace prices and zoning regulations they faced. This provides information on the cost schedule developers face. Finally, I estimate fixed costs through a nested fixed-point algorithm, extending Rust (1987) to a spatial equilibrium setting. Specifically, fixed costs are chosen so that the model best reproduces the relationship between developers' choices and the expected value of those choices. This requires repeatedly solving the full dynamic equilibrium for different candidate values of the fixed cost parameters. Indeed, the option value of future redevelopment depends on the entire future path of prices. This path is an equilibrium outcome determined by the collective actions of all developers, which, in turn, depend on the very fixed costs being estimated. I find that fixed costs are higher in denser neighborhoods and increase sharply with the size of the building to be demolished.

To estimate the parameters of the demand model, I first infer workers' preferences and firms' production functions through observed rents, commuting flows, and floorspace consumption. Accounting for worker heterogeneity and non-homothetic preferences proves essential to rationalize the patterns of floorspace demand in the data. To estimate agglomeration forces, I leverage the construction of large new buildings. Comparing areas that received construction earlier vs. later in a staggered difference-in-differences design, I estimate local elasticities of rents to floorspace supply. I find that new construction lowers the rents of nearby preexisting buildings of the same type. This effect combines a supply effect and an agglomeration effect, as new buildings change local amenities and productivity. I estimate agglomeration externalities so that the model reproduces the demand elasticities estimated in the reduced form. I find strong positive spillovers within land uses (i.e., of residential construction on residents, or commercial construction on firms), while spillovers across land uses are limited.

My model is estimated independently of my difference-in-differences estimates of the effects of upzoning. Therefore, to validate my model, I reestimate it on the subsample of

parcels that were not upzoned and then predict the effects of upzoning on parcels that were. The model’s predicted effects closely match my quasi-experimental estimates across a wide range of horizons and in different contexts, indicating that it can reliably predict the effects of zoning changes.

The estimated model allows me to simulate the city’s evolution over the coming decades under different policy scenarios. I compute counterfactuals by finding a fixed point in the paths of floorspace supplies and prices, where floorspace supplies are consistent with developers’ optimal behavior given prices and zoning, and prices each period are given by spatial equilibrium. In Section 1.6, I compare NYC’s trajectory under current regulations with scenarios in which zoning constraints are relaxed to measure how much relaxing zoning would boost construction and increase affordability.

I first quantify the aggregate effects of deregulation. The model projects that an ambitious but realistic upzoning of NYC could increase the city’s floorspace supply by 15 pp over 40 years. However, the take-up of upzoning remains limited in the medium run: at a 40-year horizon, only 18% of the newly allowed floorspace is built. Furthermore, the effects of upzoning unfold slowly: ten years after upzoning, only 23% of the reform’s long-term impact on construction has materialized. Fully removing zoning regulations yields larger increases in floorspace (+58 pp), but even in this extreme scenario, residential rents in NYC decrease only moderately (-17 pp). This is because new construction triggers migration to the city, dampening local rent reductions while contributing to diffuse rent declines in other cities. Yet, removing zoning would still yield sizable welfare gains for New Yorkers (+13 pp by 2060), with lower-income workers benefiting the most. Furthermore, upzoning substantially outperforms two alternative policies aimed at increasing construction and affordability: reducing construction costs and granting property tax breaks on new buildings. Tax breaks are particularly costly because they tend to subsidize inframarginal projects. Finally, I find that use and density zoning limits interact: the effect of relaxing both limits together is substantially larger than the sum of the effects of relaxing each one separately.

Because the fixed costs associated with redevelopment are large and increase sharply with the amount of floorspace to be demolished, the effects of relaxing zoning are highly heterogeneous over space. Relaxing zoning has very limited effects in neighborhoods where floorspace is inexpensive. There, redevelopment is usually unprofitable and therefore rare, and new buildings tend to be small. Upzoning parcels that are already densely built is also largely ineffective. Even without zoning limits, developers rarely redevelop these parcels because fixed costs typically exceed the additional rents redevelopment would yield. The neighborhoods whose growth is most distorted by zoning are those with limited density and high floorspace prices. A simpler model that ignores the large adjustment costs associated with redevelopment would not only overstate the effects of relaxing zoning, predicting nearly twice as much floorspace growth as my dynamic model would, but also suggest that removing zoning would completely reshape the city’s structure. In reality, high redevelopment costs create considerable path dependence and tightly constrain urban growth.

An alternative way to assess zoning’s impact on urban growth is to quantify its effect on floorspace supply elasticities. Under current zoning regulations, NYC’s floorspace supply is largely unresponsive to price increases, with an aggregate floorspace supply elasticity reaching only 0.13 at a 40-year horizon. Supply elasticities vary widely across neighborhoods and are essentially determined by how much of the maximum allowed density has already been built.

Removing zoning would more than quintuple the city’s overall floorspace supply elasticity and make local elasticities depend more on density, with denser areas remaining less elastic.

The large welfare losses resulting from zoning raise the question of why city planners implemented such costly regulations. In the final section of the paper, I argue that NYC’s zoning resolution may have been socially beneficial when adopted in 1961: demand for floorspace had plateaued, so zoning imposed relatively low costs while helping to curb negative externalities from industrial activity. Today, those externalities have largely faded, while floorspace prices often exceed marginal construction costs by a factor of two or three, making growth restrictions much more costly. The zoning code has, however, failed to adapt to changing economic conditions and has been much more persistent than its creators intended.

Contribution to the literature. Understanding the allocation of land uses in cities has long been a core goal of urban economics. Successive literatures have described how economic fundamentals and agglomeration shape urban density and land use, from the early studies of [Burgess \(1925\)](#) and [Hoyt \(1939\)](#) to monocentric city models ([Alonso, 1964](#); [Mills, 1967](#); [Muth, 1969](#); [Ogawa and Fujita, 1980](#); [Lucas and Rossi-Hansberg, 2002](#)) and modern quantitative approaches ([Allen and Arkolakis, 2014](#); [Ahlfeldt et al., 2015](#); [Allen, Arkolakis and Li, 2015](#); [Heblich, Redding and Sturm, 2020](#)). These frameworks typically analyze long-run steady states and assume a frictionless adjustment of land uses. However, cities change slowly, and their evolution is severely constrained by redevelopment costs and regulation. Hence, this paper focuses on the slow evolution of cities and how zoning affects their transition dynamics. This perspective is more policy-relevant than static approaches. It also allows me to quantify the adjustment costs associated with new construction, which I find are central to understanding the effects of zoning. Finally, it enables an analysis of historical persistence, which I show plays a key role in shaping city structure.

Studying these urban dynamics requires detailed data on the evolution of cities and a granular model of urban change. To that end, I build the first parcel-level panel of land use and zoning for a city, and use it to estimate a new model of redevelopment. My model builds on three modeling approaches in the literature. A first set of studies uses data from individual development projects to model the dynamic behavior of developers ([Murphy, 2018](#); [Soltas, 2024](#)). I extend these analyses by measuring how zoning and demolition costs constrain developers. A second set of articles, originating with [Ahlfeldt et al. \(2015\)](#), builds granular spatial equilibrium models that account for the complex interactions between urban neighborhoods. I extend these studies by building a model with rich heterogeneity in worker productivity and non-homothetic preferences, and by providing measures of agglomeration externalities, including novel measures of spillovers across land uses. Finally, I build on dynamic urban models that describe the progressive evolution of a city ([Hsiao, 2024](#); [Gechter and Tsivanidis, 2023](#); [Greaney et al., 2025](#); [Henderson et al., 2021](#); [Peng, 2023](#)). For tractability and due to data limitations, these studies model (re)development as an increase in the aggregate capital stock of different neighborhoods. While these models account for the slow evolution of urban areas, they cannot precisely identify the effect of zoning and demolition costs on construction, as both constraints vary widely from one parcel to the next within a neighborhood.

By evaluating how NYC would evolve in the coming decades if zoning were relaxed, I contribute to an active literature assessing the impact of land use regulations. A first set of studies, at the micro level, leverages recent zoning changes to show that relaxing land use regulation leads to additional construction ([Anagol, Ferreira and Rexer, 2024](#); [Büchler and Lutz, 2021](#); [Liao, 2022](#); [Peng, 2023](#)). Observed rezonings, however, are typically limited in scope and enacted in areas where planners believe they will be effective. The model developed in this paper can extrapolate beyond these case studies and predict the impact of various zoning reforms over extended time periods. A second set of studies, at the macro level, shows that zoning reduces the aggregate housing supply, raises prices, and causes a costly spatial misallocation of land uses ([Glaeser, Gyourko and Saks, 2005](#); [Gyourko and Molloy, 2015](#); [Ganong and Shoag, 2017](#); [Glaeser and Gyourko, 2018](#); [Hsieh and Moretti, 2019](#); [Ospital, 2022](#)). This paper microfound the aggregate supply and demand for floorspace and shows that existing spatial equilibrium models can substantially overstate the effects of relaxing zoning if they do not account for the large fixed costs of redevelopment. Both sets of studies have overwhelmingly focused on housing and density restrictions. I extend my analysis to commercial real estate and use regulations (which restrict where commercial and residential buildings can be constructed). This wider scope is valuable because commercial buildings represent a large share of the floorspace stock (35% in NYC) and their location is a key determinant of job access and commuting times.¹

I also contribute to the literature measuring housing supply elasticities. [Saiz \(2010\)](#) highlights the importance of geography in determining these elasticities across metropolitan areas, and [Baum-Snow and Han \(2024\)](#) find more variation in supply elasticities within metropolitan areas than between them, with lower elasticities near central business districts. Like [Murphy \(2018\)](#), I use a dynamic model of developers' behavior to measure supply elasticities. This allows me to provide granular elasticity estimates across different time horizons. I find wide variation in supply elasticities across neighborhoods, with zoning as their primary determinant, rather than existing construction.

Finally, a large body of research has emphasized the role of adjustment costs in explaining investment in fixed assets and labor ([Lucas, 1967](#); [Dixit, 1989](#); [Nickell, 1986](#); [Caballero, Engel and Haltiwanger, 1997](#)). These costs have both fixed and variable components. They may reflect technological constraints (e.g., installation or training costs) or stem from regulation (such as environmental or employment protection laws). Disentangling these frictions is essential to quantify misallocation in the economy ([Asker, Collard-Wexler and De Loecker, 2014](#); [David and Venkateswaran, 2019](#)). This paper contributes to this literature by carrying out such a decomposition for investment in a specific but important form of capital: buildings.

1.2 Data and Context

With over eight million inhabitants, NYC is the largest city in the United States. The total value of its real estate exceeds three trillion dollars, and while real home prices in NYC have almost doubled between 2000 and 2025, the city's growth over this period has been slow.

¹This broader focus relates my work to [Kulka, Sood and Chiumenti \(2024\)](#), who also investigate the multifaceted nature of zoning by studying the interaction between different types of regulations that limit housing density. I also find such interactions, namely between density and use restrictions.

NYC’s neighborhoods vary widely in density and income levels, providing a rich setting for analyzing urban change across a range of contexts. I assembled data from a wide range of sources to measure the behavior of developers as well as the prices and regulations they face. This section describes the key datasets I rely on, with Appendix A² providing additional details.

A parcel-level panel of NYC buildings. I track the evolution of land use in NYC at the parcel level. This granular data is essential for studying redevelopment, as developers make decisions at the parcel level and the technological and regulatory constraints they face vary widely from one parcel to the next. However, parcel-level panels of land use are not readily available. The closest existing datasets consist of repeated cross-sections of all buildings in a city. Over time, redevelopment causes some buildings to disappear from the cross-section and others to appear. To construct a panel dataset of land use, one must link new buildings with the old ones demolished to make way for them.

To do so, I rely on cadastral maps. These maps divide all of NYC into lots, each associated with a tax identifier. I overlay maps published in different years and link them using a custom polygon conflation algorithm.³ This allows me to divide NYC into about 833,000 parcels and find the property tax identifiers of the buildings located on them between 2004 and 2022.

I then associate these buildings with a wide range of characteristics. My primary source is property tax data gathered from publicly available assessor rolls, FOIA requests, and scraping individual property tax statements. I complemented this data with information from StreetEasy (Zillow’s NYC subsidiary), for which I scraped data covering 1.6 million units.

Over my period of study, about 22,000 parcels were redeveloped.⁴ To understand when redevelopment started, I match my panel with a dataset of building permits. To measure when it ended, I rely on certificates of occupancy (documents issued by the city certifying that a new building is fit for use).⁵ Construction projects are usually completed quickly, with certificates of occupancy typically granted two years after the corresponding building permit (see Appendix Figure C.2).

Real estate prices and rents. To measure floorspace prices throughout the city and over time, I rely on the NYC Department of Finance’s Property Sales Files, which document the

²This dissertation excludes the appendix materials of all three papers. For these materials, please consult my website: <https://vincent-rollet.com/>.

³This procedure (described in Appendix A.1) allows me to track parcel boundary changes (such as splits and mergers) that sometimes accompany redevelopment. To ensure a perfect tracking of all of NYC’s parcels, I manually coded 5,700 boundary changes that were not detected by the algorithm.

⁴I consider that a parcel has been redeveloped when a new building is constructed. I also tracked conversions (usually of commercial buildings to residential ones), which account for a small share of total redevelopment activity (see Appendix Figure C.1) and are therefore ignored in the quantitative analysis below. Renovations and building extensions are also excluded from the scope of this study, as the amount of floorspace added to the city through them is an order of magnitude lower than that added through new construction.

⁵Certificates of occupancy issued after July 2012 are available in a public database. To cover earlier redevelopment events, I scraped and digitized 250,000 certificates from the NYC Buildings Information System.

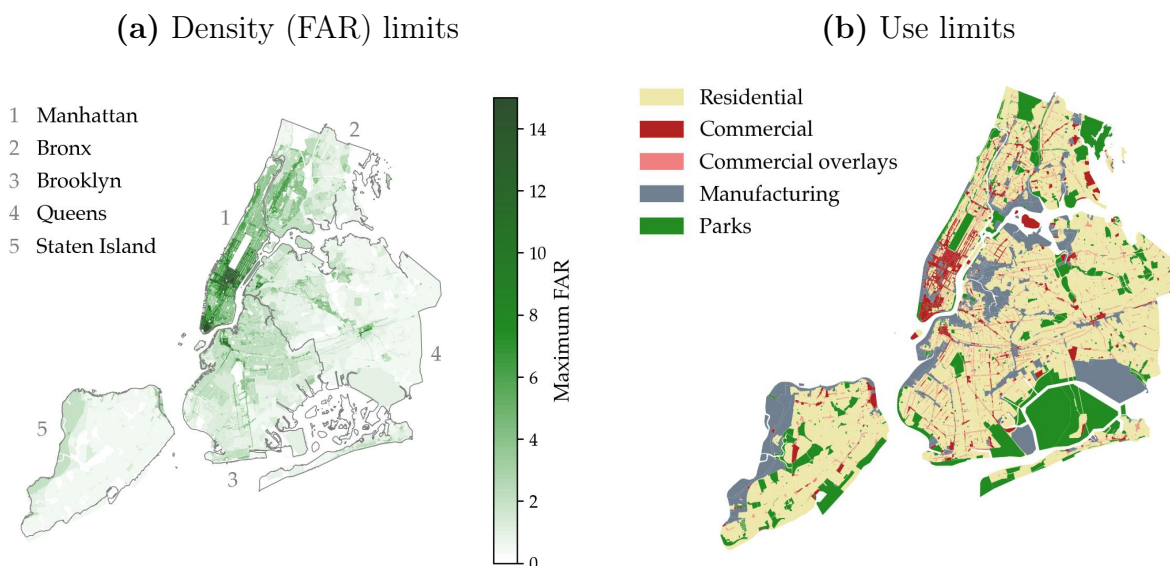
universe of real estate transactions that took place in the city between 2003 and 2022.

For a subset of buildings, I can further recover yearly rent estimates from the Notices of Property Values (NOPVs) sent annually to property owners to inform them of their property taxes. These documents are used in [Li \(2022\)](#) to build building-level rent measures covering 2003–2013. By scraping over 600,000 NOPVs issued in recent years, I extend [Li](#)’s dataset up to 2021.

Zoning constraints. I extract from NYC’s zoning resolution the regulatory constraints imposed on different parcels. The zoning map, of which an extract is presented in Appendix Figure B.1, associates each parcel with a zoning district, indicating allowed land uses and the maximum allowed size of new buildings. Density in NYC is mainly regulated through Floor Area Ratio (FAR) limits. The FAR of a parcel is defined as the total floorspace of its buildings divided by its land area (see Appendix Figure B.2 for an illustration).

NYC’s current zoning resolution was adopted in 1961 and largely reflects the city’s organization in the early 1960s. Commercial zoning districts are concentrated in the high-density neighborhoods of the lower half of Manhattan and Downtown Brooklyn, while more peripheral neighborhoods are almost exclusively zoned for residential purposes, with strictly limited densities (see Figure 1.1 and Appendix Figures B.3 and B.4).

Figure 1.1: NYC’s zoning map



Notes: This figure illustrates NYC’s zoning map. Panel (a) shows the maximum allowed densities (FARs) on parcels across the five boroughs of the city. Panel (b) shows land use districts in NYC’s zoning map. Residential districts only allow residential uses, commercial districts allow both residential and commercial uses, and manufacturing districts only allow commercial uses. Commercial overlays correspond to zones in residential areas that allow some commercial uses (e.g., stores on the first floor of buildings).

The average parcel in NYC has a built FAR of 1.1 and a maximum allowed FAR of 1.4 (see Appendix Figure C.3). A third of parcels exceed their maximum allowed FAR, often because they were built before 1961 and have been grandfathered in. Another quarter are

within 0.25 FAR points of the limit, making density restrictions binding or nearly binding for most parcels in the city.

Additional data sources. For some analyses, I further leverage smartphone data from Advan, commuting flows data from the LEHD-LODES database, commuting times from Google Maps, historical building permits from the Office for Metropolitan History, and historical sales data from the Real Estate Board of New York.

1.3 Motivating Facts

The granular datasets I have collected allow me to describe developers' behavior at the micro level. In this section, I present new facts describing how redevelopment progressively changes land uses and reduced-form evidence of zoning's effects on development. I report additional descriptive statistics in Appendix C.

Redevelopment is associated with densification. Redevelopment entails high fixed costs. Indeed, to construct a new building, existing structures must be demolished (and their future stream of rents forgone), building permits must be obtained, existing tenants must be relocated, etc. Hence, redevelopment is financially viable only when new structures are significantly more valuable than the ones they replace. A developer can increase a parcel's value by increasing the quality of the building on it. Indeed, older structures are much more likely to be redeveloped than newer ones. However, new buildings are more valuable than old ones primarily because they are larger: Figure 1.2(a) compares the FAR of new structures with the FAR of those they replaced. 96% of redevelopment events increased the amount of floorspace on the redeveloped parcel, and on average, new buildings had FARs 3.4 times higher than those they replaced.⁶

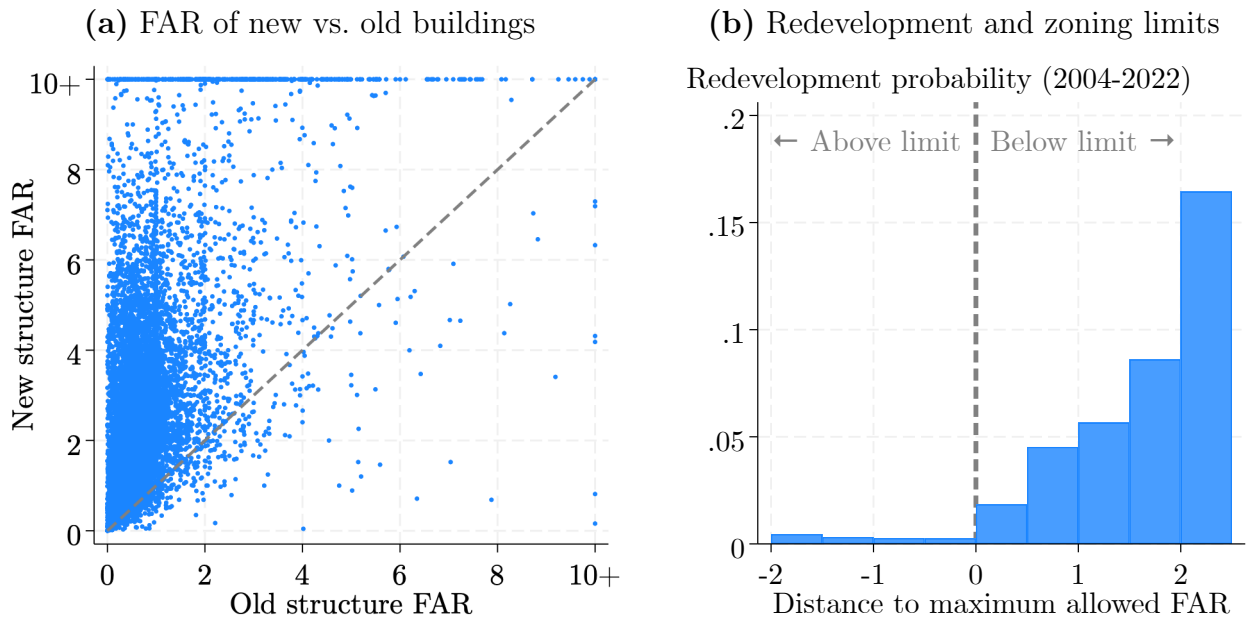
In line with this, redevelopment only takes place when zoning allows the replacement of an existing structure with a larger one. Figure 1.2(b) shows that parcels at or above the maximum allowed FAR are very rarely redeveloped. Indeed, it is unlikely that redevelopment would increase the value of these parcels enough to cover the fixed costs of redevelopment. On the other hand, buildings well below the maximum allowed FAR are much more likely to be redeveloped.

Relaxing zoning leads to progressive density increases in some areas. The extent to which zoning affects redevelopment can be directly evaluated by analyzing recently rezoned parcels. Although zoning rules tend to be persistent, several neighborhoods of NYC have been upzoned in the past decades, meaning that FAR restrictions have been substantially relaxed. These areas offer valuable case studies to investigate the effects of zoning reform.

Rezoning is politically fraught, requires approval from many stakeholders, and can take decades to pass. Therefore, when the Department of City Planning initiates a neighborhood rezoning, there is considerable uncertainty about which parcels will be affected, how their

⁶This finding echoes that of [Henderson et al. \(2021\)](#), who, using LiDAR data, show that newly built structures in Nairobi are roughly three times larger than the buildings they replaced.

Figure 1.2: Redevelopment happens when it can substantially increase a parcel’s floorspace



Notes: Panel (a) compares, for redeveloped parcels, the FAR of new structures with the FAR of the structures they replaced. The dashed line corresponds to the 45-degree line. Panel (b) shows the probability of observing a redevelopment event in the panel for different bins of the distance (in FAR points) between the FAR of the initially existing structure and the maximum FAR allowed by zoning.

FAR and use limits will change, and when those changes will take effect.⁷ While upzoned parcels are typically ripe for redevelopment and represent a selected sample, the exact timing of rezonings can reasonably be considered exogenous.

To estimate the effect of upzoning on redevelopment, I focus on parcels within areas rezoned between 2004 and 2022, whose maximum allowed FARs increased by more than 0.25.⁸ I then compare parcels that were upzoned earlier vs. later using the staggered difference-in-differences procedure of [de Chaisemartin and d’Haultfoeulle \(2024\)](#).

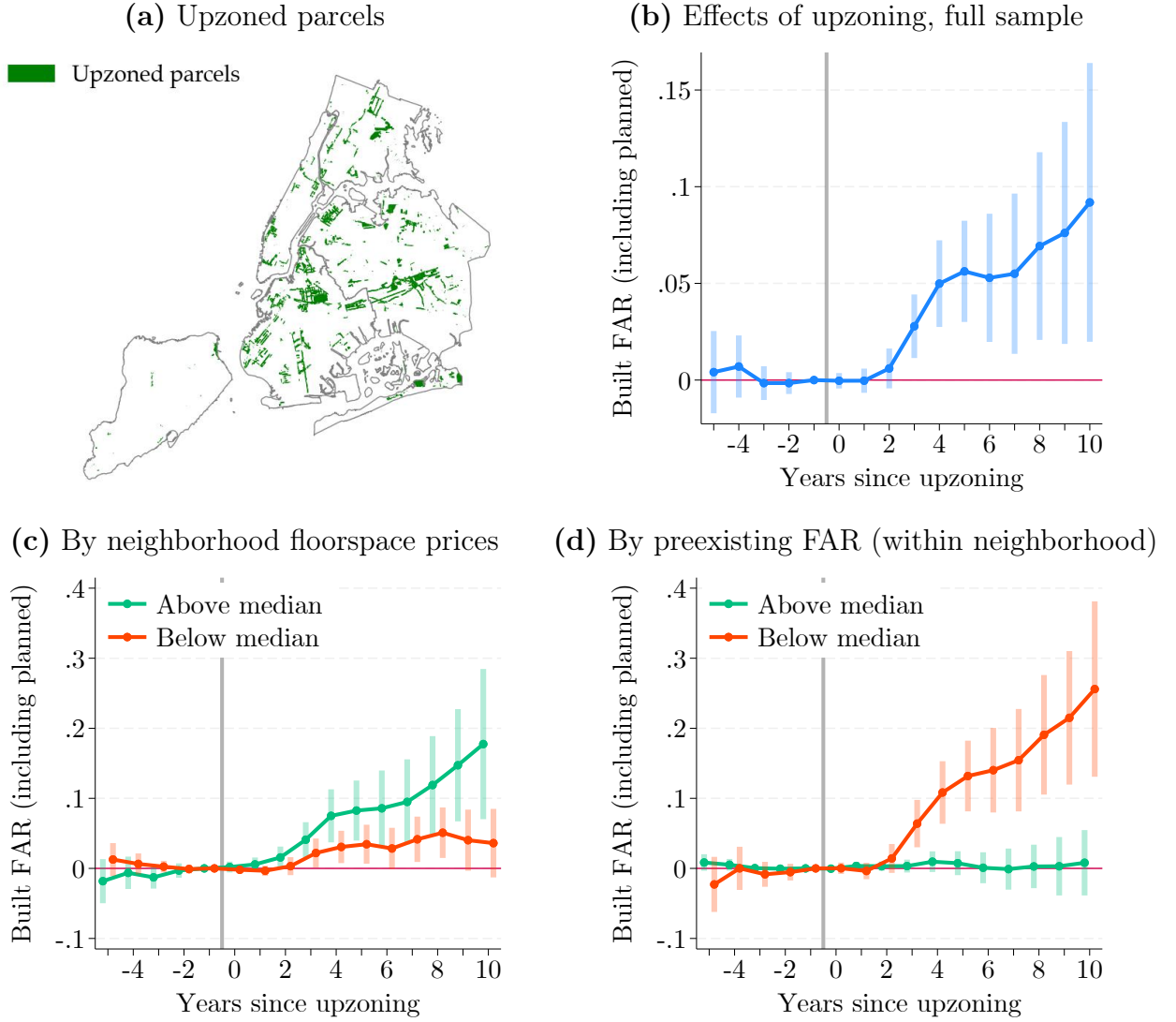
Figure 1.3(b) reports event study estimates of the effects of upzoning on built FAR. While relaxing zoning constraints stimulates development, increases in floorspace supply only materialize slowly over time. Even a decade after the policy change, take-up remains limited. On average, upzonings raise the maximum allowed FAR by one, but result in only a 0.09 increase in built FAR over ten years.

These results are striking in light of the wide gap between the price of floorspace and

⁷For instance, discussions to rezone the Gowanus Canal area began in the mid-2000s during the Bloomberg administration. Local community groups opposed rezoning proposals, fearing the changes would lead to gentrification and worsen the canal’s flooding and contamination. The rezoning was finally approved in late 2021, after nearly twenty years of negotiation.

⁸In the analysis sample for the event study, I exclude parcels that were upzoned as part of small rezonings, defined as rezonings affecting fewer than 30 parcels. These small rezonings are often requested by developers and may be processed more quickly than larger ones, making their occurrence endogenous. Large rezonings, conversely, tend to be initiated by city planners and follow a longer approval process. Approximately 10% of the redevelopment events occur on parcels upzoned as part of a large rezoning, and 1% on parcels upzoned as part of a small rezoning. Parcels in my estimation sample are mapped in Figure 1.3(a).

Figure 1.3: Effects of upzoning on built FAR



Notes: This figure presents estimates of the effects of upzoning on built FAR. Panel (a) shows parcels that were upzoned (i.e., that saw their maximum allowed FAR increase by more than 0.25) between 2004 and 2022, excluding upzonings that were part of rezonings affecting fewer than 30 parcels. Panel (b) shows the effect of upzoning on built FAR (including the planned floorspace of buildings under construction) for the full sample. In Panel (c), I split the estimation sample between neighborhoods with floorspace prices above/below the median. In Panel (d), I split the sample between parcels whose built FAR in 2004 was above/below their neighborhood's median. Treatment effects are computed using the procedure of [de Chaisemartin and d'Haultfœuille \(2024\)](#), comparing parcels upzoned earlier vs. later. Observations are weighted by parcel size, and standard errors are clustered at the city block level. On average, upzonings increased the maximum allowed FAR by one. Before rezoning, upzoned parcels had an average built FAR of 1.3.

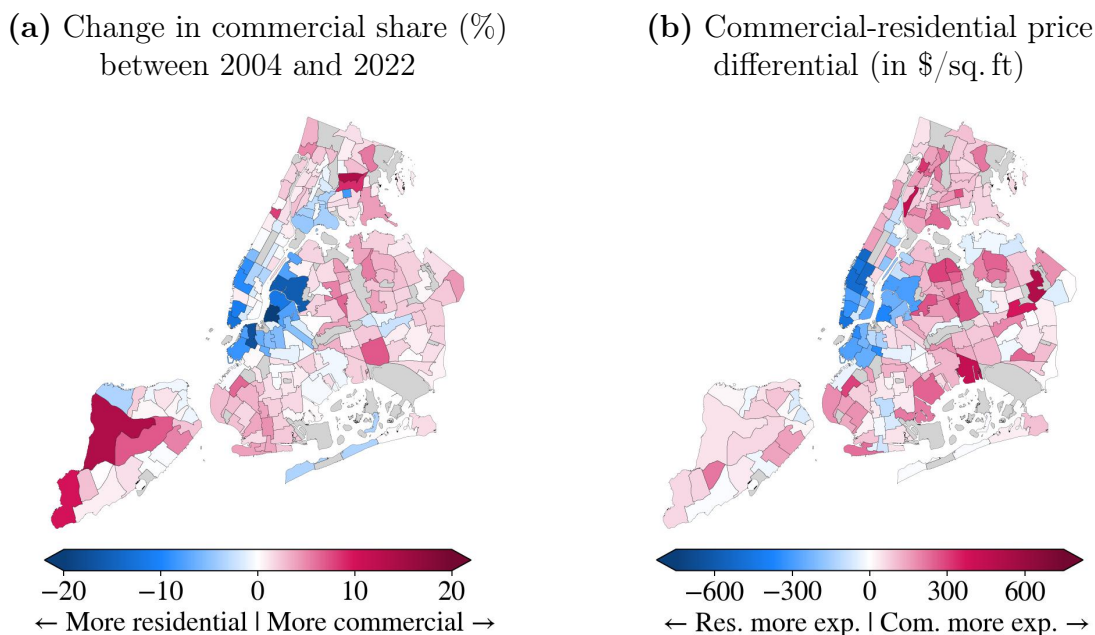
its marginal construction cost in NYC. In 2019, reported construction costs there ranged between \$160 and \$600 per sq. ft of floorspace (see, e.g., [RSMeans, 2020](#); [New York Building Congress, 2019](#)). At the same time, the price of floorspace averaged around \$700 per sq. ft, reaching much higher levels in some parts of Manhattan. This large wedge, noted by [Glaeser, Gyourko and Saks \(2005\)](#), would be arbitrated by developers if they could easily increase

the city's floorspace supply. However, even when zoning is relaxed to permit additional construction, technological constraints limit the extent to which developers can expand the supply of floorspace.

Consistent with the presence of large fixed costs of redevelopment, the effects of upzoning are highly heterogeneous. Figure 1.3(c) shows the impact of upzonings separately for neighborhoods with high vs. low floorspace prices, revealing that the effects of upzoning on construction are almost entirely driven by upzonings in expensive neighborhoods. Upzoning parcels in areas where floorspace prices are low yields little new construction; there, the reward for increasing floorspace supply is usually too low to cover redevelopment costs. Figure 1.3(d) further shows that the effects of upzoning are fully concentrated on parcels with relatively low FARs before upzoning. Upzoning parcels that already have large buildings typically does not result in additional construction, as redeveloping these structures involves forgoing a large stream of future rents and paying higher demolition costs.

Redevelopment reallocates land to its most profitable use. Land uses in NYC are becoming increasingly mixed. Figure 1.4(a) shows that, over the past decades, the city's (mostly commercial) central neighborhoods have shifted to become more residential, while its (mostly residential) peripheral neighborhoods have become more commercial.

Figure 1.4: Redevelopment reallocates land to its most profitable use



Notes: Panel (a) shows the change in the share of commercial floorspace in total floorspace between 2004 and 2022. Panel (b) shows the price difference between a sq. ft of commercial floorspace (office or retail) and a sq. ft of residential floorspace. Data is aggregated at the Neighborhood Tabulation Area level.

These city-level trends align with the financial incentives developers face. In NYC's central neighborhoods, residential floorspace is on average 30% more expensive than commercial floorspace. In contrast, in peripheral neighborhoods, commercial floorspace commands a 30% premium over residential floorspace; see Figure 1.4(b). The similarities

between the two panels show that redevelopment tends to reallocate land to its most profitable use.

These patterns reflect NYC’s broader transformation from a manufacturing hub to a service-oriented metropolis. Since 1961, the share of manufacturing in employment in New York has fallen by a factor of 10, and the central areas of the city have become quieter, less polluted, and generally more attractive for residents. Between 1950 and 2019, rents in the city center grew 2.5 times faster than in more residential neighborhoods (see Appendix Figures B.5 and B.6). Meanwhile, businesses have been gradually relocating away from the central business district.

Smartphone location data, analyzed in Appendix Figure B.7, suggests that this trend will accelerate with the rise of working from home. In the aftermath of the 2020 pandemic, visits to commercial locations in the center of NYC decreased relative to those in peripheral locations, consistent with workers spending more time in residential neighborhoods.

Regulation, however, slows this reallocation of land uses. Indeed, zoning favors the construction of commercial buildings in the city’s core and usually disallows such construction in peripheral areas (see Appendix Figure B.8). The extent to which zoning rules limiting parcels’ residential or commercial use bind can be directly tested using event studies analogous to those of Figure 1.3. In Appendix Figure G.1, I show that rezonings that newly allow a parcel to have a residential (or commercial) use spur the construction of floorspace for that use.

Remaining questions. The evidence presented in this section highlights two benefits of relaxing zoning regulations: densification and the reallocation of land to more profitable uses. However, it is insufficient to answer several key policy questions. Indeed, the upzonings we observe are in neighborhoods that were selected by planners, are often limited in scope, and can only be evaluated over short time horizons. Since the effects of upzoning vary widely and unfold over several decades, the effects of broader, more ambitious upzonings are uncertain, especially in the longer run. Moreover, the effects of upzoning on affordability remain unclear. Here, measurement is complicated by general equilibrium effects, as building more in one area affects not only local prices but also prices in other neighborhoods.

1.4 A Model of Urban Evolution

To better understand the city’s slow evolution through redevelopment and evaluate the extent to which zoning distorts this process, I build a dynamic spatial equilibrium model of the city. In the model, new floorspace is supplied by forward-looking developers whose decisions depend on prices, zoning regulations, and the characteristics of existing buildings. Workers consume residential floorspace, while firms use commercial floorspace in production.

1.4.1 Supply

The model’s supply side is designed to capture three key aspects of the redevelopment process. First, replacing old buildings with new ones entails large fixed costs, which can vary with the characteristics of the building to be demolished. Second, what developers

build is restricted by zoning and influenced by market prices. Third, as with other forms of investment, development decisions are made dynamically. Buildings deteriorate slowly over time, and the price of floorspace evolves from year to year as a function of expected future rents, leading developers to optimize when they tear down a structure to replace it with a new one.⁹

Setup. The city comprises a fixed set of parcels, each associated with a potential developer. Each year, developers decide whether to redevelop their parcel or not, taking an action a in $\mathcal{A}_{it}^{\text{allowed}} \subset \{\emptyset, R, C\}$. $\emptyset$ corresponds to not redeveloping the parcel, and R (resp., C) corresponds to building a new residential (resp., commercial) structure on it. If a developer decides to build, they choose h^{new} , the FAR of the new structure.

The set of available actions $\mathcal{A}_{it}^{\text{allowed}}$ is restricted in two ways. First, to redevelop a parcel, a developer must first buy it. Redevelopment, therefore, can happen only after a parcel is transacted on the real estate market, which happens with probability $p_i^{\text{transaction}}$ each year.¹⁰ Second, new construction must comply with zoning. If a parcel is zoned for residential uses only, a developer cannot decide to build a new commercial building on it.

Developers make decisions so as to maximize the profits they can extract from a parcel,

$$V_{it} = \max_{a \in \mathcal{A}_{it}^{\text{allowed}}} \mathbb{E}_t \left[(\pi_{ait} + \epsilon_{ait}) \cdot K_i + \frac{V_{i,t+1}(a)}{1+r} \right], \quad (1.1)$$

where π_{ait} corresponds to the payoff of taking action a on parcel i in year t , and ϵ_{ait} is a parcel-period-action-specific profit shock. To facilitate interpretation, these payoffs and shocks are measured per sq. ft of land. Hence, the total flow payoff of choosing a_{it} is $(\pi_{ait} + \epsilon_{ait}) \cdot K_i$, where K_i is the land area of parcel i . Flow payoffs are all measured relative to the status quo (choosing not to redevelop), and are given by

$$\pi_{\emptyset it} = 0, \quad \pi_{\theta it} = \max_{h^{\text{new}} \leq \bar{h}_{\theta it}} \left[\underbrace{\mathcal{P}_{it}(h^{\text{new}}, \mathbf{x}_{it}^{\text{new}}) - \mathcal{P}_{it}(h_i^{\text{old}}, \mathbf{x}_{it}^{\text{old}})}_{\text{Change in parcel value}} - \underbrace{[VC_{\theta it}(h^{\text{new}}) + FC_{it}]}_{\text{Cost of redevelopment}} \right], \quad (1.2)$$

where $\theta \in \{R, C\}$. When redevelopment occurs, developers choose the FAR of the new building h^{new} so as to maximize redevelopment profits subject to the zoning limit $\bar{h}_{\theta it}$. The developer's profit corresponds to the increase in the parcel's value brought by redevelopment minus redevelopment costs, which can be fixed or variable. The vector $\mathbf{x}_{it}^{\text{new}}$ (resp., $\mathbf{x}_i^{\text{old}}$) corresponds to the characteristics of the new (resp., old) building, including, for instance, its age, type, and condition. $\mathcal{P}(h, \mathbf{x})$ corresponds to the discounted sum of future rents derived from a building. As time passes, buildings age, and the old buildings' value $\mathcal{P}_{it}(h_i^{\text{old}}, \mathbf{x}_{it}^{\text{old}})$

⁹Many studies, including [Titman \(1985\)](#), [Quigg \(1993\)](#), and [Murphy \(2018\)](#), provide evidence of developers' forward-looking behavior.

¹⁰Buildings in NYC are usually not directly redeveloped by their owners. Rather, the redevelopment process tends to begin with a developer purchasing a building and applying for a building permit. Appendix Figure C.4 shows that the sale probability of a parcel rises sharply in the years preceding redevelopment, reaching nearly 40% in the year before permit issuance, compared to a baseline annual sale probability of just 5.6%. Overall, 73% of redeveloped parcels are sold in the year redevelopment begins or in one of the two years prior.

decreases, making redevelopment increasingly profitable. Buildings that are shorter and have less valuable characteristics are more profitable to redevelop and hence are more likely to be demolished.

Variable costs VC correspond to construction costs, which increase with the size of the new building h^{new} . Fixed costs FC are those associated with demolition, vacating existing tenants, and permitting. I allow them to vary from one parcel to the next.

I assume that developers are perfectly foresighted and price-taking: they know the path of future costs and prices, and do not condition their choices on other developers' actions. This is justified by the very large number of parcels in the city, as well as the competitive nature of the real estate development industry in NYC, which encompasses thousands of firms and has a low degree of concentration (see Glaeser et al., 2005).

Parametrization. To estimate the model, I parametrize price and cost functions. For prices, I assume that the expected value of a parcel (per sq. ft of land) is a function of its geographic location and the characteristics of the building occupying it, with

$$\mathcal{P}_{it}(h_{it}, \mathbf{x}_{it}) = \underbrace{P_{\theta nt}}_{\text{Baseline price per sq. ft}} \times \underbrace{Q(\mathbf{x}_{it})}_{\text{Building quality}} \times \underbrace{h_{it}}_{\text{Building FAR}}. \quad (1.3)$$

Here, $P_{\theta nt}$ measures the price of a sq. ft of floorspace of type θ (residential or commercial) and of average quality in neighborhood n and time t . It is the discounted sum of future rents $R_{\theta nt}$:

$$P_{\theta nt} = \sum_{k=0}^{\infty} \frac{R_{\theta n, t+k}}{(1+r)^k}. \quad (1.4)$$

This price is inflated or deflated by $Q(\mathbf{x}_{it})$, which reflects the building's quality as a function of its characteristics (including its FAR h_{it}). Finally, the quality-adjusted price per sq. ft is multiplied by the number of sq. ft of floorspace per sq. ft of land, h_{it} .

Studies estimating the production function for housing have found it is well approximated by a Cobb-Douglas aggregation of land and capital, for both small and tall structures (Epple, Gordon and Sieg, 2010; Combes, Duranton and Gobillon, 2021; Ahlfeldt and McMillen, 2018). Hence, I assume that investing k units of capital on a sq. ft of land allows the developer to build a structure of FAR $h = k^\zeta$, where ζ is the capital cost share in construction. The cost of a unit of capital is given by $\alpha_{\theta it} e^{\eta_{it}}$, where η_{it} is a construction cost shock. Hence, the variable cost function is given by

$$VC_{\theta it}(h_{it}^{\text{new}}) = \alpha_{\theta it} (h_{it}^{\text{new}})^{1/\zeta} e^{\eta_{it}}. \quad (1.5)$$

Finally, while some fixed costs of redevelopment are independent of the height of the old building, others increase with the amount of floorspace to demolish. Furthermore, demolition costs are likely higher for protected parcels (either as landmarks or because they are in historic districts) and in denser neighborhoods. Hence, I parametrize fixed costs as

$$FC_{it} = (\delta^0 + \delta^{\text{demolition}} h_{it}^{\text{old}} + \delta^{\text{density}} \bar{h}_{n(i)t}) (1 + \delta^{\text{protected}} \mathbb{1}_{it}^{\text{protected}}), \quad (1.6)$$

where δ^0 reflects the fixed cost (per sq. ft of land) associated with starting the construction of

a building on vacant land that is unprotected (i.e., not landmarked or in a historic district). The parameter $\delta^{\text{demolition}}$ measures how fixed costs grow with the FAR of the structure to demolish, δ^{density} how they grow with the average FAR of parcels in the same neighborhood $\bar{h}_{n(i)t}$, and $\delta^{\text{protected}}$ quantifies the extent to which protected parcels (with $\mathbb{1}_{it}^{\text{protected}} = 1$) are harder to redevelop.

Additional assumptions. To facilitate estimation, I make three additional assumptions. First, I assume that redevelopment is a terminal decision, i.e., that once a building has been redeveloped, it will not be redeveloped again. This assumption simplifies estimation by turning the developer’s dynamic decision process into an optimal stopping problem. It is also unlikely to substantially affect my results, as the median age of redeveloped buildings is 85 years, and I only focus on the city’s evolution in the next 40 years in counterfactuals.

Second, I assume that the profit shocks ϵ_{ait} are distributed i.i.d. according to a Type I Extreme Value distribution with a mean of zero and a scale parameter $\sigma_{it} = \sigma_{\epsilon}^0 + \sigma_{\epsilon}^{\text{demolition}} h_{it}^{\text{old}} + \sigma_{\epsilon}^{\text{density}} \bar{h}_{n(i)t}$. $\sigma_{\epsilon}^{\text{demolition}}$ and $\sigma_{\epsilon}^{\text{density}}$ are both expected to be positive, capturing the fact that the shocks facilitating or hindering redevelopment are more variable for parcels with larger buildings or in dense neighborhoods.

Finally, I make some timing assumptions. Each period, developers choose an action in $\mathcal{A}^{\text{allowed}}$ after observing profit shocks ϵ_{ait} , but they observe the cost shock η_{it} only if they decide to redevelop. After observing η_{it} , the developer chooses the amount of floorspace they want to provide. This allows me to estimate the model in steps, as described in Section 1.5.1.

1.4.2 Demand

To understand the effect of redevelopment on prices and welfare, I develop a model of the demand for floorspace, building on the quantitative spatial models of [Ahlfeldt et al. \(2015\)](#) and [Tsivanidis \(2019\)](#). These workhorse models rely on data on commuting times and floorspace prices (or land prices) but not on floorspace quantities, which are treated as structural residuals. Accordingly, these models are typically used to evaluate the effects of changing commuting times but not to predict the effects of increasing the amount of floorspace in a neighborhood. In contrast, the model developed in this section directly incorporates data on quantities of floorspace, which I can vary in counterfactuals. Matching the observed data on the amount of floorspace consumed in each neighborhood requires modeling two important features of reality: the high heterogeneity in workers’ earnings capacity and the fact that housing is a necessity good, making preferences non-homothetic. A model without these features would fail to rationalize, for instance, that the typical resident of the Upper East Side spends more than five times as much on housing as the typical resident of the Bronx.¹¹

Workers. I partition the city into a finite set of neighborhoods populated by $\mathcal{L}$ workers as well as firms that produce a single final good, the numéraire.¹² Workers consume the final

¹¹In Appendix E.1, I calibrate a simple quantitative spatial model with homogeneous agents and homothetic preferences and describe why it fails to adequately predict floorspace consumption.

¹²In the estimated model, neighborhoods correspond to NYC’s 188 Neighborhood Tabulation Areas (NTAs) as well as an additional location corresponding to the rest of the metropolitan area (see Appendix E.3

good and housing, and choose where to live and where to work. They are perfectly mobile within the city.

Each worker has a type θ and a skill level $s(\theta)$, corresponding to the number of efficiency-adjusted units of labor they can provide. The distribution of worker types is given by L , with $\mathcal{L} = \int_{\theta} L(\theta) d\theta$. A worker o living in neighborhood n and working in neighborhood m has the following Stone-Geary utility function:

$$U_{nmo} = \frac{B_n z_{no}^H z_{mo}^W}{d_{nm}} C_{nmo}^{1-\beta} (H_{nmo} - \underline{H}_n)^{\beta}, \quad (1.7)$$

where C_{nmo} is the worker's consumption of the final good, H_{nmo} is their consumption of residential floorspace, B_n is the amenity level in neighborhood n , and d_{nm} is an iceberg commuting cost. This commuting cost is a function of the commuting time τ_{nm} between neighborhoods n and m , with $d_{nm} = \exp(\kappa\tau_{nm})$. z_{no}^H and z_{mo}^W are idiosyncratic shocks capturing workers' heterogeneous preferences for different home and work locations, respectively. I assume these shocks are drawn independently from Fréchet distributions with shapes ε^H and ε^W —larger values of ε correspond to more homogeneous worker preferences. I assume the following timing of shocks: workers draw z_{no}^H , decide where to live, and then draw z_{mo}^W and choose where to work. Finally, housing is a necessity good, and workers must consume at least $\underline{H}_n$ sq. ft of housing to live in neighborhood n . Poorer households spend a larger share of their income on housing, and that share approaches β as income grows.

In counterfactuals, I measure welfare changes for workers of each type as the equivalent variation—the income gain that would be required in the initial equilibrium to make workers achieve the expected utility level they reach in the counterfactual equilibrium.

Migration and congestion. As time passes, redevelopment increases the total amount of floorspace in the city, which tends to lower residential rents and increase wages. This raises residents' expected utility and causes the city's population to grow through migration. I assume that the number of workers of each skill level in the city grows with their expected welfare, with a migration elasticity ε_M of 3 (this corresponds to a typical estimate in the literature—see, e.g., [Bryan and Morten, 2019](#); [Bilal, 2023](#); [Hornbeck and Moretti, 2024](#)).

Population growth increases congestion on transportation networks. To reflect this in the model, I assume that commuting times τ increase with the total population of the city $\mathcal{L}$, with a congestion elasticity ε_C of 0.15, on the higher end of values identified in the literature ([Akbar and Duranton, 2017](#); [Couture et al., 2018](#); [Akbar et al., 2023](#)).

Firms. Using as inputs labor L_{Cm}^{eff} and floorspace H_{Cm} , firms in each neighborhood m competitively produce a single final good that is freely traded within the city. Firms in neighborhood m have access to the following Cobb-Douglas production technology:

$$Y_m = A_m (H_{Cm})^{\alpha_m} (L_{Cm}^{\text{eff}})^{1-\alpha_m}, \quad (1.8)$$

for details). NTAs are defined by the NYC Department of City Planning as aggregates of census tracts that approximately correspond to historical neighborhoods of the city (e.g., Chinatown, Hamilton Heights). They are mapped in Figure 1.4.

where A_m is a productivity shifter and α_m governs the importance of floorspace relative to labor in production in neighborhood m . L_{Cm}^{eff} is the total amount of efficiency units of labor provided by workers in m . The wage of an efficiency unit of labor provided in m is denoted by w_m .

Floorspace. The amount of residential floorspace H_{Rn} and commercial floorspace H_{Cn} available in a neighborhood is given by the sum of the (quality-adjusted) floorspace supplied by all parcels in that neighborhood. Residential floorspace can be rented at a price R_{Rn} and commercial floorspace can be rented at a price R_{Cn} . As richer households tend to own more real estate, I consider that rental income is redistributed to workers proportionally to their wage income.¹³

Endogenous productivities and amenities. New construction affects both local amenities and productivity. Indeed, increasing the housing stock in a neighborhood may make it more attractive by fostering social interaction or encouraging public investment. However, it also increases congestion, decreasing the neighborhood's amenity value. Similarly, building commercial floorspace in a neighborhood will increase the number of people working there and make it more productive through agglomeration spillovers. Firms' activity may also directly affect local amenities. These spillovers (whether real or perceived) motivated the introduction of zoning in NYC, whose early planners separated commercial areas from residential ones to mitigate the externalities of factories and stores on nearby residents.

To account for these potential spillovers, I follow [Ahlfeldt et al. \(2015\)](#) and allow amenity and productivity levels in a neighborhood to vary with the density of residents ($\tilde{L}_{Rn}$) and jobs ($\tilde{L}_{Cn}$) there:

$$B_n = \bar{B}_n \tilde{L}_{Rn}^{\gamma_{RR}} \tilde{L}_{Cn}^{\gamma_{CR}} \quad A_m = \bar{A}_m \tilde{L}_{Rm}^{\gamma_{RC}} \tilde{L}_{Cm}^{\gamma_{CC}}. \quad (1.9)$$

Here, $\bar{B}_n$ and $\bar{A}_i$ measure the fundamental amenity and productivity attributes of neighborhoods, respectively, capturing exogenous features that make an area more productive or attractive (e.g., riverside views, access to parks, or proximity to key transportation nodes). The spillover parameters γ capture agglomeration externalities. For instance, γ_{CC} captures productivity spillovers leading firms to make other nearby firms more productive, and γ_{CR} measures spillovers of commercial activity on nearby residents.¹⁴

¹³Total rental income is given by $I_{\text{rent}} = \sum_n (R_{Rn}H_{Rn} + R_{Cn}H_{Cn})$, and the total income of worker o is therefore given by $I_o = w_ms(\theta_o)(1 + \mathcal{T}) \equiv w_ms(\theta_o) \left(1 + \frac{I_{\text{rent}}}{\mathcal{LE}[ws(\theta)]}\right)$. The budget constraint they face is $I_o \geq C_{nmo} + R_{Rn}H_{nmo}$.

¹⁴The parametrization I retain differs from that of [Ahlfeldt et al. \(2015\)](#) in two ways. First, [Ahlfeldt et al. \(2015\)](#) only consider spillovers of firms on firms and of residents on residents, i.e., both γ_{CR} and γ_{RC} are assumed to be zero. I allow these parameters to be non-zero here because spillovers across uses were a key justification for zoning. However, I estimate these parameters to be small, justifying [Ahlfeldt et al. \(2015\)](#)'s parametrization. Second, in [Ahlfeldt et al. \(2015\)](#), a location's amenity and productivity levels depend not only on the density of residents and jobs in that location, but also on the density of nearby locations. This is justified by the very granular scale of [Ahlfeldt et al. \(2015\)](#)'s analysis, where locations are city blocks. Here, the locations I consider are substantially larger, with the median neighborhood containing 135 city blocks. The existing literature and my analysis below suggest that agglomeration spillovers decay quickly over space, justifying my focus on within-neighborhood spillovers.

The variables described above (floorspace, rents, wages, productivities, and amenities, in particular) vary from one period to the next. Here, their time subscripts have been omitted for simplicity.

1.4.3 Equilibrium

A dynamic equilibrium of the model is a path of floorspace supplies ($\mathbf{H}_R, \mathbf{H}_C$), rents ($\mathbf{R}_R, \mathbf{R}_C$), prices ($\mathbf{P}_R, \mathbf{P}_C$), population and jobs distributions ($L, \mathbf{L}_R, \mathbf{L}_C, \mathbf{L}_C^{\text{eff}}$), productivities $\mathbf{A}$, amenities $\mathbf{B}$, wages $\mathbf{w}$, transfers $\mathcal{T}$, and commuting costs $\boldsymbol{\tau}$ for each period t such that:

1. The supply of floorspace on each parcel is governed by developers' profit maximization (1.1), floorspace supply in each neighborhood is the expected sum of the floorspace supplied by each of its constituent parcels, and floorspace prices are the discounted sum of rents (1.4).
2. In each period, $L, \mathbf{L}_R, \mathbf{L}_C, \mathbf{L}_C^{\text{eff}}, \mathbf{R}_R, \mathbf{R}_C, \mathbf{w}, \mathcal{T}$, and $\boldsymbol{\tau}$ are such that workers maximize utility (1.7), firms maximize profits (1.8), amenities and productivities are determined by equation (1.9), total population is governed by the migration elasticity ε_M , commuting times are governed by the congestion elasticity ε_C , and floorspace markets clear.

To find equilibrium paths, I use the iterative procedure described in Appendix F. Starting from a guess for the paths of prices and average densities in each neighborhood, I compute developers' expected path of decisions at the parcel level and aggregate them into neighborhood floorspace supplies for each year. I then solve the demand system year by year to recover rents, wages, commuting costs, amenities, and productivities that clear floorspace markets. Floorspace prices are finally computed as the discounted sum of rents, yielding updated price and density paths. An equilibrium is reached when the guessed and recovered paths coincide.¹⁵

1.5 Model Estimation

1.5.1 Supply

To characterize developers' behavior, I estimate the parameters that determine redevelopment profits in turn: building prices, variable and fixed construction cost parameters, transaction probabilities, and the dispersion of idiosyncratic profit shocks.

Parcel values. To measure the value of buildings as a function of their characteristics, I estimate (separately for residential and commercial structures) on the real estate transactions data the following hedonic model, derived from equation (1.3):

$$\log(p_s) = P_{1,\theta n} + P_{2,\theta bt} + \boldsymbol{\beta}_\theta \cdot \mathbf{x}_s + \nu_s, \quad (1.10)$$

¹⁵The tractability of this procedure benefits substantially from parallelization: within each iteration, I parallelize the computation of developers' expected decision paths across parcels.

where p_s is the price per sq.ft observed in sale s , n (resp., b) is the neighborhood (resp., borough) where the sale took place, t is the sale year, and ν_s is an idiosyncratic shock to the observed sale price. β_θ measures the extent to which different parcel characteristics in $\mathbf{x}_s$ are valued. $\mathbf{x}_s$ includes the building's FAR, age, landmark status, quality grade, construction material, and broad type (e.g., office, retail, industrial), as well as the size of the unit(s) sold.

The value of buildings $\mathcal{P}(h, \mathbf{x})$ that developers consider when making decisions is the structure's use value (the discounted sum of future rents), which does not include the option value of redevelopment. Indeed, through redevelopment, developers forgo the use value of the building they demolish, and the new building's value reflects only its use value, as redevelopment is a terminal action. The transaction prices I observe, however, likely capture both use values and option values. To ensure my estimate of $\mathcal{P}(h, \mathbf{x})$ reflects only the former, I estimate equation (1.10) on the sample of parcels at or above the maximum FAR limit, which have a near-zero probability of redevelopment (0.02% each year) and whose option value of redevelopment is hence negligible. I report estimated parameters for equation (1.10) in Appendix Table D.1. They allow me to estimate the prices entering the developers' profit function.¹⁶

Variable costs. The second component of the profit function is variable costs, parameterized in equation (1.5). Their level depends on the cost of materials, $\alpha_{\theta it} e^{\eta_{it}}$, and the rate at which they grow with FAR is governed by the capital cost share ζ .

In the absence of zoning regulations, developers would choose FARs equating the price of floorspace with its marginal cost of construction, and variable costs could be obtained by comparing the FARs chosen by developers with the floorspace prices they faced when making construction decisions. However, the presence of zoning constraints complicates estimation, as the FAR chosen by a developer may no longer equalize the marginal construction cost of floorspace with its price. In many construction projects, maximum FAR limits are indeed binding, and the chosen FAR of new buildings is often close to the zoning code's maximum FAR (see Appendix Figure C.5).¹⁷

Despite these constraints, developers' choices respond strongly to prices. Appendix Figure C.6 shows that developers facing higher floorspace prices choose FARs that are higher and closer to the zoning limit. I exploit this variation—FARs chosen by developers whose optimal decisions lie below the zoning limit, and the probability that the zoning constraint binds—to identify the parameters of the floorspace production function.

Specifically, I parametrize $\alpha_{\theta it}$ as $\alpha^0 \alpha_{\theta b} \alpha_t$, where b is the borough in which parcel i is located. I calibrate the $\alpha_{\theta b}$ and α_t parameters using engineering estimates and estimate

¹⁶To recover expected prices in levels from the coefficients of equation (1.10), estimated in logs, I assume that the price shock ν is distributed $\mathcal{N}(0, \sigma_\nu^2)$, giving $\hat{p} = \exp(\hat{P}_{1, \theta n} + \hat{P}_{2, \theta bt} + \hat{\beta}_\theta \cdot \mathbf{x} + \hat{\sigma}_\nu^2/2)$.

¹⁷Many buildings have a built FAR above the maximum allowance, as developers can exempt part of the built floorspace from the limit (e.g., basements, attics, or mechanical space) or ask for a small relaxation of the FAR limit. To account for cases where developers build above the maximum allowed FAR, I assume that when making redevelopment decisions, developers face the constraint $h_{\theta it} \leq \bar{h}_{\theta it} = (1 + b) \bar{h}_{\theta it}^{\text{code}}$, where $\bar{h}_{\theta it}^{\text{code}}$ is the maximum allowed FAR in the zoning code, and b is a shock revealed to the developer at the same time as the cost shock and distributed according to an exponential distribution with parameter λ . I estimate λ at 1.93, suggesting that in the median construction project, developers can exempt up to 26% of the floorspace they build from counting towards the FAR limit, which is consistent with the types of spaces that are excluded from the FAR limit.

the baseline cost of materials α^0 .¹⁸ Furthermore, I assume the cost shock η_{it} is distributed $\mathcal{N}(0, \sigma_\eta^2)$. This lets me compute, for each new construction project, the probability that the developer chooses any FAR as a function of the price of floorspace, the zoning limit, and variable cost parameters. I estimate these parameters using maximum likelihood (see Appendix D.1 for details).¹⁹

The parameters I recover are shown in Table 1.1(a). They allow me to trace out the variable costs faced by developers as a function of the FAR of the structure they choose to build, as well as the typical FAR that developers will choose when faced with different price levels. These functions are plotted in Appendix Figure D.1. I estimate that, in 2019, building a small structure with an FAR of 1 would cost developers \$143/sq. ft on average. A mid-rise structure, with an FAR of 4, would typically cost \$535/sq. ft, and the average construction costs for a skyscraper (with an FAR of 20) would rise to \$2,468/sq. ft.

These results are consistent with available engineering estimates. [RSMeans \(2020\)](#) reports expected construction costs for a 1-to-3-story apartment building in Manhattan in 2019 at \$185/sq. ft. The [New York Building Congress \(2019\)](#), which tracks the construction costs of larger structures, reports average construction costs between \$400 and \$600/sq. ft. To benchmark my cost estimates for very tall buildings, I recovered the building costs of skyscrapers built since 2000 (which had an average FAR of 23.9). I find wide variation in their reported construction costs, around an average of \$1,712/sq. ft (see Appendix Figure A.10). The estimated capital cost share ζ , at 0.51, is similar to the estimate of 0.54 that [Combes et al. \(2021\)](#) report for Paris, and implies that variable construction costs, which are proportional to $h^{1/\zeta}$, increase almost quadratically with FAR.

Transaction probabilities. I observe the transaction of individual parcels in the real estate sales data, allowing me to estimate $p_i^{\text{transaction}}$. To account for the fact that some parcels are more likely to be transacted than others, I predict $p_i^{\text{transaction}}$ by estimating a probit regression of parcel sale dummies on parcel characteristics (see Appendix Table D.2). I find that buildings with multiple ownership, like condominiums, tend to be transacted much less frequently. Offices also tend to be transacted less frequently, consistent with the long durations of office leases ([Gupta, Mittal and Van Nieuwerburgh, 2024](#)).

Fixed costs. With estimates of the price function $\mathcal{P}$ and the variable cost function VC in hand, I can compute the expected flow profit from redevelopment net of fixed costs: $\tilde{\pi}_{\theta it} = \mathbb{E} [\max_{h^{\text{new}} \leq \bar{h}_{\theta it}} \{ \mathcal{P}_{it}(h^{\text{new}}, \mathbf{x}_{it}^{\text{new}}) - \mathcal{P}_{it}(h_i^{\text{old}}, \mathbf{x}_{it}^{\text{old}}) - VC_{\theta it}(h_{it}^{\text{new}}) \}]$.²⁰ Intuitively, $\tilde{\pi}_{\theta it}$ is the surplus available to cover fixed costs if the parcel is redeveloped. Empirically, redevelopment probabilities are essentially zero when $\tilde{\pi} < 0$, but then increase sharply with $\tilde{\pi}$ when it is positive (see Appendix Figure D.2). Furthermore, conditional on $\tilde{\pi}$, redevelopment

¹⁸Specifically, I use the [RSMeans \(2020\)](#) time series for construction costs as well as estimates of the relative construction costs in different boroughs and for different building types. α^0 captures the cost of capital for residential buildings in Manhattan in 2019.

¹⁹While the land use panel I constructed covers 2004–2022, I sometimes lack data on the characteristics of the buildings whose construction was not completed by 2022. Hence, I restrict my sample to 2004–2019 to estimate variable and fixed costs and simulate the city’s evolution from 2019 onwards in counterfactuals.

²⁰I set the characteristics $\mathbf{x}_{it}^{\text{new}}$ of new buildings to the average characteristics of buildings of type θ constructed after 2000 (except for the building’s age, which is set to zero).

Table 1.1: Cost parameter estimates

(a) Variable cost parameters			
α^0	Baseline cost of materials	80.4	(2.3)
ζ	Capital cost share	0.51	(0.005)
σ_η	Cost shock standard deviation	1.08	(0.02)
(b) Fixed cost parameters			
δ^0	Base fixed cost	175.2	(7.7)
$\delta^{\text{demolition}}$	Demolition multiplier	1853.8	(37.6)
δ^{density}	Neighborhood density multiplier	318.5	(11.3)
$\delta^{\text{protected}}$	Protected parcels multiplier	0.77	(0.03)
(c) Profit shock parameters			
σ_ϵ^0	Base profit shock variance	73.7	(2.9)
$\sigma_\epsilon^{\text{demolition}}$	Demolition multiplier	138.8	(5.1)
$\sigma_\epsilon^{\text{density}}$	Neighborhood density multiplier	248.2	(4.9)

Notes: This table reports parameter estimates for the variable cost function (equation 1.5), the fixed costs that developers must pay if they redevelop a parcel (equation 1.6), and the σ_ϵ parameters determining the scale of the Type 1 Extreme Value distribution from which the profit shocks ϵ are drawn. All costs are reported in 2019 dollars.

is less likely on parcels that are already densely built or in denser neighborhoods, suggesting higher fixed costs of redevelopment there. I leverage this variation to estimate fixed cost parameters.

Because development decisions are dynamic, estimation requires assumptions about developers' expectations regarding the evolution of variables after 2019. I assume developers expect zoning constraints, demand parameters (fundamental productivities and amenities), and cost parameters to stay at their 2019 level. Furthermore, developers form rational expectations about the path of prices and floorspace supplies after 2019: they expect floorspace quantities to evolve according to the supply model, rents to evolve according to the demand model, and prices to correspond to the sum of future rents, discounted at the rate r , which I calibrate to 5%.

I estimate fixed costs to best match the relationship between developers' decisions and the expected profit from different choices. This requires repeatedly solving the dynamic equilibrium for different candidate values of the fixed cost parameters. Indeed, the option value of future redevelopment depends on the entire future path of prices. This path is an equilibrium outcome determined by the collective behavior of all developers, which, in turn, depends on the very fixed costs being estimated. My estimation procedure extends Rust's 1987 nested fixed point algorithm to a spatial equilibrium setting. In the inner loop, I solve developers' dynamic decision problem given prices and fixed costs. In the middle loop, I impose spatial equilibrium each period to obtain the path of neighborhood prices. In the outer loop, I compute the likelihood that the developer takes each action $(\emptyset, R, C)$ in each period and recover fixed-cost parameters by maximum likelihood.

Table 1.1(b) reports the estimated fixed cost parameters. I find that fixed costs increase sharply with the size of the building to be demolished, and are also larger in denser neighborhoods. Therefore, construction creates large adjustment costs that hinder future

redevelopment. Fixed costs are also estimated to be 77% higher for parcels that are landmarked or in a historic district.

Model validation. My model aims to accurately predict developers’ decisions on the extensive and intensive margins. On the extensive margin, I estimate the probability that a parcel will be redeveloped in any given year as a function of prices and zoning. On the intensive margin, I predict how much floorspace a developer will build if they proceed with redevelopment.

Appendix Figures D.3(a) and D.4 evaluate model fit on the extensive margin, and show that redevelopment probabilities in the model closely align with those observed in the data, both for the full sample and in different parts of the city. Appendix Figure D.3(b) focuses on the intensive margin: for redeveloped parcels, I compare the FAR chosen by developers with the model’s prediction. Reassuringly, these two variables are tightly correlated. The model also successfully predicts whether developers build a commercial building or a residential one, conditional on proceeding with redevelopment—see Appendix Figure D.3(c).²¹

In counterfactuals, I use the model to predict the evolution of NYC if zoning were relaxed in different ways. Therefore, the model will be successful only if it can correctly predict the causal effect of upzoning. To assess whether this is likely the case, I exploit the fact that my model’s estimation does not rely on Figure 1.3’s difference-in-differences estimates of the effect of upzoning on construction. More generally, upzonings only play a limited role in estimation, as upzoned parcels account for only a small share of the city’s parcels (6%) and redevelopment (10%). To validate my model, I first reestimate the model’s cost parameters using only the sample of parcels that have not been upzoned. I then predict the evolution of the upzoned parcels with and without the policy change, yielding a model-implied causal effect of upzoning. I finally compare these estimates to the quasi-experimental estimates of Figure 1.3. The two align (see Figure 1.5), indicating that the model is successful at predicting the effect of zoning changes.

1.5.2 Demand

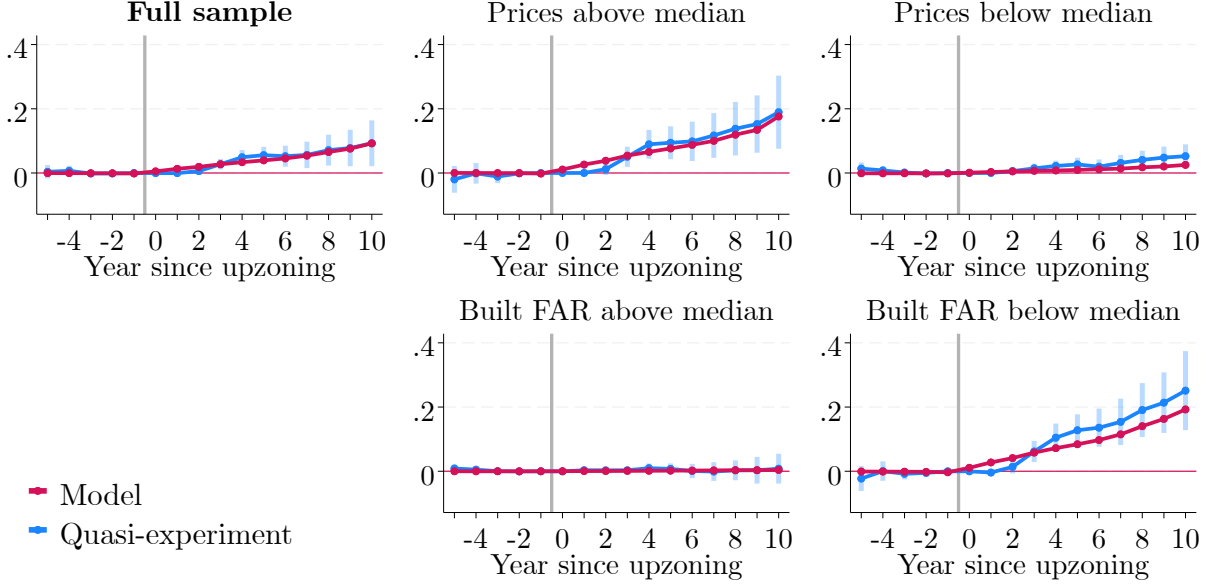
The procedure I follow to calibrate the demand model builds on the quantitative spatial economics literature (e.g., Ahlfeldt et al., 2015; Redding and Rossi-Hansberg, 2017; Tsivanidis, 2019) and is described in detail in Appendices E.2 and E.3.

First, I calibrate the distribution of worker skills to a truncated lognormal to match the observed income distribution in the ACS (after excluding households with an annual income below \$12,000). The ACS data also shows that the richest households in the NYC metropolitan area spend about 10% of their income on housing, and I therefore calibrate β to 0.1.

Second, I estimate the dispersion of workers’ idiosyncratic preferences ε^W for work locations using data on commuting patterns in NYC. I extract from the LEHD Origin-Destination Employment Statistics (LODES) dataset the number of workers living and

²¹In Appendix Figure D.5, I conduct a corresponding out-of-sample exercise: I reestimate the model using the first half of my panel (2004–2011), and then predict the evolution of the city over the second half of my panel (2012–2019). The developers’ predicted behavior on the intensive and extensive margins again aligns with their observed behavior.

Figure 1.5: Model-implied vs. quasi-experimental effects of upzoning on built FAR



Notes: This figure compares the event study coefficients from Figure 1.3—which measure the effect of upzoning on built FAR, including floorspace under construction—with the model’s predicted effects. To generate these predictions, I reestimate the model’s cost parameters using only parcels that were not upzoned, and then simulate the impact of the observed upzonings.

working in each neighborhood ($\mathbf{L}_R$ and $\mathbf{L}_C$), as well as commuting flows between them. I associate these flows with commuting times τ taken from Google Maps. This allows me to estimate the commuting elasticity through a gravity equation following Ahlfeldt et al. (2015)—I find $\kappa\varepsilon^W$ to be approximately 0.044.²² Assuming that the commuting cost parameter κ is equal to 0.01 (following Ahlfeldt et al., 2015, and Tsivanidis, 2019), I find a preference heterogeneity parameter ε^W equal to 4.4.

Third, I jointly calibrate wages $\mathbf{w}$, amenity levels $\mathbf{B}$, subsistence levels of housing $\mathbf{H}$, and the dispersion of idiosyncratic preferences for home locations ε^H . Intuitively, wages can be estimated using the spatial distribution of jobs. Areas that attract many commuters are inferred to be high-wage locations, especially if they are far away from where workers live. Amenity levels can be derived from residential prices and the population distribution: neighborhoods that attract many residents despite high residential prices are inferred to provide high amenity levels. Subsistence levels of housing are estimated using data on residents’ average spending on housing in each location—higher levels of $\mathbf{H}$ are associated with higher housing consumption. Finally, the dispersion of idiosyncratic preferences for home locations is calibrated from the observed sorting of workers across neighborhoods. In the absence of idiosyncratic preferences, workers would perfectly sort across neighborhoods as a function of their skill level, with high-skill workers systematically choosing high-amenity locations. The more dispersed idiosyncratic preferences are, the less pronounced such sorting patterns become. I calibrate ε^H to 2.9 to match the observed variation in average incomes

²²Following Dingel and Tintelnot (2020), I estimate the commuting elasticity using a Poisson Pseudo-Maximum Likelihood (PPML) estimator, which corrects for the bias associated with origin-destination neighborhood pairs with no commuters.

across neighborhoods.

Fourth, calibrated wages, along with the supply and price of commercial floorspace in each location, allow me to recover the parameters of the production function: location-specific productivities $\mathbf{A}$ and floorspace shares α .

Spillovers. The last parameters to calibrate are those governing agglomeration spillovers. One way to learn about these spillovers is to study the effects of new construction on the rents in nearby buildings. For example, suppose that constructing a commercial building in an area decreases the rents of nearby residential units. This would suggest that commercial activity induces negative externalities on nearby residents.

Therefore, to estimate γ , I identify large construction events, which I define as new buildings that increased the amount of residential or commercial floorspace within 500 ft by more than 10% and by more than 20,000 sq. ft. After isolating these events, I draw 500-ft-radius disks around them and track the rent levels of preexisting buildings within these disks.²³ Data on rents is available for 290 residential construction events and 112 commercial construction events. Appendix Figure E.2 shows their location in the city and Appendix Figure E.3 illustrates the construction of the disks used as the unit of analysis.

Comparing disks that received construction earlier vs. later allows me to evaluate the effect of new construction on rents. In Figure 1.6, I use the procedure of [de Chaisemartin and d’Haultfoeuille \(2024\)](#) to measure the effect of new construction on the quantity and price of floorspace.²⁴

Upon the completion of a large structure, the amount of floorspace within a disk rises sharply. Over time, this leads to a drop in rents in nearby structures of the same type. However, constructing one type of floorspace has no discernible effect on the other type of floorspace.

The procedure described above is similar to that proposed by [Li \(2022\)](#), who estimated the effects of constructing large residential towers on residential rents within 500 ft. My finding that increasing the housing stock lowers residential rents is consistent with hers, as well as those of the broader literature ([Pennington, 2021](#); [Asquith et al., 2023](#)). I further find that these price effects decay quickly over space (they are not statistically significant beyond 500 ft—see Appendix Figure E.5), echoing previous findings ([Arzaghi and Henderson, 2008](#); [Ahlfeldt et al., 2015](#); [Pennington, 2021](#); [Li, 2022](#); [Baum-Snow et al., 2024](#); [Blanco and Sportiche, 2024](#)).

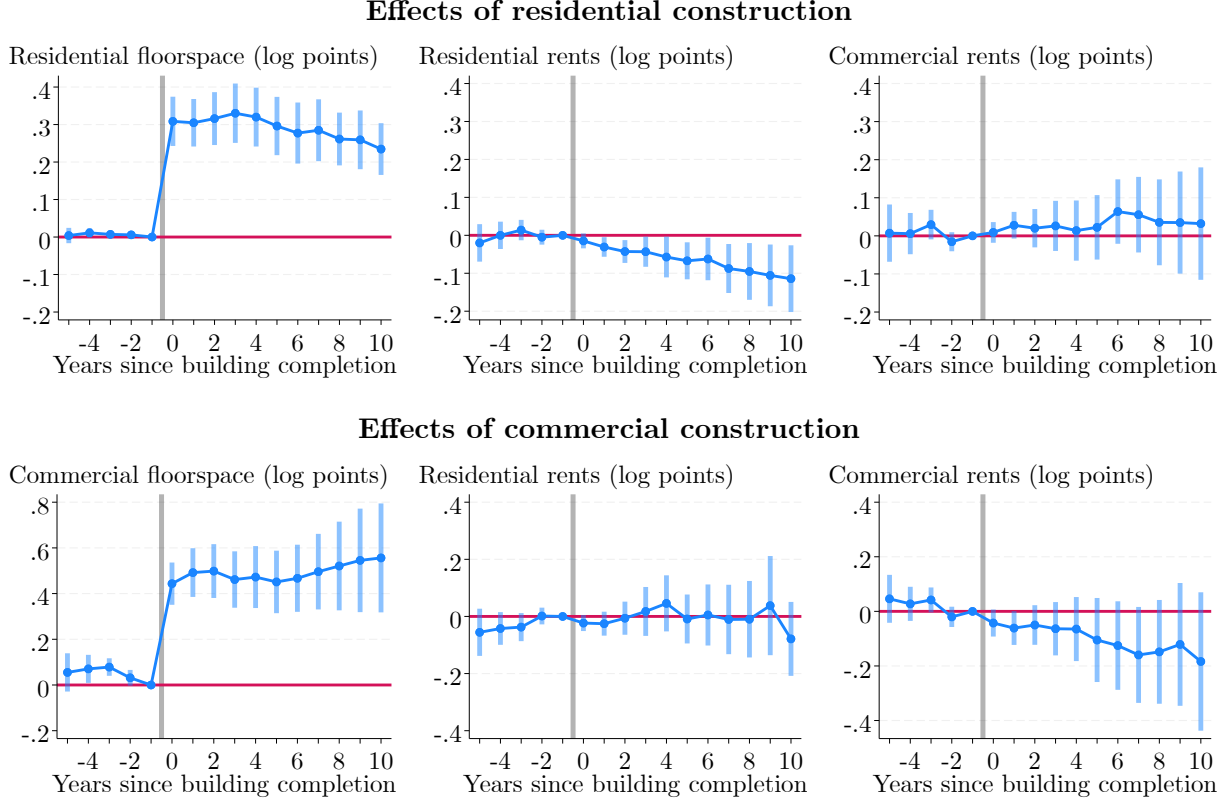
The event studies of Figure 1.6 provide reduced-form estimates of the elasticities of local residential/commercial rents to the local residential/commercial floorspace supply, which I denote by ε^{RR} , ε^{RC} , ε^{CR} , and ε^{CC} . For instance, the elasticity of residential rents to residential floorspace supply, ε^{RR} , is estimated at -0.42.

In the absence of agglomeration externalities, increasing the housing stock in an area leads to a decline in residential rents there. Negative externalities from residents on other residents ($\gamma^{RR} < 0$) will amplify the magnitude of this price response, while positive externalities

²³When computing rent levels, I exclude buildings with rent-stabilized units, as they might not appropriately reflect market conditions.

²⁴In dense areas of the city, the disks used as the observation in this analysis sometimes overlap. I exclude such overlapping events from the event study, and show in Appendix Figure E.4 that the results of Figure 1.6 do not meaningfully change when these overlapping disks are included.

Figure 1.6: Effects of new construction on floorspace quantities and rents



Notes: This figure shows event studies estimating the effects of new construction projects on the quantities and rent prices of floorspace in a 500-ft-radius disk around the new building. In the top panel, I study the effects of constructing a new residential building that increases residential floorspace within 500 ft by more than 10% and by more than 20,000 sq. ft. I show effects on floorspace quantities as well as on the residential and commercial rents of preexisting buildings. The bottom panel shows analogous effects for new commercial buildings. Treatment effects are computed using the procedure of [de Chaisemartin and d'Haultfœuille \(2024\)](#), comparing areas that witnessed construction earlier vs. later. In regressions, I include one observation per event and year, and cluster standard errors at the neighborhood level.

($\gamma^{RR} > 0$) will dampen it—with very large positive values of γ^{RR} , supply increases can even cause local price increases. Increasing the residential stock in a neighborhood also affects firms there, by making it easier for them to attract workers (and inducing rises in commercial rents). Positive externalities of residents on firms ($\gamma^{RC} > 0$) can lead to larger price increases.

More generally, there exists a mapping between values of γ and values of ε (illustrated in Appendix Figure E.6), which can be recovered through simulation: for any set of values for γ , I can simulate in the model the price effects of increasing floorspace quantities. I therefore calibrate γ such that the model matches the estimated reduced-form elasticities ε . The results of this indirect inference procedure are presented in Table 1.2.

Amenity levels in a neighborhood increase with the number of its residents, and its productivity grows with the number of jobs there. The agglomeration elasticities I find (0.11 and 0.07) align with existing estimates: the corresponding elasticities in [Ahlfeldt et al. \(2015\)](#), for instance, are 0.16 and 0.07. My results suggest small externalities across land uses.

Table 1.2: Spillover parameter estimates

Parameter	Interpretation	Calibrated value	Targeted elasticity
γ^{RR}	Effect of residents on amenities	0.11	$\varepsilon^{\text{RR}} = -0.42$
γ^{RC}	Effect of residents on productivity	0.03	$\varepsilon^{\text{RC}} = 0.14$
γ^{CR}	Effect of jobs on amenities	-0.03	$\varepsilon^{\text{CR}} = -0.03$
γ^{CC}	Effect of jobs on productivity	0.07	$\varepsilon^{\text{CC}} = -0.28$

Notes: This table reports parameter estimates for the spillover parameters γ . They are calibrated through indirect inference to match the reduced-form elasticities ε , estimated using the event studies of Figure 1.6. ε^{XY} measures the effect of increasing the supply of type-X floorspace (residential or commercial) on type-Y rents (also residential or commercial). I measure these elasticities using the average of the coefficients for the last three periods of the event studies. While the γ parameters are jointly calibrated to match all reduced-form elasticities, the value of γ^{XY} is mostly determined by the value of the corresponding reduced-form elasticity ε^{XY} (see Appendix Figure E.6).

Model validation. The calibrated demand model successfully matches a set of untargeted moments of the data, as shown in Appendix Figures E.7 and E.8. First, the model accurately replicates the Engel curve for housing. Second, it successfully predicts the sorting of workers across neighborhoods, with a correlation of 0.92 between the average neighborhood income in the model and that in the ACS. Third, the model reliably predicts commuting flows between neighborhood pairs. Fourth, the calibrated shares of floorspace in production α average 0.18 (with an interquartile range of 0.14 to 0.22), matching typical estimates in the literature (Valentinyi and Herrendorf, 2008; Ahlfeldt et al., 2015). Furthermore, neighborhoods whose commercial floorspace is primarily used for manufacturing or storage have higher calibrated values of α than neighborhoods where commercial floorspace is mostly office or retail space. Fifth, the calibrated subsistence housing levels $\underline{H}$ average 220 sq. ft (with an interquartile range of 166 to 260 sq. ft), about half the size of a studio, and corresponding well to what would plausibly be the minimum quantity of housing a person could consume. Moreover, the estimated $\underline{H}$ are positively correlated with the average unit size in each neighborhood, reflecting that workers tend to consume more housing in neighborhoods with larger housing units. Sixth, model-derived amenities B are tightly correlated with residents’ reported satisfaction with their neighborhood’s cleanliness, access to parks, and public safety, among other amenities (see Appendix Table E.1).

1.6 Measuring the Effects of Relaxing Zoning Regulations

I use the estimated model to evaluate the effects of relaxing zoning on construction and prices. To do so, I simulate NYC’s growth until 2060 under several zoning scenarios, assuming that fundamental amenities and productivities in each location stay fixed at their 2019 levels. To compute counterfactuals, I find a fixed point in the joint transition paths of floorspace and prices: floorspace is given by developers’ optimal behavior given prices and zoning, and period-by-period prices are yielded by spatial equilibrium.

As a baseline, I consider a status quo scenario in which zoning regulations stay at their 2019 levels. I then compare this scenario to an upzoning in which regulations are substantially relaxed near transit stations. The reform I consider increases the maximum

allowed FAR of parcels within 0.25 miles of a transit station to 6 (both for commercial and residential uses) and further increases the allowed FAR of parcels within 0.5 miles of a station to 4.²⁵ This policy change is realistic but ambitious, increasing the maximum allowable floorspace citywide by 59%. In an extreme counterfactual, I also consider fully removing zoning regulations across the city.²⁶ Finally, I compare the predictions of my dynamic model with a benchmark model that ignores the adjustment costs associated with redevelopment. This allows me to measure the extent to which these technological constraints restrict urban change beyond zoning.

In Section 1.6.1, I describe the aggregate effects of relaxing zoning. Section 1.6.2 then shows how these effects vary from one neighborhood to the next. Section 1.6.3 investigates the distributional effects of upzoning, and Section 1.6.4 compares upzoning to alternative policies promoting construction and affordability. In Section 1.6.5, I evaluate the extent to which zoning and historical persistence shape city structure. Finally, in Section 1.6.6, I show how my model can be used to measure neighborhood-level supply elasticities and how zoning shapes them.

1.6.1 What are the aggregate effects of relaxing zoning?

Effects on construction. In the status quo scenario, NYC continues to grow slowly, with its floorspace supply expanding by 13% and its population increasing by 8% between 2019 and 2060 (see Figure 1.7). I estimate that upzoning would substantially boost the city’s growth, increasing floorspace supply by 15 pp and its population by 12 pp. This reform is projected to add 786 million sq.ft of floorspace to the upzoned areas by 2060. While this is considerable, it only corresponds to 18% of the 4.3 billion sq.ft of additional floorspace allowed by the zoning change. This highlights that, in many cases, simply allowing more floorspace will not yield more construction, as upzoning does not raise expected profits from redevelopment sufficiently to overcome the fixed costs of rebuilding. The upzoning policy I consider delivers about a quarter of the boost in construction that would be obtained from completely removing zoning. Indeed, in a no-zoning counterfactual, NYC’s floorspace supply grows by 58 pp and its population by 46 pp. Removing zoning progressively expands the supply of floorspace by encouraging the redevelopment of more land (an extensive margin) and increasing the FARs of the new buildings constructed on that land (an intensive margin). Both margins play important roles, though the intensive margin has a slightly greater impact.²⁷

²⁵Parcels with higher FAR allowances at baseline are unaffected by the policy change. The upzoned parcels are primarily in Brooklyn and Queens and account for 37% of NYC’s developable land (see Appendix Figure C.7).

²⁶In the upzoning and no-zoning counterfactuals, I do not change regulations on parcels with landmarks and those in historic districts or flood zones.

²⁷Between 2020 and 2060, the share of NYC’s land that is redeveloped is twice as high in the no-zoning scenario than under the status quo (12% vs. 23%), and the average FAR of newly built structures is 2.5 times higher (4.0 vs. 1.6). Appendix Figures G.2 and G.3 show the distribution of the FAR of new buildings under both scenarios.

Figure 1.7: Effects of relaxing zoning

Change by 2060, relative to 2019 (%)



Notes: This figure shows the simulated evolution of NYC by 2060. In the status quo scenario, zoning regulations remain at their 2019 level. In the upzoning counterfactual, I upzone parcels located within 0.5 miles of a transit station (see Appendix Figure C.7). In the no-zoning counterfactual, I remove all zoning restrictions (excluding parcels that are landmarked, in a historic district, or in a flood zone). In the no-zoning and no-migration counterfactual, I remove all zoning restrictions and set the migration elasticity to zero, such that the population of the NYC metro area remains at its 2019 level. Finally, I compare the results of my model to those of a frictionless benchmark, in which the amount of floorspace in each neighborhood is determined by equating the price of floorspace with its marginal cost of construction, and where the price of residential floorspace is equal to that of commercial floorspace.

Effects on prices and welfare. While relaxing zoning yields large floorspace increases, its effects on rents are moderate—in the no-zoning counterfactual, they fall by 17 pp by 2060. This is because an increase in floorspace supply triggers migration to NYC, dampening local rent reductions while improving affordability in other cities. In a scenario with no net migration to NYC, the effects of removing zoning on rents would be fully concentrated in NYC and much larger (with a 32 pp drop in rents by 2060). This scenario can be interpreted as one in which utility in the rest of the country increases at the same rate as in NYC, for instance due to nationwide changes in zoning policy. Overall, my results suggest that while upzoning can produce sizable rent decreases, these price effects are very diffuse.

Beyond rent decreases, relaxing zoning also increases New Yorkers' welfare through higher wages (+4 pp in the no-zoning counterfactual—see Appendix Figure G.4). This is because upzoning increases the amount of commercial floorspace firms can use, and migration increases productivity through agglomeration effects. Increases in population density also benefit workers through higher amenity levels. Furthermore, relaxing zoning lowers the separation of land uses in the city.²⁸ This tends to reduce commuting times, although this effect is almost perfectly offset by increased congestion resulting from the city's growth.

²⁸New construction also shrinks the gap between the price of residential and commercial floorspace described in Figure 1.4. However, these price differentials remain sizable in the medium run even in the absence of zoning.

Overall, my model predicts that by 2060, removing zoning regulations would increase New Yorkers' welfare by 13 pp relative to the status quo, a substantial gain.

Role of use vs. bulk regulations. Zoning restricts both the use and density of new buildings. In Appendix Figure G.5, I explore the effect of both types of regulation on the city's trajectory. I find that relaxing only use limits would not substantially boost construction or increase welfare. Indeed, new construction in the city is primarily constrained by density limits, as most buildings are near or above the FAR limit, and redevelopment is generally only profitable when upward growth is permitted. Use limits are not irrelevant; rather, they interact with density limits, and relaxing both leads to significantly more construction than relaxing FAR limits alone.

Transition path. Appendix Figure G.6 shows the evolution of key variables under different zoning scenarios. The effects of relaxing zoning unfold very slowly. Only 23% of the long-run effect of upzoning on construction has materialized after 10 years, and half after 27 years (see Appendix Figure G.7).²⁹ Accordingly, it takes time for upzoning to deliver lower rents. However, the policy change causes an immediate drop in floorspace prices as agents correctly anticipate lower future rents.

Comparison with a benchmark model ignoring adjustment costs. If developers could frictionlessly adjust the amount of capital invested in each parcel, each parcel would, absent zoning constraints, reach an FAR that equalizes the marginal construction cost of floorspace with its price. Using this logic to evaluate the effects of removing zoning constraints would vastly overstate their true impact, predicting that the amount of floorspace in NYC would increase by 126% if zoning were abolished (see Figure 1.7). Even if zoning is a strong constraint on development and drives an important gap between floorspace prices and construction costs, removing zoning could not fully close that wedge. Technological constraints, i.e., the important costs associated with increasing the amount of floorspace on a parcel, are responsible for a large part of the disconnect between prices and marginal costs.³⁰

1.6.2 Where is zoning a binding constraint?

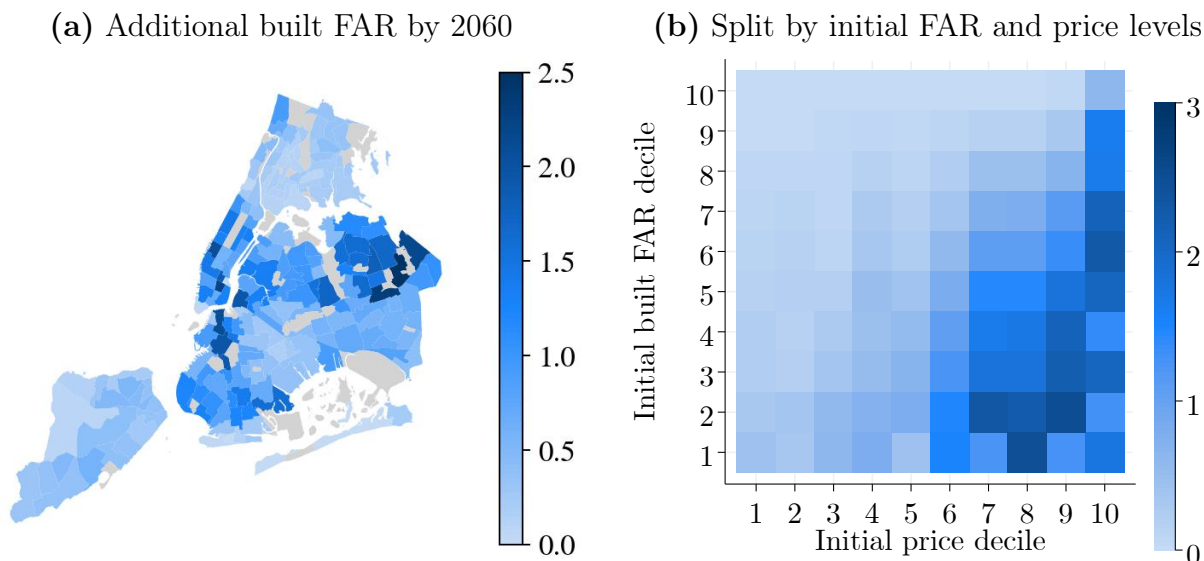
Figure 1.8(a) shows neighborhood-level effects of removing zoning on floorspace growth by 2060. Interestingly, the effects of relaxing zoning vary widely from one neighborhood to the next. These differences are easily rationalized by the basic economics of redevelopment. In Figure 1.8(b), I show the average effect of removing zoning on the additional FAR built by 2060 for different deciles of floorspace price and initial FAR levels. In neighborhoods where floorspace is inexpensive, redevelopment is usually unprofitable and therefore rare. The

²⁹Here, I measure the long-term impact of upzoning as its effect by 2100.

³⁰The large adjustment frictions associated with demolition I find are consistent with studies of historical fires that destroyed large portions of cities (e.g., [Hornbeck and Keniston, 2017](#); [Siodla, 2015](#)). Such studies find that burned areas become substantially denser than comparable unburned areas in the decades following the disaster. By paying the fixed cost associated with demolition, such fires pave the way for redevelopment, allowing for the construction of new, taller structures.

parcels that are redeveloped there tend to be vacant or minimally developed (see Appendix Figure G.9), and new buildings tend to be smaller, often below the zoning limit. Hence, zoning only moderately distorts the growth of neighborhoods where prices are low. In NYC, this suggests that upzoning most parts of the Bronx would not produce much additional growth.³¹

Figure 1.8: Heterogeneous effects of removing zoning on additional built FAR by 2060



Notes: The left panel shows the effect of removing zoning on the amount of floorspace built by 2060 in each neighborhood. In the right panel, I compute for each parcel the expected effect of removing zoning on FAR built by 2060, and show the average effects of the policy change for each decile of floorspace price and initial built FAR.

Similarly, parcels that are already densely built are unlikely to be redeveloped, as the fixed costs associated with doing so are particularly high. Hence, zoning limits do not usually distort the amount of floorspace supplied there. Accordingly, my model predicts that removing zoning would only slightly boost the growth of Midtown Manhattan despite the high floorspace prices there.

The areas that would grow the most in the absence of zoning are those with high prices but low density levels, as the fixed costs of redevelopment there are low and expected developer profits are high. This is why I find that removing zoning would boost floorspace supply the most in Western Brooklyn and Northern Queens, which have low-to-medium density levels, relatively high prices, and where most parcels are already built at or above the maximum allowed FAR. Removing zoning would also lead to large floorspace increases in some areas of Manhattan, notably Hell's Kitchen and the East Village, where many buildings have four stories or fewer despite extremely high prices.

Overall, the distortive effects of zoning are spatially concentrated. In the no-zoning counterfactual, 39% of the additional floorspace is built on 10% of the upzoned area (see

³¹Appendix Figure C.8 maps floorspace prices across NYC's neighborhoods. Floorspace prices are lowest in the Bronx and highest in the lower half of Manhattan. In Appendix Figure G.10, I show the share of parcels at or above the zoning limit for different price and density levels.

Appendix Figure G.11). If rezoning neighborhoods is politically costly, policymakers wishing to favor construction may want to focus their efforts on the areas where upzoning is likely to lead to important shifts in supply.³² This appears to have been the behavior of planners in NYC over the past decades: I find that uniformly increasing a neighborhood’s maximum allowed FAR by 1.0 raises built FAR by 0.03 on average over the subsequent 10 years, substantially less than the average effect of 0.09 reported in Figure 1.3 (see Appendix Figure G.13). This suggests that the recently upzoned parcels were ripe for redevelopment and strongly responsive to policy changes.

These results have two important implications for the external validity of studies of zoning changes. First, measuring the effects of past rezonings may be an imperfect guide to predicting the impact of potential future zoning changes. Upzoned neighborhoods are selected based on their suitability for redevelopment, and upzoning other neighborhoods in the same city might lead to less new construction. Second, while relaxing zoning constraints would substantially boost floorspace in NYC, doing so would likely have more limited effects in less expensive cities. In Appendix Figure G.14, I show how NYC would evolve in the coming decades if its floorspace price levels were comparable to those of Miami (with 50% lower prices) or Chicago (67% lower prices). In these scenarios, NYC’s future growth would be very limited, and removing zoning would lead to only small increases in the city’s overall floorspace supply.

1.6.3 Who benefits from relaxing zoning?

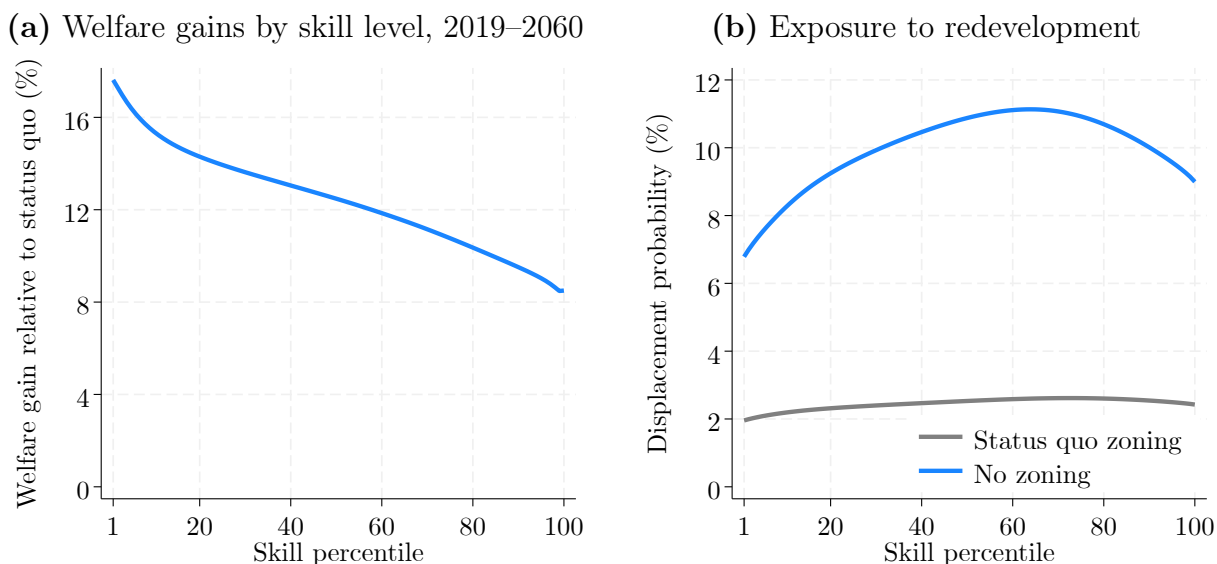
In Panel (a) of Figure 1.9, I show how the welfare gains of removing zoning would be distributed across workers’ skill distribution. I find larger gains for lower-income workers, who disproportionately benefit from rent decreases (see Appendix Figure G.15).³³ Indeed, poorer workers spend a larger fraction of their income on housing, making rent decreases more consequential for them. Furthermore, increasing housing supply in the city leads to larger rent drops in neighborhoods with lower amenity levels, where poorer households tend to live. This is because increasing the housing supply in high-amenity neighborhoods allows workers living in lower-amenity neighborhoods to move there and climb the neighborhood quality ladder. This lowers housing demand in low-amenity neighborhoods, which leads to lower rents there. This mechanism, illustrated in Appendix Figure G.16, is a neighborhood analog to the vacancy chain mechanism typically described in housing economics, where the supply of new high-quality housing units triggers a chain of moves along the housing quality ladder, eventually lowering the price of the low-quality housing units in which poorer

³²In Appendix Figure G.12, I describe a counterfactual where I remove zoning on parcels with built FARs below the 60th percentile in neighborhoods with floorspace prices above the 60th percentile. This policy upzones 14% of NYC’s developable land and delivers 49% of the floorspace gains of the no-zoning scenario.

³³The welfare changes I report do not include profits from redevelopment. I do not redistribute these profits to workers in the model to better compare my counterfactuals with the frictionless benchmark, where there are no profits from redevelopment. The model predicts that in the status quo scenario, redevelopment profits would average \$3.9 billion per year over 2020–2060. This corresponds to \$448 per worker, or 0.9% of New Yorkers’ income in 2019. In the no-zoning counterfactual, redevelopment profits grow to \$13 billion per year, corresponding to \$1,242 per worker or 2.5% of New Yorkers’ income in 2019. These estimates are in line with the overall size of the construction sector in NYC, with construction spending in the city totaling \$60.6 billion in 2019 ([Office of the New York State Comptroller, 2021](#)).

households live (Mast, 2023; Bratu, Harjunen and Saarimaa, 2023; Kindström and Liang, 2024).

Figure 1.9: Distributional effects of removing zoning



Notes: Panel (a) shows the welfare gains from removing zoning (by 2060) for different worker types. Panel (b) displays, for each worker type, the average share of residential floorspace in their initial neighborhood that will be redeveloped. If, within a neighborhood, the redevelopment probability of a parcel is uncorrelated with the skill level of the people living there, this share corresponds to the probability that a worker’s initial dwelling will be demolished by 2060.

Another informative measure of the potential distributional consequences of upzoning is workers’ exposure to redevelopment. While redevelopment eventually benefits households through lower rents and a more efficient allocation of land uses, they may be forced to move because their initial housing unit is redeveloped or otherwise suffer in the short run from the disamenities associated with construction.

Using the model, I can predict the share of the residential floorspace in each neighborhood that will be redeveloped by 2060—these shares are mapped in Appendix Figure G.17. I can then compute, for each worker type, the average share of residential floorspace in their initial neighborhood that will be redeveloped. This share, a proxy for the probability that a worker will be displaced by redevelopment, is plotted in Figure 1.9(b). While there is overall limited heterogeneity in workers’ exposure to redevelopment across types, workers at the upper-middle of the income distribution tend to be more exposed to redevelopment, consistent with the greater likelihood of redevelopment in more expensive neighborhoods.³⁴

³⁴Workers at the very top of the skill distribution are less exposed to redevelopment as they tend to live in the dense neighborhoods of Manhattan, where buildings have a low probability of being redeveloped due to their high existing FAR.

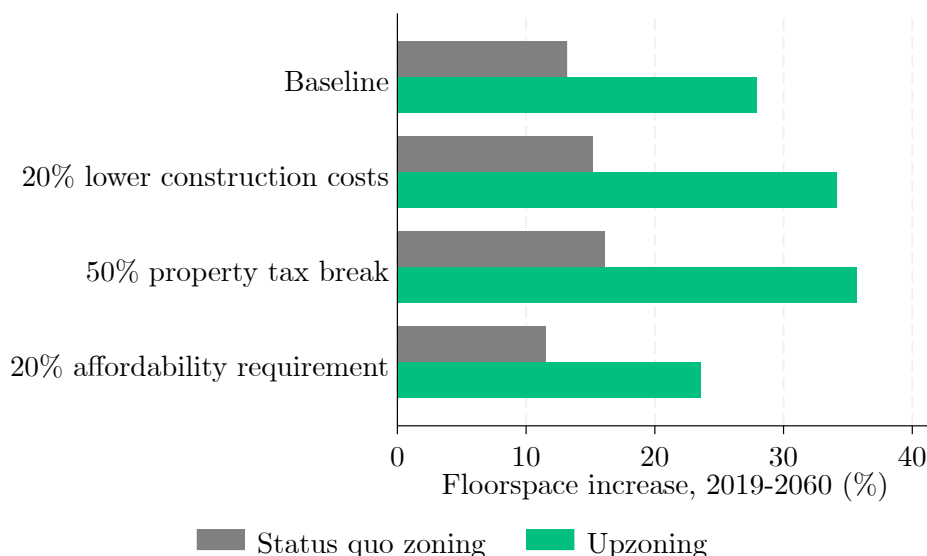
1.6.4 How efficient are policies favoring construction and affordability?

Upzoning is only one of several policies hailed as a means to encourage construction and increase affordability (Lebret, Liu and Valentin, 2025). Over the past decades, construction costs have increased much more rapidly than overall inflation, contributing to the housing affordability crisis. Policymakers are therefore exploring ways to reduce these costs, for instance by simplifying construction regulations or promoting modular and prefabricated buildings. Cities also routinely incentivize new construction with property tax breaks on newly built structures. Finally, to directly promote affordable housing construction, municipalities are increasingly adopting inclusionary zoning policies. These require or incentivize developers to reserve a portion of newly built units—typically 10–30%—as affordable housing.

To evaluate these alternative policies, I consider three additional counterfactuals. First, I examine a scenario in which variable construction costs are reduced by 20%, erasing the growth of inflation-adjusted construction costs over 2004–2019. Second, I consider a policy that reduces the property taxes on new buildings from 2% to 1%. Third, I analyze a mandatory inclusionary zoning policy requiring that 20% of all newly built floorspace be rented at half the market rate.

Because these policies are sometimes used in conjunction with upzoning, I evaluate their effect on floorspace growth over 2019–2060 both in a status quo zoning scenario and under the upzoning considered above. Figure 1.10 reports my results.

Figure 1.10: Effects of alternative policies on floorspace growth



Notes: This figure shows the effects of three policies (lower construction costs, lower property taxes on new structures, and mandatory inclusionary zoning) on floorspace supply growth between 2019 and 2060, both under status quo zoning and the upzoning described at the start of Section 1.6.

Under status quo zoning, lowering construction costs by 20% increases the amount of

floorspace added to the city by 2060 by 15%, relative to the baseline. A generous tax break policy has a slightly larger effect on new construction (+22%), but is particularly costly, as property taxes are reduced for all new structures, including those that would have been built in the absence of the subsidy. I find that on average, forgoing \$1,000 of property taxes (in net present value) leads to the construction of only 1.4 additional sq. ft of floorspace. In other words, incentivizing the construction of a 1,000 sq. ft unit typically requires forgoing \$720,000 of property tax revenue.³⁵ Both policies are much less effective than zoning reform: upzoning increases the amount of floorspace added to the city over 2020–2060 by 112%.³⁶

Under a mandatory inclusionary zoning policy, NYC’s floorspace growth over 2020–2060 is reduced relative to the baseline scenario (+12% instead of +13%). This highlights the fundamental tradeoff posed by such policies: mandating affordable housing construction enables some households to rent at below-market prices but discourages developers from building, thereby reducing the overall housing supply. Between 2020 and 2060, the inclusionary zoning policy leads to the construction of 73 million sq. ft of affordable floorspace, but lowers the total amount of floorspace in the city by 400 million sq. ft. In other words, for each sq. ft of affordable floorspace created by the policy, the overall available floorspace drops by 5.5 sq. ft.

1.6.5 How do zoning and historical persistence shape city structure?

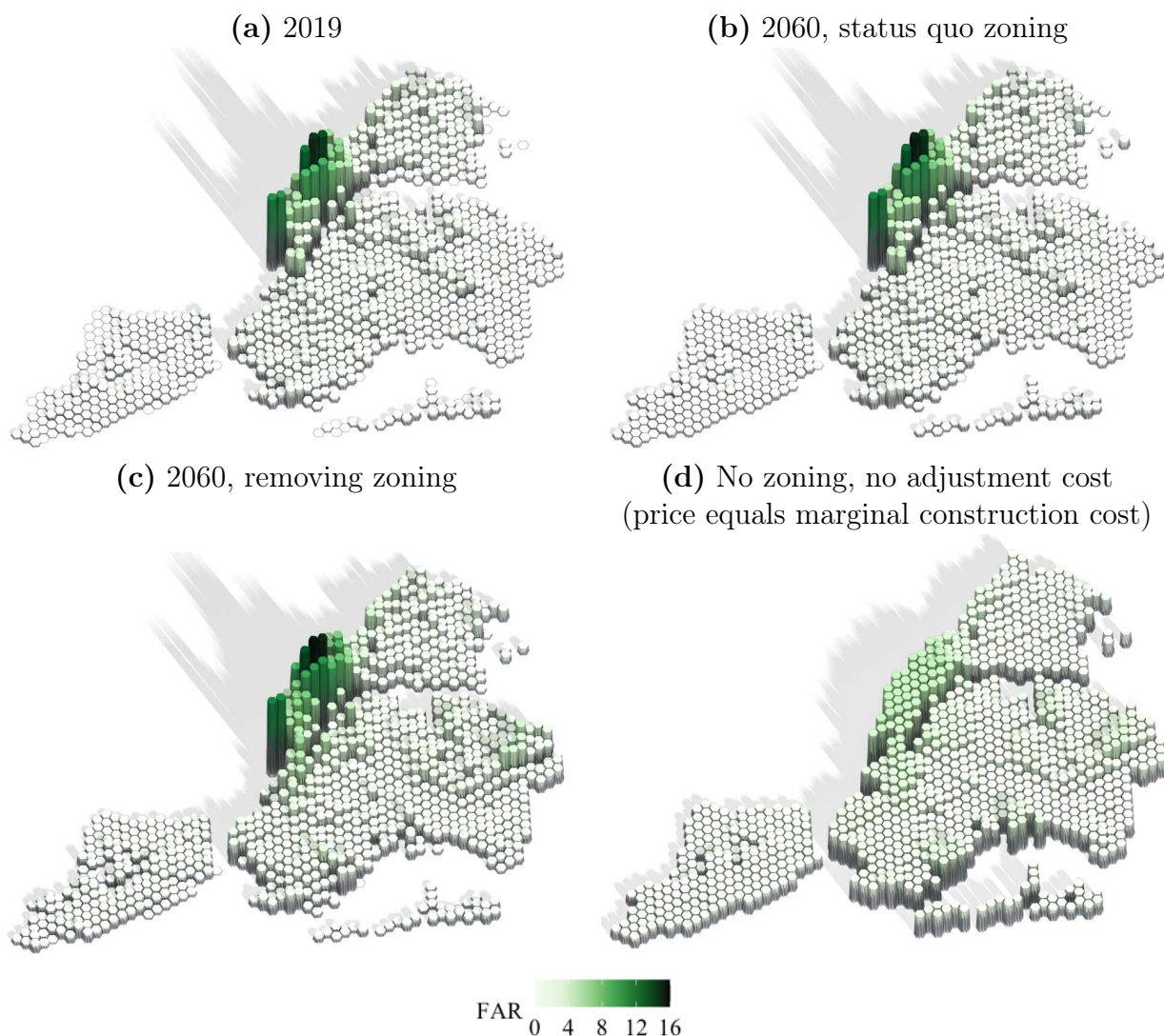
To evaluate the extent to which zoning and technological constraints shape city structure, I consider the density profile of NYC in different scenarios. Figure 1.11(a) shows the distribution of built FAR in NYC in 2019, with most of the city’s density concentrated in Manhattan. My model predicts that if zoning remained unchanged, NYC’s structure would stay stable in the next decades—see Figure 1.11(b). Eliminating zoning constraints would stimulate development in Brooklyn and Queens, leading to a more gradual decline in density as one moves away from the city center, as shown in Figure 1.11(c).

Using a model that assumes away adjustment costs yields a vastly different prediction of the effects of removing zoning on NYC’s structure. Figure 1.11(d) shows the distribution of density in NYC that equalizes floorspace prices with its marginal construction cost, displaying a much smoother density gradient. One neighborhood in which the two approaches diverge the most is Midtown Manhattan. The dynamic model predicts that this very dense area would see its floorspace grow by 7% by 2060 if zoning were removed. In the frictionless benchmark, where the height of buildings can adjust costlessly up or down, Midtown instead shrinks by 61% (see Appendix Figure G.8). Indeed, large floorspace increases in the peripheral neighborhoods of NYC lower prices in the city center, leading to lower equilibrium heights of buildings there.

³⁵Such a high estimate is consistent with previous findings: [Soltas \(2024\)](#) finds that the typical fiscal cost of incentivizing the construction of an additional low-income housing unit in NYC is \$1.6 million.

³⁶Cost reductions and tax breaks also appear to be complementary to upzoning. Under upzoning, reducing construction costs and tax breaks lead to larger increases in floorspace than under status quo zoning (+22% and +28%, respectively). Upzoning also slightly increases the efficiency of tax breaks by making it less costly to incentivize the construction of additional floorspace.

Figure 1.11: Counterfactual FAR profiles of NYC



Notes: This figure shows the FAR profile of NYC for different counterfactual scenarios. Panel (a) shows FAR levels in 2019. Panel (b) (resp., c) shows expected FAR levels in 2060 if zoning were to remain at its 2019 level (resp., was removed). Panel (d) shows the distribution of FAR that equalizes the construction cost of the marginal sq. ft of floorspace with its price. In this figure, I aggregate data to hexagonal grid cells using Uber's H3 indexing system.

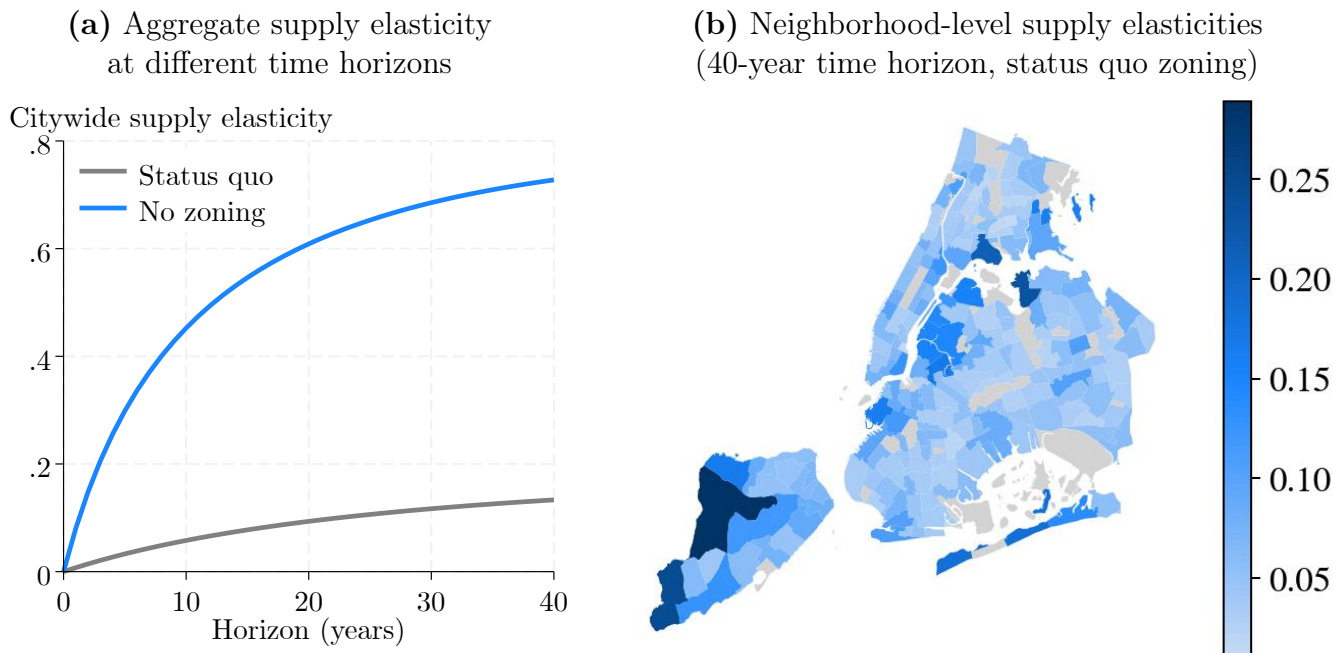
This shows that taking into account the technological constraints to redevelopment is key to producing credible counterfactuals. It also shows that historical persistence is essential in explaining cities' structure.

1.6.6 How does zoning affect floorspace supply elasticities?

Another way to assess zoning's impact on NYC's evolution is by measuring its effect on the floorspace supply elasticity, which determines how much the city grows as more households and firms want to locate there. One strength of the model developed in this paper is that it allows me to quantify the effect of demand shocks on construction, and therefore supply

elasticities, at a granular scale and across various time horizons.³⁷ Figure 1.12(a) shows NYC’s overall supply elasticity at different horizons, and Figure 1.12(b) shows how supply elasticities vary across neighborhoods.

Figure 1.12: Floorspace supply elasticities



Notes: Panel (a) plots estimates of NYC’s overall floorspace supply elasticity at different time horizons, both under status quo zoning and no zoning. Panel (b) maps neighborhood-level supply elasticities at a 40-year horizon, under status quo zoning.

I estimate that under current zoning regulations, NYC’s overall supply elasticity reaches 0.13 at a 40-year horizon. These estimates are substantially smaller than those typically estimated at the metropolitan area level, consistent with the results of [Baum-Snow and Han \(2024\)](#), who show that supply elasticities sharply decrease close to cities’ central business districts. While supply elasticities are somewhat lower in denser neighborhoods, they correlate much more strongly with zoning limits. Appendix Figure G.18(a) illustrates the tight correlation between a neighborhood’s estimated supply elasticity and the share of the maximum FAR allowed by the zoning code (called the zoning envelope) that has already been built.

I estimate that removing zoning constraints would more than quintuple the city’s floorspace supply elasticity, which would reach 0.73 over 40 years. In that counterfactual scenario, neighborhood-level supply elasticities are strongly negatively correlated with existing density levels—see Appendix Figure G.18(b).

Even in the absence of zoning, estimated supply elasticities are also much lower than what standard static models would suggest, as these models assume that the capital stock in

³⁷The floorspace supply elasticity at horizon h is defined as the increase in floorspace (in pp) observed h years after a permanent shock to fundamental amenities and productivities that raises residential and commercial rents by 1% on impact.

a neighborhood can be frictionlessly adjusted up or down. For instance, the parameterization of [Ahlfeldt et al. \(2015\)](#) implies a floorspace supply elasticity of three everywhere in the city.

1.7 Why did NYC’s planners impose such costly regulations?

The substantial welfare cost imposed by NYC’s stringent zoning is reflected in the large gap between the price of floorspace and its marginal construction cost. In 2019, building an additional housing unit in NYC cost about \$400,000, while the average home sale price approached \$1 million. For such construction to reduce welfare, it would need to generate a net negative externality of at least \$600,000—an implausibly large effect. If anything, the evidence points in the opposite direction: new development in NYC seems to create positive externalities. Therefore, when zoning prevents the construction of a new housing unit, it typically results in a welfare loss of several hundred thousand dollars. This raises the question of why urban planners impose such costly policies. I argue that the inadequacy of NYC’s zoning code stems from the economic conditions at the time of its initial adoption, combined with the strong persistence of zoning policy, which is more enduring than planners often anticipate.

When NYC’s current zoning resolution was drafted in the late 1950s, population growth had come almost to a halt and land values were below their historical trend (see Appendix Figure B.9). At the time, urban planners believed the city had nearly reached its ultimate size, and their main aim was to guide NYC’s future growth, which they believed would be limited, particularly by promoting tower-in-the-park architecture and further separating commercial development from residential areas. Such separation of land uses was partly justified by the negative externalities of manufacturing, which accounted for approximately 30% of employment in NYC until the 1960s ([Harrison, Ballard & Allen, 1950](#); [Voorhees Walker Smith & Smith, 1958](#)).³⁸

In the mid-20th century, floorspace prices in NYC were roughly equal to marginal construction costs.³⁹ Therefore, small negative externalities could make new construction welfare-decreasing and justify zoning restrictions. Furthermore, as new construction did not yield substantial surplus, limitations on floorspace supply were not particularly socially costly. Hence, when the 1961 resolution was adopted, its potential costs were limited and the existence of a large manufacturing sector made zoning potentially beneficial.

Economic conditions in NYC have changed dramatically since the 1960s. With floorspace prices rising well beyond construction costs, the costs of zoning have increased significantly.

³⁸While new industrial construction is now rare in NYC, I find some evidence suggesting that it decreases nearby residential rents (see Appendix Figure G.19), consistent with such land uses inducing negative externalities.

³⁹In 1950, construction costs were around \$10/sq.ft (or \$10,000 per dwelling unit) for small structures, and twice as large for taller structures ([U.S. Department of Labor, 1950](#); [RSMMeans, 2020](#); [Willis, 1995](#); [Engineering News-Record, 2025](#); [Office for Metropolitan History, 2023](#)). In the 1950 Census, the average reported value of a housing unit in NYC was approximately \$12,500, or about \$13 per square foot. Real estate transaction data from [Real Estate Board of New York \(1950\)](#) suggests that properties typically traded at lower prices, around \$5/sq. ft.

Furthermore, the potential benefits of the 1961 zoning resolution have likely faded as the externalities associated with commercial land uses have evolved. In fact, contemporary planners typically favor mixed-use developments and oppose tower-in-the-park projects, in striking contrast to their predecessors. Yet, despite these radical shifts, the numerous amendments to NYC’s zoning resolution since 1961 have mainly brought minor changes, and the city’s current zoning map almost perfectly aligns with its 1960s version (see Appendix Figure B.10).

Appendix Section B.1 provides a historical perspective on NYC’s zoning and shows that the issues that led planners to establish various zoning regulations often disappeared in subsequent decades. Yet, these regulations persist far longer than planners anticipate or intend. Such persistence is not unique to NYC,⁴⁰ and does not seem to stem from urban planners’ resistance to change but rather from political constraints, as the rezoning process is slow, contentious, and vulnerable to obstruction by officials and interest groups with a stake in the status quo.

1.8 Conclusion

Cities can expand outward or upward. In the 1990s, 80% of rapidly growing urban areas were doing so mostly outward, by spreading. In the 2010s, this figure had shrunk to 28% (Frolking et al., 2024), as many cities reached a mature stage. Redevelopment has therefore become an increasingly important phenomenon in urban areas, allowing them to increase their floorspace supply and to reallocate land to its most profitable use.

This paper provides a framework for analyzing this process, leveraging the detailed and increasingly available microdata describing urban land uses. Tracking the city’s evolution at the parcel level allows modeling and separating two important barriers to redevelopment: the substantial costs associated with tearing down existing structures to replace them with new ones, and zoning restrictions on the use and size of new buildings.

This framework can be used to evaluate the effects of transit upgrades, property taxation, or structural transformation. In this paper, I use it to study a pressing policy question: the effects of zoning on construction and affordability. I find that redevelopment is usually unprofitable in inexpensive or densely developed neighborhoods, severely hindering their ability to change regardless of the stringency of zoning. Zoning regulations primarily distort development in areas with high floorspace prices and low density. In NYC, I show that the targeted upzoning of these neighborhoods can substantially boost floorspace supply. However, the effects of upzoning take time to materialize, and its impact on rents is diffuse. Local upzonings may thus initially appear disappointing to policymakers aiming to improve affordability, but over time and in aggregate, they yield large gains.

⁴⁰See Shertzer, Twinam and Walsh (2018) and Twinam (2018) for case studies of Chicago and Seattle.

Bibliography

- Ahlfeldt, Gabriel M and Daniel P McMillen, “Tall Buildings and Land Values: Height and Construction Cost Elasticities in Chicago, 1870–2010,” *Review of Economics and Statistics*, 2018, 100 (5), 861–875.
- , Stephen J Redding, Daniel M Sturm, and Nikolaus Wolf, “The Economics of Density: Evidence From the Berlin Wall,” *Econometrica*, 2015, 83 (6), 2127–2189.
- Akbar, Prottoy and Gilles Duranton, “Measuring the Cost of Congestion in a Highly Congested City: Bogotá,” 2017.
- , Victor Couture, Gilles Duranton, and Adam Storeygard, “Mobility and Congestion in Urban India,” *American Economic Review*, 2023, 113 (4), 1083–1111.
- Allen, Treb and Costas Arkolakis, “Trade and the Topography of the Spatial Economy,” *The Quarterly Journal of Economics*, 2014, 129 (3), 1085–1140.
- , – , and Xiangliang Li, “Optimal City Structure,” *Yale University, Mimeograph*, 2015.
- Almagro, Milena and Tomás Domínguez-Iino, “Location Sorting and Endogenous Amenities: Evidence From Amsterdam,” *Econometrica*, 2025, 93 (3), 1031–1071.
- , Eric Chyn, and Bryan A Stuart, “Neighborhood Revitalization and Inequality: Evidence from Chicago’s Public Housing Demolitions,” 2024.
- Alonso, William, “Location and Land Use; Toward a General Theory of Land Rent,” 1964.
- Anagol, Santosh, Fernando V Ferreira, and Jonah M Rexer, “Estimating the Economic Value of Zoning Reform,” 2024.
- Arzaghi, Mohammad and J Vernon Henderson, “Networking Off Madison Avenue,” *The Review of Economic Studies*, 2008, 75 (4), 1011–1038.
- Asker, John, Allan Collard-Wexler, and Jan De Loecker, “Dynamic Inputs and Resource (Mis)allocation,” *Journal of Political Economy*, 2014, 122 (5), 1013–1063.
- Asquith, Brian J, Evan Mast, and Davin Reed, “Local Effects of Large New Apartment Buildings in Low-Income Areas,” *Review of Economics and Statistics*, 2023, 105 (2), 359–375.
- Barr, Jason M, *Building the Skyline: The Birth and Growth of Manhattan’s Skyscrapers*, Oxford University Press, 2016.
- Bassett, Edward Murray, *Stores in Residence Zones* 1926.
- Baum-Snow, Nathaniel and Lu Han, “The Microgeography of Housing Supply,” *Journal of Political Economy*, 2024, 132 (6), 1897–1946.
- , Nicolas Gendron-Carrier, and Ronni Pavan, “Local Productivity Spillovers,” *American Economic Review*, 2024, 114 (4), 1030–1069.
- Bayer, Patrick and Christopher Timmins, “On the Equilibrium Properties of Locational Sorting Models,” *Journal of Urban Economics*, 2005, 57 (3), 462–477.
- Bilal, Adrien, “The Geography of Unemployment,” *The Quarterly Journal of Economics*, 2023, 138 (3), 1507–1576.

- Blanco, Hector and Noémie Sportiche**, “Local effects of bypassing zoning regulations,” 2024.
- Bratu, Cristina, Oskari Harjunen, and Tuukka Saarimaa**, “JUE Insight: City-Wide Effects of New Housing Supply: Evidence from Moving Chains,” *Journal of Urban Economics*, 2023, 133, 103528.
- Brooks, Leah and Zachary Liscow**, “Infrastructure Costs,” *American Economic Journal: Applied Economics*, 2023, 15 (2), 1–30.
- Bryan, Gharad and Melanie Morten**, “The Aggregate Productivity Effects of Internal Migration: Evidence from Indonesia,” *Journal of Political Economy*, 2019, 127 (5), 2229–2268.
- Büchler, Simon and Elena Catharina Lutz**, “The Local Effects of Relaxing Land Use Regulation on Housing Supply and Rents,” *MIT Center for Real Estate Research Paper*, 2021, (21/18).
- Burgess, Ernest W**, *The Growth of the City: An Introduction to a Research Project* 1925.
- Caballero, Ricardo, Eduardo Engel, and John Haltiwanger**, “Aggregate Employment Dynamics: Building from Microeconomic Evidence,” *American Economic Review*, 1997, pp. 115–137.
- Combes, Pierre-Philippe, Gilles Duranton, and Laurent Gobillon**, “The Production Function for Housing: Evidence from France,” *Journal of Political Economy*, 2021, 129 (10), 2766–2816.
- Couture, Victor, Gilles Duranton, and Matthew A Turner**, “Speed,” *Review of Economics and Statistics*, 2018, 100 (4), 725–739.
- David, Joel M and Venky Venkateswaran**, “The Sources of Capital Misallocation,” *American Economic Review*, 2019, 109 (7), 2531–2567.
- de Chaisemartin, Clément and Xavier d’Haultfœuille**, “Difference-in-Differences Estimators of Intertemporal Treatment Effects,” *The Review of Economics and Statistics*, 02 2024, pp. 1–45.
- Dingel, Jonathan I and Felix Tintelnot**, “Spatial Economics for Granular Settings,” 2020.
- Dixit, Avinash**, “Entry and Exit Decisions Under Uncertainty,” *Journal of Political Economy*, 1989, 97 (3), 620–638.
- Engineering News-Record**, “Building Cost Index History,” 2025.
- Epplé, Dennis, Brett Gordon, and Holger Sieg**, “A New Approach to Estimating the Production Function for Housing,” *American Economic Review*, 2010, 100 (3), 905–924.
- Frolking, Steve, Richa Mahtta, Tom Milliman, Thomas Esch, and Karen C Seto**, “Global Urban Structural Growth Shows a Profound Shift from Spreading Out to Building Up,” *Nature Cities*, 2024, pp. 1–12.
- Ganong, Peter and Daniel Shoag**, “Why Has Regional Income Convergence in the US Declined?,” *Journal of Urban Economics*, 2017, 102, 76–90.
- Gechter, Michael and Nick Tsivanidis**, “Spatial Spillovers from High-Rise Developments:

- Evidence from the Mumbai Mills,” 2023.
- Glaeser, Edward and Joseph Gyourko**, “The Economic Implications of Housing Supply,” *Journal of Economic Perspectives*, 2018, 32 (1), 3–30.
- Glaeser, Edward L, Joseph Gyourko, and Raven Saks**, “Why Is Manhattan So Expensive? Regulation and the Rise in Housing Prices,” *The Journal of Law and Economics*, 2005, 48 (2), 331–369.
- Greaney, Brian, Andrii Parkhomenko, and Stijn Van Nieuwerburgh**, “Dynamic Urban Economics,” 2025.
- Gupta, Arpit, Vrinda Mittal, and Stijn Van Nieuwerburgh**, “Work from Home and the Office Real Estate Apocalypse,” 2024.
- Gyourko, Joseph and Raven Molloy**, “Regulation and Housing Supply,” in “Handbook of Regional and Urban Economics,” Vol. 5, Elsevier, 2015, pp. 1289–1337.
- Harrison, Ballard & Allen**, *Plan for Rezoning the City of New York* 1950.
- Heblich, Stephan, Stephen J Redding, and Daniel M Sturm**, “The Making of the Modern Metropolis: Evidence From London,” *The Quarterly Journal of Economics*, 2020, 135 (4), 2059–2133.
- Henderson, J Vernon, Tanner Regan, and Anthony J Venables**, “Building the City: From Slums to a Modern Metropolis,” *The Review of Economic Studies*, 2021, 88 (3), 1157–1192.
- Hirt, Sonia A**, *Zoned in the USA*, Cornell University Press, 2015.
- Hornbeck, Richard and Daniel Keniston**, “Creative Destruction: Barriers to Urban Growth and the Great Boston Fire of 1872,” *American Economic Review*, 2017, 107 (6), 1365–1398.
- **and Enrico Moretti**, “Estimating Who Benefits from Productivity Growth: Local and Distant Effects of City Productivity Growth on Wages, Rents, and Inequality,” *Review of Economics and Statistics*, 2024, 106 (3), 587–607.
- Hoyt, Homer**, *The Structure and Growth of Residential Neighborhoods in American Cities*, Federal Housing Administration, 1939.
- Hsiao, Allan**, “Sea Level Rise and Urban Adaptation in Jakarta,” 2024.
- Hsieh, Chang-Tai and Enrico Moretti**, “Housing Constraints and Spatial Misallocation,” *American Economic Journal: Macroeconomics*, 2019, 11 (2), 1–39.
- Kindström, Gabriella and Che-Yuan Liang**, “Does New Housing for the Rich Benefit the Poor? On Trickle-Down Effects of New Homes,” 2024.
- Kulka, Amrita, Aradhya Sood, and Nicholas Chiumenti**, “Under the (Neighbor)Hood: Understanding Interactions Among Zoning Regulations,” 2024.
- Lebret, Daniel, Crocker H. Liu, and Maxence Valentin**, “Carrot and Stick Policies and the Affordability Puzzle: Evidence from (Re)development in New York City,” 2025.
- Li, Xiaodi**, “Do New Housing Units in your Backyard Raise your Rents?,” *Journal of Economic Geography*, 2022, 22 (6), 1309–1352.

- Liao, Hsi-Ling**, “The Effect of Rezoning on Local Housing Supply and Demand: Evidence from New York City,” 2022.
- Lucas, Robert E**, “Adjustment Costs and the Theory of Supply,” *Journal of Political Economy*, 1967, 75 (4, Part 1), 321–334.
- and **Esteban Rossi-Hansberg**, “On the Internal Structure of Cities,” *Econometrica*, 2002, 70 (4), 1445–1476.
- Mast, Evan**, “JUE Insight: The Effect of New Market-Rate Housing Construction on the Low-Income Housing Market,” *Journal of Urban Economics*, 2023, 133, 103383.
- Mills, Edwin S**, “An Aggregative Model of Resource Allocation in a Metropolitan Area,” *The American Economic Review*, 1967, 57 (2), 197–210.
- Murphy, Alvin**, “A Dynamic Model of Housing Supply,” *American Economic Journal: Economic Policy*, 2018, 10 (4), 243–67.
- Muth, Richard F**, “Cities and Housing: The Spatial Pattern of Urban Residential Land Use,” 1969.
- New York Building Congress**, “Construction Outlook Update: New York City Construction Costs,” 2019.
- New York City Board of Estimate and Apportionment**, “Development and Present Status of City Planning in New York City,” 1914.
- , “Final Report of the Commission on Building Districts and Restrictions to the Committee on the City Plan,” 1916.
- Nickell, Stephen J**, “Dynamic Models of Labour Demand,” *Handbook of Labor Economics*, 1986, 1, 473–522.
- Office for Metropolitan History**, “New Building Permits, Manhattan 1900–1986,” 2023. Accessed February 2023 at <https://www.metrohistory.com/permit-search>.
- Office of the New York State Comptroller**, “The Construction Industry in New York City: Recent Trends and Impact of COVID-19,” 2021. Available at <https://www.osc.ny.gov/files/reports/osdc/pdf/report-3-2021.pdf>.
- Ogawa, Hideaki and Masahisa Fujita**, “Equilibrium Land Use Patterns in a Nonmonocentric City,” *Journal of Regional Science*, 1980, 20 (4), 455–475.
- Ospital, Augusto**, “Urban Policy and Spatial Exposure to Environmental Risk,” 2022.
- Peng, Xuequan Elsie**, “The Dynamics of Urban Development: Evidence from Land Use Reform in New York,” 2023.
- Pennington, Kate**, “Does Building New Housing Cause Displacement? The Supply and Demand Effects of Construction in San Francisco,” 2021.
- Quigg, Laura**, “Empirical Testing of Real Option-Pricing Models,” *The Journal of Finance*, 1993, 48 (2), 621–640.
- Real Estate Board of New York**, “Open Market Sales, Manhattan Real Estate,” 1950.
- Redding, Stephen J and Esteban Rossi-Hansberg**, “Quantitative Spatial Economics,” *Annual Review of Economics*, 2017, 9, 21–58.

- RSMeans**, *Square Foot Costs* 2020.
- Rust, John**, “Optimal Replacement of GMC Bus Engines: An Empirical Model of Harold Zurcher,” *Econometrica*, 1987, pp. 999–1033.
- Saiz, Albert**, “The Geographic Determinants of Housing Supply,” *The Quarterly Journal of Economics*, 2010, *125* (3), 1253–1296.
- Shertzer, Allison, Tate Twinam, and Randall P Walsh**, “Zoning and the Economic Geography of Cities,” *Journal of Urban Economics*, 2018, *105*, 20–39.
- Siodla, James**, “Razing San Francisco: The 1906 Disaster as a Natural Experiment in Urban Redevelopment,” *Journal of Urban Economics*, 2015, *89*, 48–61.
- Soltas, Evan J**, “The Price of Inclusion: Evidence from Housing Developer Behavior,” *Review of Economics and Statistics*, 2024, *106* (6), 1588–1606.
- Titman, Sheridan**, “Urban Land Prices Under Uncertainty,” *The American Economic Review*, 1985, *75* (3), 505–514.
- Tsivanidis, Nick**, “Evaluating the Impact of Urban Transit Infrastructure: Evidence From Bogotá’s TransMilenio,” 2019.
- Twinam, Tate**, “The Long-Run Impact of Zoning: Institutional Hysteresis and Durable Capital in Seattle, 1920–2015,” *Regional Science and Urban Economics*, 2018, *73*, 155–169.
- U.S. Department of Labor**, “Construction Annual Review,” 1950. Bulletin No. 1047.
- Valentinyi, Akos and Berthold Herrendorf**, “Measuring Factor Income Shares at the Sectoral Level,” *Review of Economic Dynamics*, 2008, *11* (4), 820–835.
- Voorhees Walker Smith & Smith**, *Zoning New York City* 1958.
- Willis, Carol**, *Form Follows Finance: Skyscrapers and Skylines in New York and Chicago*, Princeton Architectural Press, 1995.

Chapter 2

Electoral Turnovers

Publication note: This chapter is based on Benjamin Marx, Vincent Pons, and Vincent Rollet, “Electoral Turnovers,” *The Review of Economic Studies*, 92(5), 3306–3339, 2025, <https://doi.org/10.1093/restud/rdae108>.

2.1 Introduction

In most societies, voting incumbents out is the only mechanism allowing citizens to peacefully replace their country’s leadership. For this reason, electoral turnovers are fundamental to democracy, which [Przeworski \(1991\)](#) characterized as “a system in which parties lose elections.” Despite this, economists have given little attention to the consequences of turnovers at the national level. Assessing the costs and benefits of turnovers is particularly relevant to current debates prompted by democratic backsliding and the rise of populism in many countries ([Levitsky and Ziblatt, 2019](#); [Gurieiev and Papaioannou, 2022](#)).

This paper asks how power transitions caused by national elections shape country performance. [Jones and Olken \(2005\)](#) showed that, in autocracies, new leaders appointed after the death of their predecessor can change the growth trajectory of nations. Turnovers could give new impetus to a country’s performance by bringing to power new leaders facing stronger incentives to perform ([Persson and Tabellini, 2002](#); [Ashworth, 2005](#); [Ferraz and Finan, 2011](#)). At the same time, the loss of political experience ([Alt et al., 2011](#)), the personnel instability ([Akhtari et al., 2022](#)), and the policy uncertainty ([Alesina et al., 1996](#); [Horowitz et al., 2009](#)) induced by turnovers could be detrimental to economic performance.

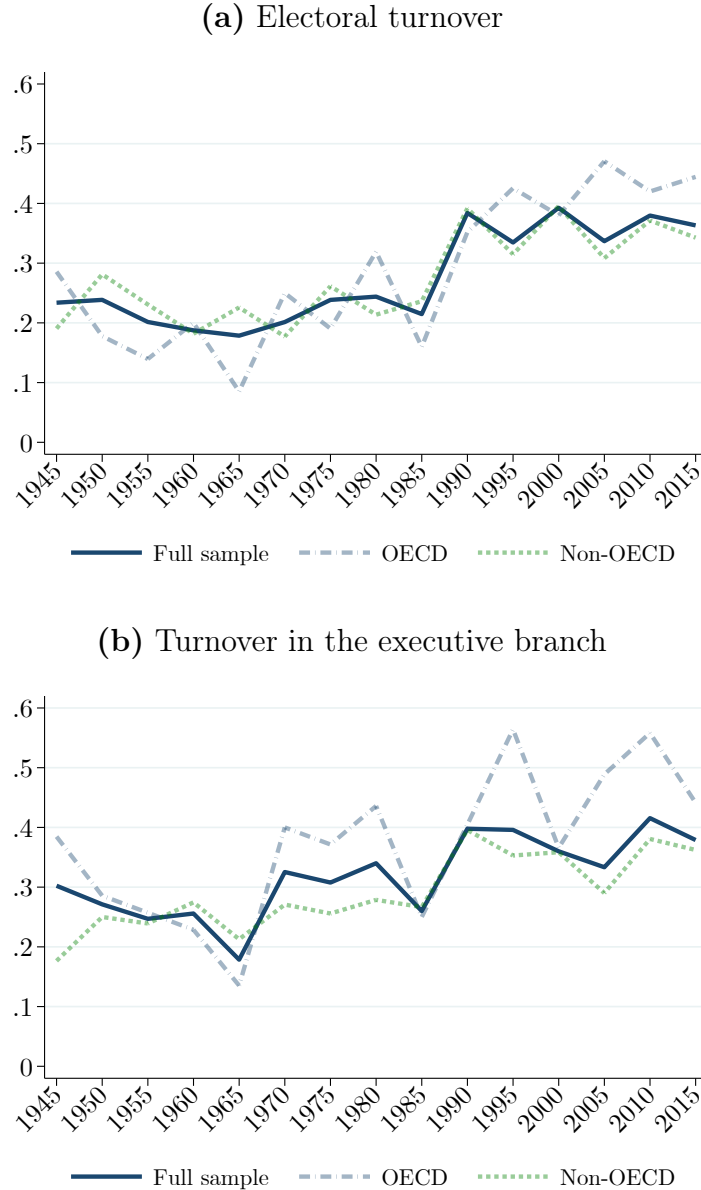
To explore the impacts of power transitions caused by elections, we build a new dataset of national election results. Our dataset includes all presidential and parliamentary elections held around the world between 1946 and 2018. We estimate the impact of electing a challenger versus that of reelecting the incumbent on several dimensions of country performance, using a regression discontinuity design (RDD) across countries. Electoral turnovers are not random events and in particular may be more likely to occur after an economic downturn, making it difficult to attribute post-election differences in performance to the electoral outcome ([Brender and Drazen, 2008](#); [Fair, 2009](#); [Nunn et al., 2018](#)). By focusing on close elections in which the incumbent narrowly won or lost, our empirical

strategy seeks to address this concern. Our paper is among the first to implement a close-elections RDD in a cross-country setting, a demanding empirical exercise which required assembling the largest possible database of national election results through a systematic process of identifying and validating available data sources.¹

We define an electoral turnover as an electoral defeat of the incumbent candidate or party, namely an election where the candidate of the incumbency fails to secure a plurality of votes (in presidential elections) or a plurality of seats (in parliamentary elections). We identify an incumbent candidate or party across 2,488 national elections, including 1,817 parliamentary elections and 671 presidential elections. These elections constitute the main sample for our analysis. Figure 2.1 shows the worldwide prevalence of electoral turnovers since 1945. Importantly, our analysis distinguishes between *electoral turnovers* (electoral defeats of the incumbent, which may or may not lead to a change of leadership) and changes in the leadership of the executive branch resulting from elections. We refer to the latter as *executive turnovers*.

¹One exception is Girardi (2020), who estimates the stock market impacts of partisanship across 758 national elections. Others have implemented close elections RDDs using data from local elections across multiple countries (e.g., Anagol and Fujiwara, 2016; Granzier et al., 2023).

Figure 2.1: Share of elections with a turnover



Notes: This figure plots the share of elections in our regression sample associated with a turnover for each half-decade since 1945. Panel a focuses on electoral turnovers and panel b on turnovers in the executive branch. We define electoral turnovers and executive turnovers in Section 2.3.1 and Section 2.3.5, respectively.

Our identification strategy assumes that elections won by incumbents are ex ante comparable to elections won by challengers. One important concern involves the potential ability of incumbents to manipulate election results. Under such manipulation, any effects of turnovers on performance could be driven by unobservable differences between elections that lead to a turnover, and those that do not. We present various identification checks to validate our empirical strategy, including the density test from Cattaneo et al. (2018) to detect electoral manipulation in close elections, and the permutation test from Canay and

Kamat (2017). We also show that a range of baseline covariates are continuous across the RD cutoff, and we obtain similar results using the randomization inference procedure from Cattaneo et al. (2015). With these identification checks and our discussion of the challenges associated with estimating a RDD across countries, we extend to a cross-country setting the literature discussing the validity of RDDs in close elections (e.g., Eggers et al., 2015). While local elections have been the focus of close-elections RDDs, national elections are allegedly more consequential for the economic and democratic trajectory of countries, as national leaders have more policy tools at their disposal than local leaders.

Turnovers potentially affect a large number of outcomes. We first focus on four economic indicators that are natural measures of a country’s performance (GDP per capita growth, inflation, unemployment, and trade intensity). We then ask whether the economic impacts of turnovers affect citizens’ welfare, which we quantify in terms of human development. Finally, we explore whether turnovers (or the lack thereof) induce a deterioration of democratic quality, as a result of instability or institutional capture by reelected incumbents. To minimize the risk of finding false positives, we follow Kling et al. (2007) to compute an index of economic performance combining the four economic outcomes above, as well as a general index of performance combining economic performance with human development and democratic quality. To construct these indicators, we identified outcomes from a variety of administrative sources, prioritizing sources with the best reliability and coverage.² Our main outcomes of interest are the changes in our measures of country performance over a typical four-year election cycle. For ease of interpretation, we normalize these outcomes such that estimates can be interpreted in terms of their standard deviation.

We first show that electoral turnovers improve performance along several dimensions, resulting in a 0.27 standard deviation (SD) improvement in economic performance and a 0.28 SD increase in our general index of performance. These positive effects are large in magnitude, robust to a range of robustness checks, and materialize gradually over time. The largest effects of turnovers are on measures of trade intensity and macroeconomic discipline, such as inflation, rather than GDP per capita growth. These effects tend to be larger in settings where governments face weaker constraints, namely after presidential elections, when checks and balances on the executive are weaker, when more power is vested in the executive leader, and in less democratic regimes. However, the point estimates are positive in all these subsamples, and differences across subsamples are generally non-significant. We then show that the effects of electoral turnovers are partly driven by *executive* turnovers in both presidential and parliamentary systems. In the latter, a defeat of the incumbent party discontinuously increases the probability of an executive turnover. Having documented this relationship, we show that executive turnovers improve a country’s index of performance by 0.34 SD. Overall, these results consistently show that power transitions caused by elections improve country performance.

One question raised by our approach is whether the positive impacts of turnovers only hold for close elections or extend to all national elections. Incumbents who struggle to win reelection may be worse performers than the average incumbent, and weak incumbents may

²Our focus is on outcomes measured one to four years after the election. We do not study the short-run response of financial markets, which has been explored by papers studying the left-right divide (Snowberg et al., 2007; Girardi, 2020).

encourage the strategic entry of high-quality challengers (Gordon et al., 2007; Ban et al., 2016). We conduct several exercises to show that the positive effect of turnovers extends to elections which are less close. First, we show that OLS estimates of the impacts of turnovers are also positive, even when we consider decisively won elections. Second, we look at a subsample of national elections involving “unlucky incumbents,” namely elections conducted in a context of unfavorable global oil prices capturing adverse economic circumstances outside the incumbent’s control (Arezki et al., 2022). Our main results largely hold in such contexts. Third, we implement the procedure from Angrist and Rokkanen (2015) to show that the effects of turnovers remain positive and sizeable away from the RD cutoff, and thus are not just driven by election closeness. Finally, there is little evidence that the quality of elected leaders (proxied by subsequent electoral performance) jumps discontinuously after a turnover. Our exploration of the role of quality is motivated by the following observation. By focusing on close national elections, we may be comparing low-quality incumbents (elected on the left-hand side of the threshold) to high-quality challengers (elected on the right-hand side of the threshold). While the quality of individual candidates varies smoothly across the RD threshold, any potential jump in the quality of *elected leaders* could therefore contribute to the effects of turnovers on performance. We additionally show that challengers elected in close elections are not disproportionately more left-wing, more populist, less liberal, or younger than reelected incumbents.

We highlight two main mechanisms behind these shifts in performance: policy change and improved governance. First, the effects of turnovers on country performance appear driven in part by greater international openness as well as changes in domestic policies. We find positive effects of turnovers on trade diversification and on country-level measures of globalization from Gygli et al. (2019), suggesting that electoral turnovers are accompanied by an effort to intensify exchanges with the rest of the world. On the domestic front, the effect of electing challengers on a measure of change in government intervention is 0.21 SD. Second, we show that turnovers reduce perceived corruption and improve the quality of governance. These effects increase over time: challengers become relatively less corrupt than reelected incumbents over the course of the subsequent term.³ Building on insights from the literature on political agency (Besley, 2007), we conclude that mechanisms related to reputation and reelection concerns likely contribute to explain our results, together with a general erosion of power taking place after many years spent in office. Turnovers improve performance by helping to replace poorly incentivized and low-performing incumbents with new national leaders facing greater incentives to prove themselves to their electorate. This can also explain why the effects of turnovers are slightly larger close to the RD threshold: narrowly reelected incumbents may be more likely to expect an impending electoral defeat and thus face lower incentives to perform, relative to incumbents reelected with a wide margin.

By providing the first estimates of the impact of national electoral turnovers, our paper relates to the literature on the economic impacts of democracy. Others have shown that democratization affects the pace of policy reforms (Papaioannou and Siourounis, 2008; Giuliano et al., 2013) and economic growth (Rodrik and Wacziarg, 2005; Acemoglu et al.,

³One explanation for this could be constitutional term limits, which are generally more binding for reelected incumbents than for first-term leaders. Term-limited leaders likely face fewer incentives to perform well (Ferraz and Finan, 2011; Fourniaies and Hall, 2021). However, we show that our results are unlikely to simply be driven by term limits (see Section 2.5.2).

2019). We show that democracy is not a sufficient condition to deliver good performance: the latter also requires a competitive electoral system allowing citizens to periodically replace their country’s leadership. Furthermore, many regimes classified as hybrid or populist also hold elections (Guriev and Treisman, 2019). We focus on a phenomenon, electoral turnovers, which occurs across different types of political regimes.⁴ By studying the impact of power transitions induced by elections, our results extend beyond the literature on the impacts of democratization.

Our results can also be interpreted in light of Olson (1984)’s seminal argument, itself echoing Hegel (1820), that stable societies eventually experience institutional sclerosis and economic stagnation, unlike societies that undergo deep structural changes as a result of wars and revolutions. While our analysis focuses primarily on power transitions occurring within a given set of institutions and political regime, it is possible that electoral turnovers also trigger a reshuffling of the governing elite and impede the formation of “distributional coalitions” which, in Olson’s view, could undermine economic efficiency.

Finally, we build on the literature studying the impacts of turnovers in local governments. Previous studies have shown that political renewal can improve performance by enhancing politicians’ incentives (Ferraz and Finan, 2011). Other evidence suggests that political turnovers may engender excessive bureaucratic turnover and foster patronage in public sector organizations, with detrimental effects on performance (Colonnelli et al., 2020; Akhtari et al., 2022). Local turnover in bipartisan systems can also affect fiscal policy (Carlini et al., 2023) and labor market outcomes (Beland, 2015). Our results obtained across a global sample of national elections suggest that the benefits of turnovers outweigh their costs at that level. This is an important result given that many dimensions of policy (e.g., trade or monetary policy) are exclusively determined at the national level. In line with this result, we find no evidence of adverse effects on bureaucratic quality in our sample, perhaps because newly elected leaders are more successful in maintaining an impartial and rigorous administration.

The paper proceeds as follows. We describe our data and empirical strategy in Sections 2.2 and 2.3, respectively. Section 2.4 presents our main results and Section 2.5 discusses mechanisms. Section 2.6 concludes.

2.2 Key Data Sources

Our dataset combines the results of all presidential and parliamentary elections conducted worldwide since World War II with detailed data on economic performance, policy outcomes, leader characteristics, and regime types. This section describes our approach; the Data Appendix provides additional details.

2.2.1 Data on Elections, Leaders, and Institutions

Elections and electoral results. Our complete dataset contains 3,985 national elections, including 1,023 presidential elections and 2,962 parliamentary elections. The latter include all

⁴Importantly, electoral turnovers are not associated with more democratization episodes or democratic reversals, and they also improve country performance when the election does not coincide with a regime change.

unicameral parliaments and the lower chamber of bicameral parliaments in both presidential and parliamentary systems. To construct this database, we first identify all national elections held between 1946 and 2018 using the Varieties of Democracy (V-Dem) database (Coppedge et al., 2021) as our primary source. We complement V-Dem with the Parliaments and Governments (PARLGOV) database, the Manifesto Project, several books by Dieter Nohlen and coauthors, the Database of Political Institutions (DPI), the Global Elections Database, and the Constituency-Level Elections Archive. Data Appendix A describes this data construction.

We then search for the results of each election. For this step, we rely on the aforementioned sources as well as Adam Carr’s Psephos election archive, the African Elections Database, the European Elections Database, the Political Database of the Americas, the Inter-Parliamentary Union PARLINE database, the International Institute for Democracy and Electoral Assistance (IDEA) database, the International Foundation for Electoral Systems (IFES) election guide, and the National Archives presidential elections database. We prioritize sources which cover more elections, show fewer inconsistencies, and have been used more often in previous work.⁵ Appendix Figure B.1 shows a comparison between our sample and preexisting databases as well as the number of elections collected from each source.

We collect data on vote shares for presidential elections and seat shares for parliamentary elections. We then systematically check data consistency within each source (Data Appendix A.5). In collecting parliamentary election results, we take into account the existence of ex ante coalitions officially formed before the election, as described in Section 2.3.1 and Data Appendix A.4. Overall, we retrieve election results for 97% of presidential elections and 97% of parliamentary elections identified in the first step.

Leaders and their parties. We link election results with information on leaders and political parties. First, we identify a head of state and a head of government for each country-year in our data (Data Appendix B). We then associate parties in our database of election results with parties in V-Parties, another dataset provided by V-Dem and containing various measures of party ideology (Data Appendix C).

Institutions and regimes. We also retrieve data on the regimes and rules under which elections take place. First, we divide the modern history of each country into regimes, allowing us to understand the role of each election (Data Appendix D). Second, we determine whether each election led to the nomination of a head of state, a head of government, or none of the two. Data Appendix E describes the rules used to determine the role of each election, as well as data quality checks. Third, we rely on the Comparative Constitutions Project (Elkins et al., 2021) to identify term limits (Data Appendix F).

⁵Academic sources are lacking or incomplete for 12.7% of elections. In these cases, we collect the results from Wikipedia.

2.2.2 Data on National Outcomes

In our analysis, we estimate the impact of electoral turnovers on measures of country performance falling into three broad categories: economic performance, human development, and democratic quality.⁶

To measure economic performance, we use four authoritative indicators: GDP per capita growth from the Penn World Tables (Feenstra et al., 2015),⁷ CPI inflation from the IMF, the unemployment rate from the ILO, and trade intensity, defined as the total value of imports and exports divided by GDP (measured by the World Bank). We selected these data sources after careful consideration of other available sources for each outcome, as they provide the most comprehensive coverage in terms of the number of national elections covered in our data. Nonetheless, in the Appendix, we probe the robustness of our results to using alternative data sources. To avoid results being driven by outliers (e.g., hyperinflation episodes or terms of trade swings), all components of the economic performance index are winsorized at the 3rd and 97th percentiles. We check the robustness of our results to different ways of winsorizing and to trimming. For human development, we use the Human Development Index (HDI) from the UNDP. Finally, we rely on V-Dem’s measures of the quality of democracy, including deliberative, egalitarian, liberal, participatory, and electoral democracy. We use the simple average of these five measures (which all vary between 0 and 1) to quantify the quality of democracy. Appendix Table B.1 indicates the number of elections for which we have data on each outcome. Appendix Table C.1 shows that the fraction of observations with missing data is not significantly affected by turnovers.

Data Appendix G provides a detailed description of the rules we used to construct our dataset of outcomes. Given the scope of our analysis, this data collection required making appropriate choices in terms of which sources to prioritize and how to define our outcomes of interest. While we aimed to cover as many countries as possible, we acknowledge the inherent limitations and shortcomings associated with combining data for such a large sample. First, our data coverage across countries and years is imperfect and varies across outcomes. For example, the HDI is only available from 1990 onwards. Second, measurement quality is also heterogeneous across countries and years. Third, we are unable to observe some important outcomes. For instance, since we generally only have limited information on the specific policies implemented by elected candidates, we focus in our analysis on several measures of government intervention, which are most consistently measured across our sample. Notwithstanding these limitations, our data coverage allows us to estimate the effects of turnovers on a wide range of outcomes and across a large sample of national elections between 1946 and 2018.

⁶Appendix Tables D.1 to D.4 report results for a broader set of outcomes than those considered in our baseline analysis.

⁷We use version 9.0 of the Penn World Tables, which addresses the data issues highlighted by Johnson et al. (2013) (see Feenstra et al., 2015). In Appendix Table D.1, we show robustness to using alternative sources to measure GDP growth.

2.3 Empirical Framework

This section presents our empirical strategy. We first define the key concepts of incumbency and electoral turnovers (Section 2.3.1). We then describe our sample (Section 2.3.2) and the construction of our outcome variables (Section 2.3.3). Section 2.3.4 presents our main empirical specification. Finally, we describe the alternative specification used to estimate the impact of turnovers in the executive branch (Section 2.3.5).

2.3.1 Defining Electoral Turnovers

Our analysis estimates the impacts of electoral turnovers using a RDD. To set up this design, we must identify which candidate or party represents the incumbency in each election. We then define electoral turnovers as a defeat of the incumbent candidate or party. This section describes how we construct these key variables for each type of election.

Presidential elections. In presidential elections, the incumbent is the individual or party which effectively held power at the time the election took place. To account for caretaker governments and transition periods, we use a flexible definition: we define the incumbent as the leader who held executive power for a period of at least 365 days in the two-year period before the election. The incumbent party is analogously defined as the party which held executive power for at least 365 days during the same period. Panel a of Appendix Table B.2 illustrates the implementation of this rule.

To account for all cases where the incumbent competes or has a clear designated successor, we consider that the incumbency is represented by: (1) the incumbent leader, if the leader is personally competing (56.3% of cases); (2) the candidate of the incumbent party, if the leader is not personally competing (39.4% of cases); (3) the candidate unambiguously designated as the representative of the ruling government, if neither the country leader nor any candidate from their party are competing (4.3% of cases).⁸ Elections in which we cannot define a candidate of the incumbency are excluded from the analysis.

We then construct a treatment variable T equal to 1 if the incumbent candidate loses the election. The running variable X is equal to the margin of victory of the best ranked challenger, which is the difference between this challenger's and the incumbent's vote share. When the election features a runoff, we use the second round results to construct the running and treatment variables. If the incumbent did not compete in the second round, we do not define a running variable. For indirect presidential elections (including U.S. presidential elections), we use electoral college vote shares to define the running variable. Furthermore, we exclude elections where one candidate ran unopposed, elections which were not the last presidential election in the calendar year, indirect presidential elections which could easily be manipulated, as well as various types of inconsequential elections (Appendices A.1 and A.3 provide additional details).⁹ We checked the validity of our key variables through an

⁸When the election features two rounds, we check whenever possible that this support was expressed before the first round.

⁹In Appendix Figure C.1, we verify that electoral turnovers are not associated with a significantly different probability of exclusion from the sample following an inconsequential election. In Appendix Table D.5, we further show that our main results are robust to including inconsequential elections in our sample.

independent audit encompassing all elections with a running variable X between -15 and +15 percentage points and a subset of other elections (see Data Appendix H.4).

Parliamentary elections. In parliamentary elections, the incumbent party is the party which secured a plurality of seats in the previous election. Our definition is based on the results of the previous election because the available data do not systematically document how the composition of parliaments varies between national elections, for example through by-elections. Furthermore, note that only some parliamentary elections lead to the designation of a leader of the executive branch. Our baseline definition is based on seat shares, because the relative seat shares obtained by different parties may matter in and of themselves, independently of who controls the executive. This definition allows us to include parliamentary elections which are not associated with the designation of a member of the executive and which may nonetheless be impactful, such as the elections to the lower chamber of parliament in presidential systems (e.g., the U.S. House of Representatives).

We set the treatment variable T equal to 1 if the incumbent party fails to secure again a plurality. The running variable X is equal to the margin of victory of the best-ranked opposition party, i.e., the difference between the seat share of this party and the incumbent party. We again drop elections in which the incumbent party ran unopposed or obtained 100% of the seats, as well as elections to constitutional assemblies without any legislative power and elections where a fraction of parliamentary seats are appointed rather than elected (see Appendix A.1). Finally, we do not define a treatment variable in 23 elections where the incumbent and challenger parties obtained exactly the same number of seats.

As with presidential elections, we account for cases where the incumbent party has a clear successor. We also account for the existence of coalitions. We collected systematic evidence to identify coalitions and distinguish those officially formed before the election (ex ante) from those formed after the election (ex post). To compute seat shares, we group together parties belonging to the same ex ante coalition but we keep as separate the members of ex post coalitions, since these are endogenous to election results. Therefore, the candidate of the incumbency in parliamentary elections is: (1) the incumbent party when it participates in the election, or alternatively the coalition that the incumbent party is part of; (2) the party or coalition unambiguously designated as the representative of the ruling government if the incumbent party is not competing. As with presidential elections, we exclude elections in which we cannot define an incumbent party. When identifying the previous election, we exclude inconsequential elections, constitutional assembly elections, and elections that are not the last parliamentary election in a calendar year. We also impose that the previous election took place no more than ten years earlier. Additional details regarding the determination of the incumbent party are provided in Appendix A.4.

2.3.2 Sample Description

Overall, we are able to identify an incumbent candidate or party and to define a running and treatment variable in 671 presidential elections and 1,817 parliamentary elections. Accordingly, our main analysis sample includes a total of 2,488 national elections across

201 countries.¹⁰ 59% of these elections are conducted under regimes classified as liberal or electoral democracies by V-Dem, and 26% of elections take place in OECD countries. Close to the threshold, when the running variable is comprised between -5 and 5 percentage points, the fractions of elections in democracies and in OECD countries increase to 84% and 39%, respectively. Figure 2.1 (a) shows the share of elections featuring an electoral defeat of the incumbency. Appendix Figure B.2 shows the geographic distribution of all elections we identified over the period, all elections included in our main analysis, and all elections with a turnover.

2.3.3 Outcome Variables

We explore how turnovers affect national outcomes. To compare the level of an outcome Y before and after an election E taking place in country c and year t_E , we compute the difference between the average level of Y in the four years following the election and the level of Y in the year before the election:

$$\underbrace{\Delta Y_E}_{\text{Improvement over the election cycle}} = \underbrace{\left(\frac{1}{k} \sum_{\tau=1}^k Y_{c,t_E+\tau} \right)}_{\text{Post-election average of the outcome}} - \underbrace{Y_{c,t_E-1}}_{\text{Pre-election value of the outcome}}$$

This definition allows us to control for large differences in levels across countries and time periods and to increase the precision of our estimates. In our baseline analysis, we use $k = 4$ since the modal distance between elections of the same type in a country is four years. To make estimates comparable across outcomes, we standardize the ΔY_E . In the Appendix, we report a wide range of checks verifying the robustness of our results to using different values of k and to replacing the pre-election value of the outcome with the average over the last three years before the election.

Building indices. Electoral results potentially affect a large number of outcomes, which raises the problem of multiple testing. To minimize the risk of overrejecting the null hypothesis and to gain statistical power, we group the ΔY_E constructed above for our four economic outcomes (GDP per capita growth, inflation, unemployment, and trade intensity) into an index equal to the unweighted average of the four standardized variables, following Kling et al. (2007). When a component is missing, we do not include it in the average (i.e., we do not impute a value). Furthermore, we adjust the sign of the components such that higher values of the index reflect better outcomes. Thus, inflation and unemployment enter negatively in the index. We use the same method to aggregate this economic index and the standardized ΔY_E of human development and democracy into the general index of country performance.

¹⁰Our sample includes a few autonomous territories which are not United Nations members (e.g., Greenland and Puerto Rico).

2.3.4 Regression Discontinuity Estimation

Effects of an electoral defeat of the incumbent. We estimate the effects of an electoral defeat of the incumbent with the following RD equation, using one observation per election:

$$\Delta Y_E = \alpha + \beta_1 X_E + \beta_2 X_E T_E + \gamma T_E + \varepsilon_E, \quad (2.1)$$

where X_E , the running variable, is the victory margin of the best-ranked challenger and $T_E = 1(X_E > 0)$, as described in Section 2.3.1. ΔY_E measures the difference in outcomes between the post-election average and the pre-election value (see Section 2.3.3). Equation (2.1) is estimated with the non-parametric method of Calonico et al. (2014). Using this method, we report the standard RD point estimate γ and the robust standard error as well as the p-value associated with the robust confidence interval for γ .¹¹ In our baseline estimation, we use the MSE-optimal bandwidth from Calonico et al. (2014). When looking at the general index of country performance, this bandwidth includes 858 national elections, corresponding to a bandwidth size of 14.4 percentage points.¹²

In our exploration of mechanisms, we also estimate alternative versions of equation (2.1) where we use $|\Delta Y_E|$ as the outcome (i.e., the absolute value of the difference in outcomes between the post-election average and the pre-election value). Finally, we compare the RD point estimates of γ in equation (2.1) with the corresponding estimates obtained via OLS, without controlling for the running variable X_E . In this case, the point estimates do not have a causal interpretation and may instead capture other factors which are endogenous to the occurrence of an electoral turnover.

Interpretation. To measure the change in performance caused by an electoral turnover, the ideal experiment would be a randomization of the electoral outcome across many elections. Without a randomized experiment, a simple comparison of outcomes between elections won by challengers and those won by incumbents would fail to identify the effect of turnovers. There are at least two sources of endogeneity to consider. First, unobserved variables correlated with incumbents' reelection prospects might affect future performance. Second, reelected incumbents may be systematically different from defeated incumbents, and elected challengers may be systematically different from defeated challengers.

Our empirical strategy addresses the first concern, as economic conditions and other variables that may affect a country's performance after the election can be expected to be identical on both sides of the RD threshold. In Section 2.4.1, we check that this is indeed verified in our data. By contrast, OLS estimates of equation (2.1) would not eliminate this bias. To some extent, the RD design also addresses the second concern as it ensures that the characteristics of reelected and defeated incumbents vary smoothly at the threshold, and likewise for challengers. This does not preclude the possibility that the characteristics

¹¹In Appendix Table D.6, we show that our results are robust to clustering standard errors at the country $\times$ year level.

¹²This bandwidth is well within the range of the bandwidths used in other close election RDDs. Note that the exact size of the MSE-optimal bandwidth from Calonico et al. (2014) depends on the dependent variable considered. Across our main outcomes, the number of elections included in the bandwidth varies between 562 and 1,192 (see Table 2.1).

of *elected leaders* might vary discontinuously at the threshold, a possibility we explore in Section 2.4.5.

Another aspect to consider when interpreting estimates of γ in equation (2.1) involves differences in the characteristics of incumbents reelected in close elections compared to those who decisively win. Close elections may feature incumbents who are negatively selected, and weak incumbents might incentivize high-quality challengers to compete. By focusing on close elections, we may therefore be comparing “weak” incumbents with “strong” challengers. Furthermore, closely reelected incumbents likely face different incentives, relative to incumbents who are decisively reelected. On the one hand, closely elected leaders may have lower legitimacy and stronger incentives to perform. On the other hand, a close victory margin might signal to these leaders a high probability of losing the subsequent election, which could lead to discouragement effects and a *de facto* term limit. Whichever hypothesis holds has implications for the interpretation of our main results: more negative sorting and worse incentives for closely reelected incumbents—relative to incumbents who decisively win reelection—imply relatively larger estimates of the effects of turnovers close to the cutoff. We return to this discussion in Section 2.4.4.

2.3.5 Turnovers in the Executive Branch

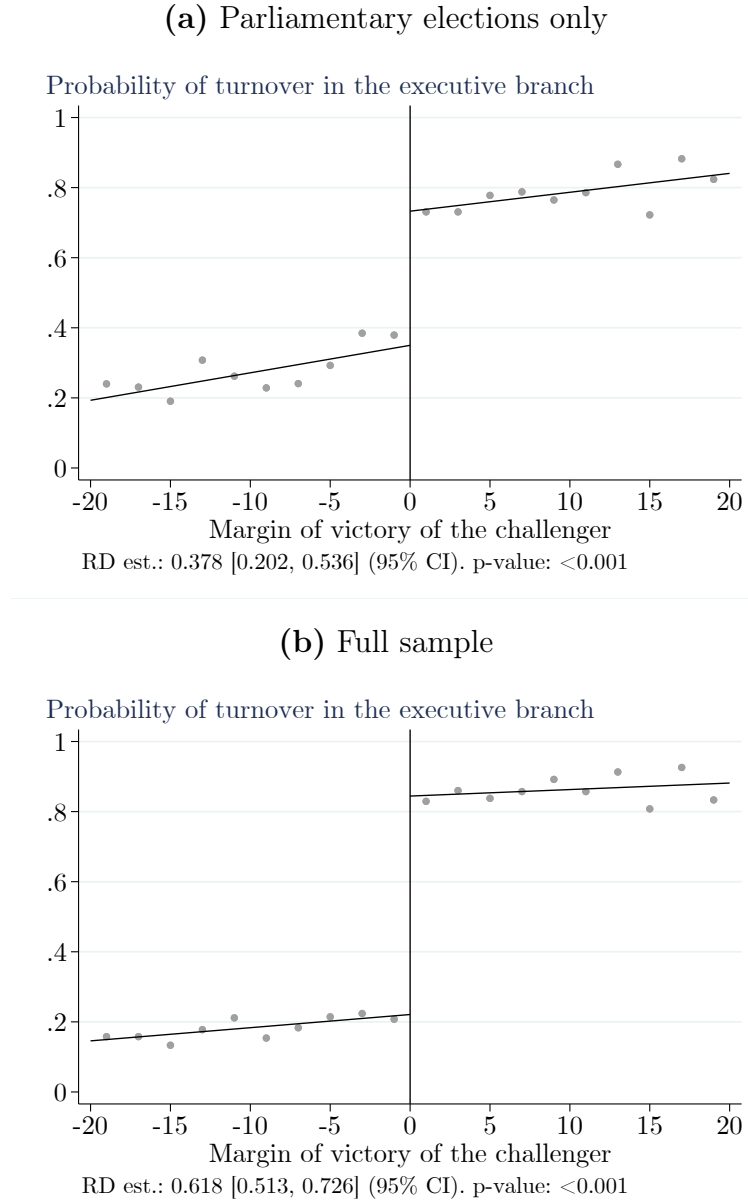
Electoral turnovers often, but do not always trigger a change of executive leader. While in presidential elections, electoral and executive turnovers generally coincide, the same may not be true for parliamentary elections. First, only slightly over half of the parliamentary elections in the sample lead (in a constitutional sense) to the designation of an executive leader. Second, in these elections, electoral turnovers usually trigger an executive turnover, but this is not always the case. For example, an incumbent party that loses a parliamentary election may be able to retain executive power by forming a different coalition after the election. To determine whether an election led to an executive turnover, we define a country’s leader and leading party before (resp. after) the election as the individual and party which led the executive during a period of at least 365 days in the two years before (resp. following) the election. Panel b of Appendix Table B.2 illustrates this rule, and Data Appendix H provides further details.

We then define an executive turnover T^x as the nomination of a new executive leader. This variable is set to 0 (meaning that an executive turnover did *not* take place) if: (1) the leader before and after the election are identical; (2) the leading party before and after the election are identical; or (3) the leading party before the election did not compete and instead supported the leading party after the election. If none of these conditions holds, we set T^x as equal to 1. If we are unable to define a leading party before the election (e.g., because the leader was an independent) or after the election, we do not define T^x . Figure 2.1 (b) shows the evolution of the frequency of turnovers in the executive branch since 1945.

Figure 2.2 (a) shows the impact of electoral turnovers on executive turnovers in parliamentary elections that lead to the designation of an executive leader. We use T^x as the dependent variable in a specification in the form of equation (2.1). We observe a large upward jump (of 38 percentage points) at the threshold, indicating that executive turnovers are much more likely to occur when the challenger party obtained slightly more seats than the incumbent party. This result generalizes a finding from Fujiwara and Sanz (2020) to

the entire world and reflects the norm that the party with a plurality of seats generally has priority over the formation of a new government. Naturally, we observe an even larger jump (of 62 percentage points) when we also include presidential elections (Figure 2.2 (b)). Thus, the effects of electoral turnovers on country performance are likely to be mediated in part by executive turnovers.

Figure 2.2: Effect of an electoral turnover on the probability of turnover in the executive branch



Notes: This figure plots the probability of a turnover in the executive branch depending on the margin of victory of the challenger. Turnovers in the executive branch are defined in Section 2.3.5. We restrict the sample to elections leading to the designation of a member of the executive, with panel a being further restricted to the sample of parliamentary elections. Each grey dot represents the probability of a turnover in the executive branch in a 2 percentage points bin and the lines represent a linear fit on each side of the discontinuity. At the bottom of each graph, we report the non-parametric RD estimate from Calonico et al. (2014), with the robust 95% confidence interval in brackets, as well as the robust p-value associated with the robust confidence interval for γ in equation (2.1).

Some of our analyses seek to isolate the effects of executive turnovers. There, we restrict the sample to presidential and parliamentary elections leading to the designation of an

executive leader. These analyses use a distinct running variable, X^x , equal to the margin of victory of the best ranked challenger over the leader or leading party before the election. To estimate the effect of executive turnovers, we rely on a fuzzy RDD in which T^x is instrumented with the assignment variable $A^x = \mathbb{1}(X^x > 0)$.¹³ This specification relies on the following exclusion restriction: the defeat of the leading party only affects outcomes through the higher probability of an executive turnover. Since this assumption is unlikely to hold in some settings, we also report reduced-form estimates in addition to the fuzzy RDD results. The RD, first stage, and reduced-form equations are shown in Appendix A.2.

Examples. To illustrate the definitions above, we provide some examples below and in Appendix A.3:

- **1977 Indian parliamentary elections.** The Indian National Congress (INC) had won by a landslide in the preceding 1971 elections, and its leader Indira Gandhi was Prime Minister before the election. In 1977, the alliance centered around the INC won 34.7% of the seats, while the Janata alliance won 63.6% of the seats. After the election, Morarji Desai of the Janata party became Prime Minister. For this election, we have $X = X^x = 29.0\%$ and $T = T^x = 1$ (there was an electoral defeat of the incumbency and a turnover in the executive branch).
- **2000 U.S. presidential election.** The incumbent leader was Bill Clinton from the Democratic party, which was represented by Al Gore in the election. Gore lost with 266 out of the 538 electoral college votes (49.4%) to George W. Bush, who won 271 votes (50.4%). For this election, we have $X = X^x = 0.9\%$ and $T = T^x = 1$ (there was an electoral defeat of the incumbency and a turnover in the executive branch).
- **2007 French presidential election.** The incumbent leader was Jacques Chirac from the UMP party. He did not compete in the election, but the UMP candidate (Nicolas Sarkozy) did and won with 53.1% of the vote. The runner-up, Segolène Royal, obtained 46.9% of the vote. Here, the running variable is $X = X^x = -6.2\%$ and the treatment is $T = T^x = 0$ (there was an electoral win of the incumbency and no turnover in the executive branch).
- **2011 Danish parliamentary election.** The Venstre party won a plurality of seats in 2007, and the incumbent head of government (Lars Løkke) was from the Venstre. The Venstre party won a plurality of seats in the 2011 elections (with 26.3% of seats), and the Social Democrats party ranked second (with 24.6% of seats), but the head of government after the election was Helle Thorning-Schmidt from the Social Democrats. Here, we have $X = X^x = -1.7\%$. However, we have $T = 0$ and $T^x = 1$ (there was an electoral victory of the incumbency and a turnover in the executive branch).

¹³For parliamentary elections it is possible to have $X^x \neq X$. Indeed, when we estimate the effects of electoral turnovers, the incumbent party is defined using the results of the previous election while when we focus on executive turnovers, the leading party is defined based on which party held executive power in the two-year period before the election. Furthermore, it is possible to have $T^x \neq T$, for instance because the head of government appointed following a parliamentary election is not necessarily affiliated with the party which won a plurality of seats.

2.4 Main Results

This section presents our main results. We first report identification checks and placebo tests to verify the validity of our RDD (Section 2.4.1). We then present our main results showing a positive effect of turnovers on country performance (Section 2.4.2), as well as robustness checks (Section 2.4.3). Section 2.4.4 discusses how our results extend to elections which are less close, and Section 2.4.5 explores how turnovers relate to changes in leader characteristics.

2.4.1 Identification Checks

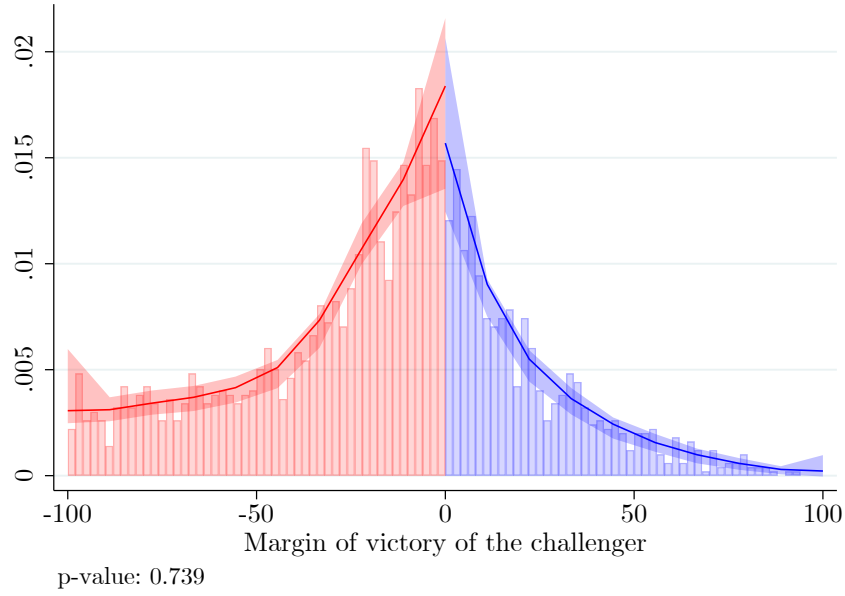
Implementing a RDD in a sample of close national elections raises many challenges, including a concern of sorting at the threshold. Incumbents may be able to manipulate election results, in a way that would systematically benefit them and hurt challengers. If this occurred, we would observe a discontinuous drop in the density of our running variable across the threshold (McCrary, 2008).

To address these concerns, we first implement the density test from Cattaneo et al. (2018). Figure 2.3 reports this density test for all elections (panel a), presidential elections (panel b), and parliamentary elections (panel c). We find no evidence of manipulation in the full sample (p-val. = 0.739) and in the subsamples of presidential and parliamentary elections (p-val. = 0.177 and 0.710, respectively).¹⁴

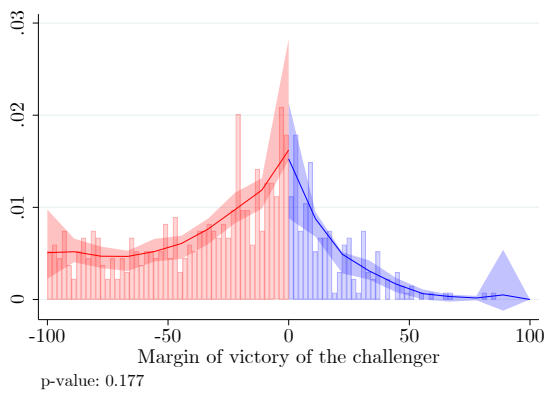
¹⁴In parliamentary elections, bunching at the threshold could be more likely to occur in small parliaments, where it may be easier for a party to win a plurality of seats by manipulating results or exerting additional effort in tangential constituencies. However, the median number of seats in parliaments in our sample is relatively high (127 seats), and we continue to find no evidence of manipulation of the running variable when focusing on parliamentary elections with fewer than 60 seats, which corresponds to the 25th percentile of the distribution (Appendix Figure C.2 (f)). Appendix Table D.8 further shows that the impact of electoral turnovers on performance is robust to excluding parliamentary elections with fewer than 60 seats.

Figure 2.3: Density tests

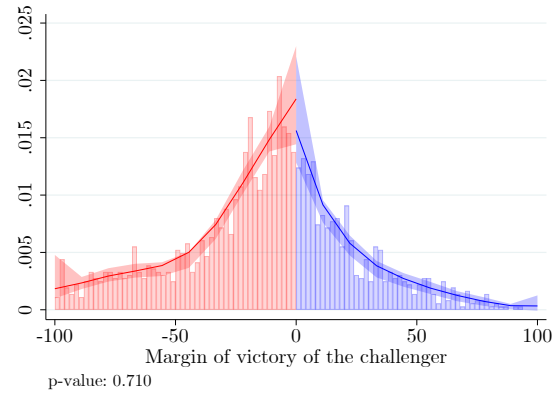
(a) Full sample



(b) Presidential elections



(c) Parliamentary elections



Notes: In this figure, we implement the density test from [Cattaneo et al. \(2018\)](#) using the margin of victory of the challenger as running variable. P-values for this test are reported below each graph, and we plot the density of the running variable on both sides of the cutoff. Panel a includes all elections in our sample, and panels b and c restrict the sample to presidential and parliamentary elections, respectively.

Furthermore, in Appendix Figure C.2, we find no evidence of manipulation among elections in democracies, elections in autocracies, and elections assessed as free and fair by V-Dem (panels a, b, and c). However, the density test fails for elections assessed as not free and fair (panel d). We keep these elections in the main sample because the negative jump in the density of the running variable could be due to endogenous retrospective coding instead of actual manipulation: experts may be more likely to rate an election as not free and fair because it was won by the incumbent. Reassuringly, Appendix Table D.7 shows

that our main results are robust to dropping these elections from the analysis.¹⁵

We then present a variety of placebo tests showing that the treatment has no impact on our outcomes of interest measured in pre-election years. Appendix Tables C.2 to C.4 report these tests with outcomes measured in levels for years $t_E - 1$, $t_E - 2$, and $t_E - 3$, while Appendix Tables C.5 and C.6 report these tests in year-on-year differences. In Appendix Table C.7, we show that turnovers are uncorrelated with decade and region dummies. We also find no jump at the threshold for the time elapsed since the last treatment, the running variable in the previous election, and the value of the treatment in the previous election (Appendix Figure C.3). We further fail to reject the null hypothesis of continuity of the distribution of a large set of pre-election covariates at the cutoff, using [Canay and Kamat \(2017\)](#)’s permutation test, either in our full sample or in the subsamples of presidential, parliamentary, free and fair, and non free and fair elections (see Appendix Table C.8).

2.4.2 Main Results: Effects of Electoral and Executive Turnovers

We now study the consequences of electoral turnovers. Table 2.1 presents RD estimates of the effect of γ from equation (2.1). Our outcomes of interest are the standardized index of economic performance (combining GDP per capita growth, inflation, unemployment, and trade), human development, democracy, and the general index of country performance. Figure 2.4 presents the corresponding RD plots.

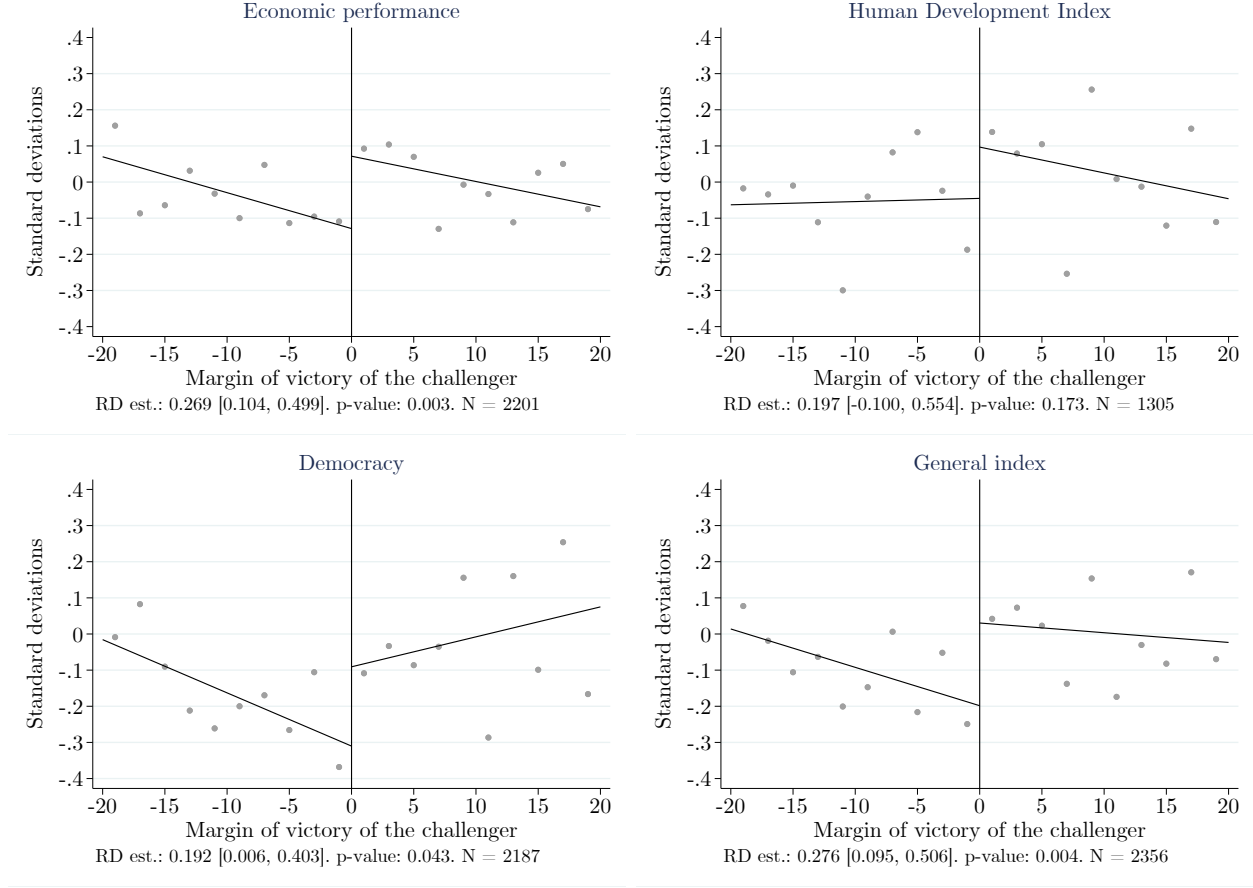
Table 2.1: Effects of electoral turnovers on country performance

	(1) Econ. perf.	(2) GDP p.c. gr.	(3) (Minus) Inflation	(4) (Minus) Unemp.	(5) Trade	(6) HDI	(7) Democ.	(8) General index
El. turn.	0.269*** (0.101)	0.044 (0.155)	0.430** (0.192)	0.221 (0.170)	0.249** (0.127)	0.197 (0.167)	0.192** (0.101)	0.276*** (0.105)
p-val	[0.003]	[0.841]	[0.012]	[0.101]	[0.028]	[0.173]	[0.043]	[0.004]
N	2201	1815	1887	1331	1767	1305	2187	2356
N eff.	763	827	723	670	763	565	1194	861
Band.	13.5	19.5	14.8	21.6	17.4	18.0	23.8	14.4

Notes: This table reports RD estimates corresponding to equation (2.1) for our measures of country performance, expressed in standard deviation terms. We report local linear regression estimates from [Calonico et al. \(2014\)](#), robust standard errors in parentheses, the p-value associated with the robust confidence interval in brackets, the number of observations in the sample and in the bandwidth, and the MSERD-optimal bandwidth. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

¹⁵In addition, we obtain similar results when focusing on elections following a previous free and fair election (Appendix Table D.9). The logic behind this sample restriction is that the assessed fairness of the previous election predicts the fairness of the present election (with a 0.72 correlation) but it should not be endogenous to the present treatment.

Figure 2.4: Effects of electoral turnovers on country performance



Notes: This figure reports RD plots corresponding to equation (2.1). The dependent variables are a standardized index of economic performance (combining GDP growth, inflation, unemployment, and trade), human development, democracy, and a general index of performance combining all these components. The grey dots are sample means across two-percentage-point bins of the running variable. See Section 2.3.3 for details on the construction of each outcome and data sources. At the bottom of each graph, we report the local linear regression estimate from Calonico et al. (2014), with the robust confidence interval in brackets, as well as the robust p-value associated with the robust confidence interval for γ in equation (2.1).

Electoral turnovers improve country performance along all dimensions, although estimates are statistically significant for only some of these dimensions. As shown in Table 2.1, an electoral defeat of the incumbent results in a 0.27 SD improvement in economic performance. This is mainly driven by a decrease in both inflation and unemployment (respectively significant at the 5% level and not significant, with a p-value of 0.101) and an increase in trade intensity (significant at the 5% level).¹⁶ The increase in trade intensity is, in turn, associated with increased openness and greater trade diversification. We find an increase in Herfindahl indices of countries' trade partners' shares for imports and exports (Appendix Table D.2) and we also measure positive effects on country-level measures of

¹⁶Turnovers increase trade intensity measured in the World Bank data, our preferred data source, by 0.25 SD (significant at the 5% level). We find effects of 0.06 SD and 0.18 SD (not significant, and significant at the 10% level) when using data from CEPII and the WTO (see Appendix Table D.2).

globalization from Gygli et al. (2019), suggesting that electoral turnovers are accompanied by an effort to intensify exchanges with the rest of the world. The effect of turnovers on GDP per capita growth is positive but small in magnitude and non-significant. We also estimate positive and large effects of turnovers on human development (0.20 SD), but this point estimate falls short of statistical significance at conventional levels.¹⁷ Finally, our general index of country performance increases by 0.28 SD after an electoral turnover, which is significant at the 1% level. In Appendix Table D.10, we also report estimates measured in natural units. These estimates can be compared to the mean and standard deviations of the outcomes of interest (Appendix Table B.3).

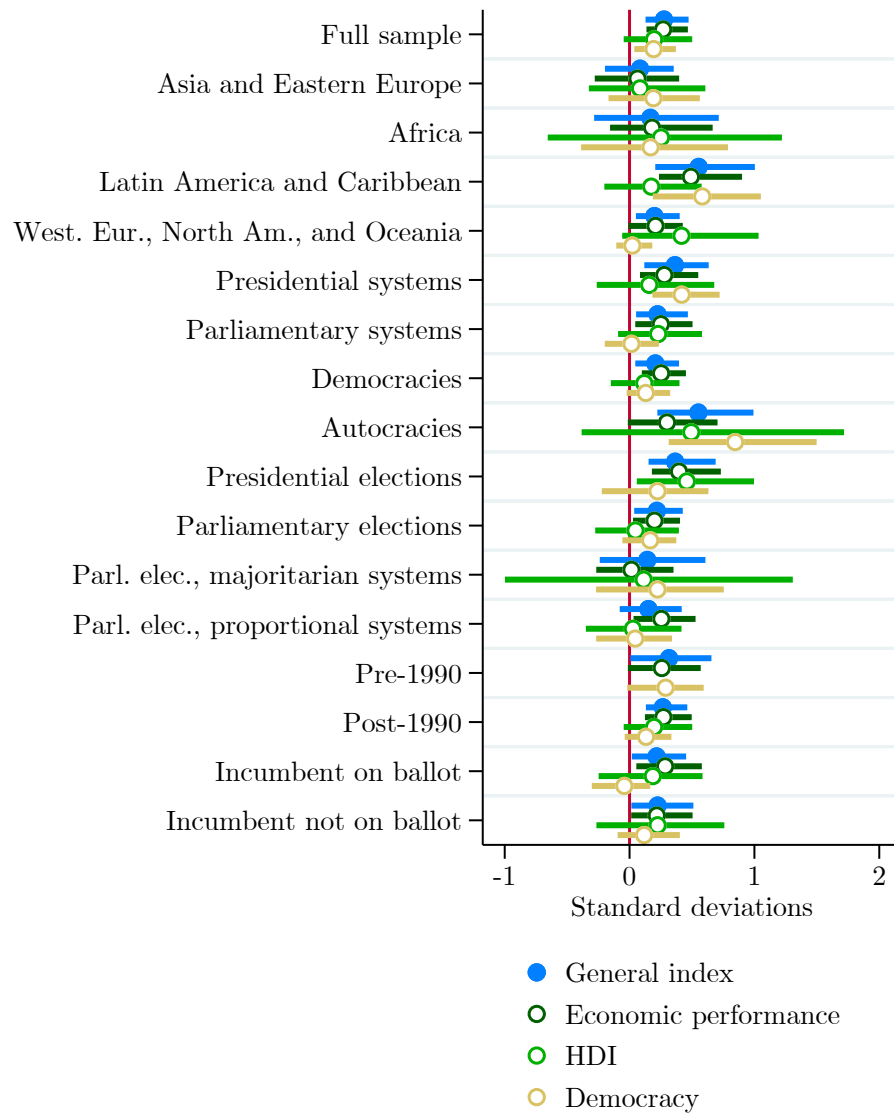
The positive effect of turnovers on democracy is also large (0.19 SD) and significant. We interpret this estimate with much caution because experts responsible for the retrospective evaluation of democratic quality may be influenced ex post by the occurrence of a turnover. Nonetheless, the effect of turnovers on the general index of performance remains large and significant (0.35 SD, see Appendix Table D.8) when excluding democracy from the construction of this index. In addition, in Appendix Table D.4, we find consistent effects on democratic quality across a wide range of independent sources and indicators, including dichotomous measures of democratic quality from the academic literature (Cheibub et al., 2010; Boix et al., 2018), and indicators which consider other dimensions than electoral democracy and are thus less likely to be directly affected by the occurrence of an electoral turnover.¹⁸ Furthermore, we test whether turnovers affect the likelihood of observing large changes in democratic quality in the period following the election, which may reflect specific events that are distinct from the election itself and capture the subsequent evolution of democratic institutions, as perceived by experts. We find a negligible effect on the probability of a large positive shock but a substantial negative effect of -5.4 percentage points, which is significant at the 10% level, on the probability of a large negative democratic shock, suggesting that turnovers prevent democratic backsliding that would occur under reelected incumbents. Finally, we find positive effects using sources which code democratic quality in real time (Freedom House) and thus cannot be affected by endogenous retrospective expert coding or recall bias.

Heterogeneity. Figure 2.5 shows the positive effects of turnovers across a variety of subsamples, including elections held across different world regions, before and after 1990, and under different types of political regimes and electoral systems. Table 2.2 reports heterogeneous effects along three key dimensions. First, looking at presidential and parliamentary elections separately, we find positive and significant effects in both subsamples, with slightly larger effects for presidential elections (columns 2 vs. 3).

¹⁷The UNDP's Human Development Index is the geometric mean of three components measuring income, life expectancy, and education. Because income is similar to GDP per capita, which we include separately in our economic performance index, we show effects on each of these three components separately in Appendix Table D.3. All point estimates are positive and the effect on the geometric mean of life expectancy and education is larger, if anything, than the baseline effect.

¹⁸Specifically, we find effects of 0.13 to 0.20 SD on all five independent dimensions of democracy included in V-Dem's overall index: electoral democracy, egalitarian democracy, liberal democracy, participatory democracy, and deliberative democracy.

Figure 2.5: Effects of turnovers on country performance across subsamples



Notes: This figure plots RD estimates and 90% robust confidence intervals of the effects of electoral turnovers on our general index of country performance across subsamples, including different regions, regime types, election types, time periods, and whether the individual incumbent was nominally on the ballot or not.

Table 2.2: Heterogeneity analysis

	Election type			Incumbent on ballot		Checks & balances	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Baseline	Pres.	Parl.	Yes	No	High	Low
Economic performance	0.269*** (0.101)	0.396*** (0.168)	0.200* (0.115)	0.285** (0.160)	0.218* (0.150)	0.159 (0.108)	0.357** (0.171)
GDP p.c. growth	0.044 (0.155)	-0.020 (0.239)	0.087 (0.200)	-0.285 (0.242)	0.074 (0.265)	-0.030 (0.152)	0.120 (0.243)
(Minus) Inflation	0.430** (0.192)	1.093*** (0.445)	0.166 (0.186)	0.425** (0.233)	0.293 (0.247)	0.312* (0.183)	0.530** (0.299)
(Minus) Unemployment	0.221 (0.170)	0.422 (0.330)	0.154 (0.211)	0.833*** (0.343)	-0.048 (0.269)	0.173 (0.320)	0.269 (0.261)
Trade	0.249** (0.127)	0.237* (0.175)	0.215 (0.163)	0.354* (0.226)	0.187 (0.237)	0.249 (0.186)	0.205 (0.213)
HDI	0.197 (0.167)	0.459* (0.285)	0.048 (0.204)	0.187 (0.253)	0.225 (0.311)	0.228 (0.272)	0.148 (0.240)
Democracy	0.192** (0.101)	0.224 (0.259)	0.165 (0.131)	-0.042 (0.142)	0.117 (0.151)	0.050 (0.060)	0.298* (0.194)
General index	0.276*** (0.105)	0.365*** (0.163)	0.219* (0.118)	0.217* (0.132)	0.224* (0.151)	0.150** (0.077)	0.332** (0.176)
N	2356	653	1703	960	515	739	1399

Notes: This table reports estimated effects of electoral turnovers for different subsamples. Each estimate corresponds to a separate regression. Column (1) reports results for the full sample. Columns (2) and (3) report results for the subsamples of presidential and parliamentary elections, respectively. Column (4) (resp., 5) shows results on the subsample of elections in which the candidate of the incumbency was the incumbent leader themselves (resp., someone else). Column (6) (resp., 7) shows results for the subsample of elections for which checks and balances on the year before the election were above (resp., below) the median, computed among close elections (i.e., elections with a running variable below 15pp in absolute value). The subsamples of columns (4) and (5) only include elections leading to the designation of a leader in the executive branch. Data Appendix I explains how we identify the candidate of the incumbency. Checks and balances are measured as the average of two V-Dem indices: the judicial constraints on the executive index and the legislative constraints on the executive index. Using the method of [Clogg et al. \(1995\)](#), we cannot reject the equality of the estimates for the general index for presidential and parliamentary elections (p-val. = 0.468), for the presence or absence of the incumbent leader on the ballot (p-val. = 0.972), and for high and low checks and balances (p-val. = 0.343). We obtain broadly consistent results when running a parametric regression in which we include the interaction between the treatment and the dimension of heterogeneity. We report the number of observations in the regressions of the table's last row, where the outcome is the general index of country performance. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Second, we distinguish between cases when the incumbent leader is nominally on the ballot or not (columns 4 vs. 5). This is ex ante ambiguous. On the one hand, individual incumbents may have accumulated valuable experience, and replacing them may be costly, thus reducing the positive effects of turnovers. On the other hand, reelection incentives may be stronger for candidates who represent the incumbency but are distinct from the outgoing leader, thus mitigating the adverse effects of keeping the incumbent party in power. Overall, we find very similar estimates in both subsamples.

Third, the positive effects of turnovers are larger in contexts with weaker checks and balances, measured using the average of two V-Dem indices (judicial and legislative constraints on the executive), in columns 6 and 7. Appendix Table E.1 further shows that

turnovers matter more when the executive faces fewer internal and external constraints, namely when more power is vested in the executive leader (see Data Appendix D.2)¹⁹ and when the country is less exposed to globalization, which we quantify using trade intensity, and which can constrain economic policy through trade agreements, currency unions, and the free movement of capital. For each dimension of heterogeneity, we consider the value of the variable in the year before each election, compute the median among elections for which the running variable is under 15 percentage points in absolute value, split the sample between elections above and below the median, and estimate equation (2.1) separately in each subsample. Differences across these subsamples, however, are generally non-significant.

In Appendix Table E.3, we split our sample between democracies and autocracies and between OECD and non-OECD countries. Overall, turnovers have a positive effect across these subsamples. Finally, we assess how the effects of turnovers vary with the tenure of the incumbent. Successive turnovers in a short timeframe may generate instability (e.g., see Gratton et al., 2015), and the risk of power erosion may be higher when incumbents have been in office for a long time, leading to larger effects of turnovers. On the other hand, long-tenured incumbents may have accumulated more experience, making it costly to replace them. In line with this second prediction, we find that the effects of turnovers tend to be larger when less time has elapsed since the last turnover (Appendix Table E.3, column 5 vs. 6) and when there was a turnover in the previous election (column 7 vs. 8). However, these differences may be driven by other differences in the composition of these two samples and they are not statistically significant.

Dynamic Effects. We further assess how the effects of electoral turnovers evolve over time. We estimate the following RD equation for each year after the election, i.e., each value of $\tau \in \{-2, 0, 1, 2, 3, 4, 5\}$:

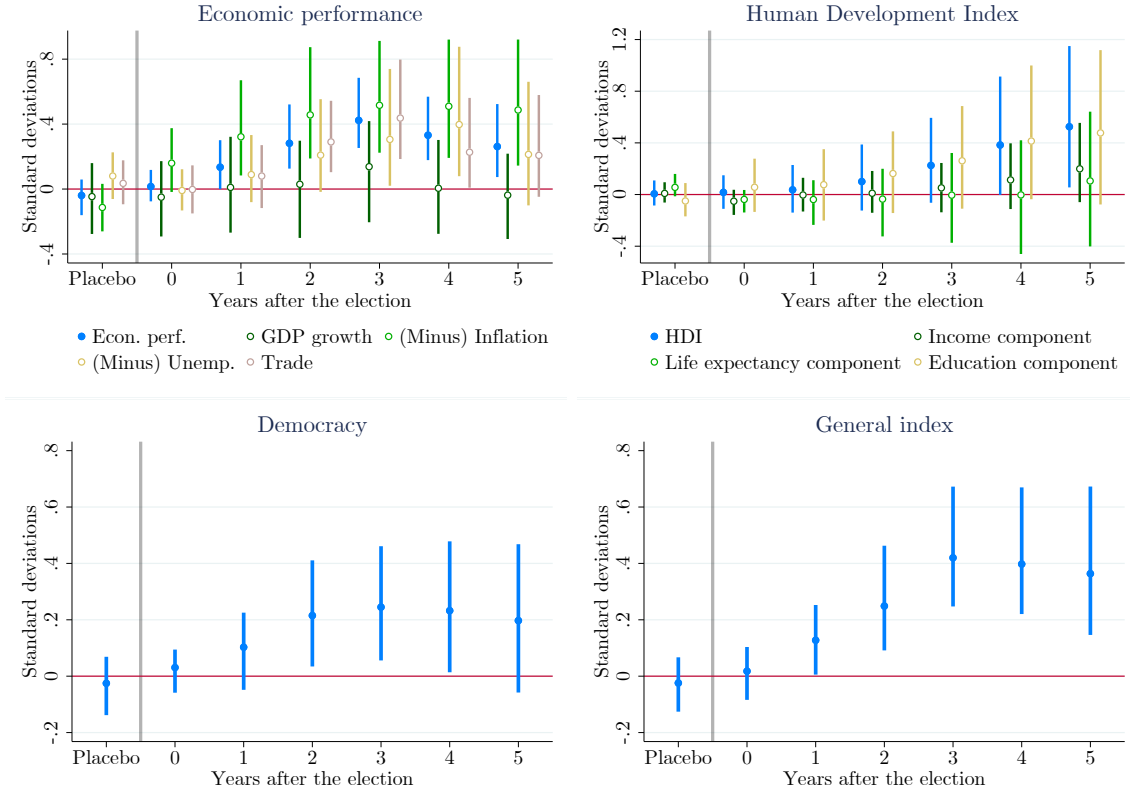
$$Y_{c,tE+\tau} - Y_{c,tE-1} = \alpha_\tau + \beta_{1,\tau}X_E + \beta_{2,\tau}X_ET_E + \gamma_\tau T_E + \varepsilon_{E,\tau} \quad (2.2)$$

Figure 2.6 reports the estimated γ_τ 's. The estimates of γ_{-2} correspond to the effect of turnovers on outcomes measured two years before the election and can be interpreted as placebo tests. These estimates are small in magnitude and non-significant, as expected. Furthermore, we find that the effects of turnovers are initially small, but increase over time. This is especially the case for economic performance and the overall index of performance, which increase gradually until the third year after a turnover. Appendix Figure E.1 corroborates these findings with separate RD plots for the general index of performance and for each year after the election. Overall, these dynamic patterns indicate that it takes a few years for electoral turnovers to impact country-level outcomes.²⁰ Consistent with these dynamic results, using data from Carlin et al. (2019), we find that turnovers improve the approval ratings of the elected leader after a few years spent in office (see Appendix Figure E.3).

¹⁹This measure can only be defined when the election leads to the designation of a member of the executive.

²⁰Turnovers could also affect the timing of the next election, which would affect the interpretation of estimates in this section. However, there is no significant effect on the number of years until the next election (Appendix Figure E.2, panel a).

Figure 2.6: Dynamic effects of electoral turnovers on country performance



Notes: This figure reports RD point estimates and 90% robust confidence intervals for the γ_τ in equation (2.2), with $\tau \in \{-2, 0, 1, 2, 3, 4, 5\}$, for all our main outcomes. Placebo refers to the point estimate obtained for $\tau = -2$. We use the procedure of [Calonico et al. \(2014\)](#) for estimation, and all outcomes are measured in standard deviations.

Executive Turnovers We further explore the impacts of turnovers in the executive branch, using the fuzzy RDD approach described in Section 2.3.5. Table 2.3 shows the first stage, second stage, reduced form, and OLS results, and Appendix Figure E.4 shows reduced form RD plots. The effects of executive turnovers are similar to those of electoral turnovers. They are slightly less precisely estimated since the exclusion of parliamentary elections that do not lead to the designation of a member of the executive decreases the sample size. Nevertheless, to the extent that the effects of electoral turnovers are driven by leadership changes in the executive branch, one would expect executive turnovers to have larger effects than electoral turnovers. Our results are generally consistent with this expectation: executive turnovers increase the index of economic performance and the general index of performance by 0.37 and 0.34 SD respectively, as compared to effect sizes of 0.27 and 0.28 SD for electoral turnovers.²¹

²¹Remember that the sample of elections used to estimate the effects of executive turnovers is a subset of the main sample since it only includes elections that lead to the designation of a leader of the executive branch. Therefore, the fact that the fuzzy RDD effects of executive turnovers are larger than the effects of electoral turnovers is not mechanical.

Table 2.3: Effects of turnovers in the executive branch on country performance

	(1) Econ. perf.	(2) GDP p.c. gr.	(3) (Minus) Inflation	(4) (Minus) Unemp.	(5) Trade	(6) HDI	(7) Democ.	(8) General index
Panel a: Fuzzy RDD (Effects of an executive turnover)								
Ex. turn.	0.367*** (0.161)	-0.026 (0.212)	0.398* (0.252)	0.811*** (0.333)	0.494*** (0.196)	0.521** (0.260)	0.058 (0.160)	0.343*** (0.154)
p-val.	[0.005]	[0.977]	[0.055]	[0.005]	[0.003]	[0.023]	[0.868]	[0.009]
N	1582	1288	1378	962	1240	940	1529	1677
N eff.	674	689	676	444	553	459	831	723
Band.	16.7	23.1	19.7	18.8	16.9	20.5	22.6	17.2
Panel b: Reduced form (Effects of a defeat of the leading party before the election)								
El. defeat	0.274*** (0.118)	-0.015 (0.166)	0.315* (0.188)	0.594*** (0.235)	0.354*** (0.142)	0.361** (0.187)	0.027 (0.109)	0.256** (0.113)
p-val.	[0.007]	[0.986]	[0.051]	[0.004]	[0.004]	[0.034]	[0.960]	[0.011]
N	1582	1288	1378	962	1240	940	1529	1677
N eff.	571	628	576	385	517	429	739	635
Band.	13.4	20.9	15.9	15.5	15.3	18.5	19.8	14.3
Panel c: First stage. Effects of a defeat of the leading party before the election on executive turnover								
El. defeat	0.697*** (0.068)	0.745*** (0.062)	0.685*** (0.069)	0.691*** (0.077)	0.678*** (0.075)	0.687*** (0.082)	0.660*** (0.064)	0.684*** (0.067)
p-val.	[<0.001]	[<0.001]	[<0.001]	[<0.001]	[<0.001]	[<0.001]	[<0.001]	[<0.001]
N	1582	1288	1378	962	1240	940	1529	1677
N eff.	674	689	676	444	553	459	831	723
Band.	16.7	23.1	19.7	18.8	16.9	20.5	22.6	17.2
Panel d: OLS (Effects of an executive turnover)								
Ex. turn.	0.084*** (0.032)	0.127** (0.055)	0.071 (0.053)	-0.070 (0.068)	0.155*** (0.055)	-0.010 (0.063)	0.046 (0.042)	0.073** (0.035)
p-val.	[0.009]	[0.021]	[0.180]	[0.305]	[0.005]	[0.879]	[0.275]	[0.037]
N	1582	1288	1378	962	1240	940	1529	1677

Notes: This table reports estimated effects of turnovers in the executive branch for the sample of elections leading to the appointment of a leader in the executive branch. In panel a, we report fuzzy RDD estimates of the effects of executive turnovers, using as assignment variable the defeat of the leading party before the election and turnover in the executive branch as treatment. We show estimates of γ in equation (A.1) – see Appendix A.2. In panel b, we report reduced form estimates of γ^r in equation (A.3) – see Appendix A.2, corresponding to the effects of an electoral defeat of the leading party before the election. We use the margin of victory of the best ranked challenger of the leading party before the election as the running variable. In panel c, we report estimates of $\tilde{\gamma}$ in equation (A.2), corresponding to the first stage of panel a. In panel d, we report OLS estimates of the effect of executive turnovers. Details about the definition of the leading party before the election can be found in Section 2.3.5. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Examples. Recent world history provides examples illustrating the positive impacts of turnovers. Appendix G discusses case studies from four national elections held in Brazil (2014), Germany (2005), Israel (1992), and the U.S. (1992). The 2014 Brazilian presidential election was won by the incumbent president, Dilma Rousseff from the Workers’ Party, with a 3% margin. Rousseff’s reelection was quickly followed by a deterioration of economic performance driven by rapidly rising inflation (Appendix Figure G.1) and the unfolding of a major corruption scandal. Rousseff’s tenure ended with a controversial impeachment procedure in 2016. By contrast, the close election victory of the CDU-CSU in Germany’s 2005

federal election, which led to Angela Merkel’s appointment as the new German chancellor, fostered important social policy reforms and improvements in overall performance (Appendix Figure G.2). The turnovers induced by the 1992 U.S. presidential election and the Labour party’s 1992 electoral victory in Israel similarly delivered notable benefits. In the first case, the new U.S. President Bill Clinton managed during his first term to restore budget balance, to keep interest rates low, to promote an economic expansion, and to implement a major tax reform—the introduction of the earned income tax credit—with beneficial impacts on the economy as a whole (Appendix Figure G.3). In the latter case, the new Israeli Prime Minister Yitzhak Rabin (who previously also governed between 1974-77) achieved marked improvements in economic performance induced by major investments in infrastructure and education funded by defense spending cuts (Appendix Figure G.4).

2.4.3 Robustness

In the Appendix, we show that our results are robust to numerous specification changes and alternative ways of constructing our outcome variables. Here we provide a brief overview of these checks.

Changing the post-election period and the pre-election baseline. Our main results in Table 2.1 and Figure 2.4 compare the post-election average of each outcome to the value of the same outcome measured in the year before the election. Instead of using the average of the four post-election years to measure post-election outcomes, we also estimate equation (2.1) including three, five, seven, and ten years in the post-election period. We further show that the results are robust to adding the year of the election to the post-election period. Appendix Tables D.11 to D.15 report these checks separately for each outcome of interest and for the general index of country performance, constructed with and without the democracy index. In addition, we show in Appendix Table D.16 that these results are robust to using the average of the three pre-election years instead of the pre-election year as the baseline period. Namely, we define $\Delta Y_E = \left(\frac{1}{4} \sum_{\tau=1}^4 Y_{c,t_E+\tau}\right) - \left(\frac{1}{3} \sum_{\tau=1}^3 Y_{c,t_E-\tau}\right)$.

Winsorizing. To avoid results being driven by extreme events (e.g., hyperinflation episodes), changes in measurement, or data errors, we winsorize the components of the economic performance index (GDP per capita growth, inflation, unemployment, and trade) at the 3rd and 97th percentiles. Appendix Tables D.11, D.14, and D.15 show that our results are unchanged when we winsorize less (at the 1st and 99th percentiles) or more (at the 5th and 95th percentiles), and that our results are robust to trimming instead of winsorizing.

RD bandwidth and specification. We check the robustness of our results to five deviations from the baseline RD specification choices in Calonico et al. (2014). First, in Appendix Tables D.11 to D.15, we show that including geographical region and decade fixed effects does not affect our estimates. Appendix Table D.17 shows that controlling for pre-election outcomes also does not change our results, though it slightly increases the precision of our estimates. Second, in Appendix Tables D.11 to D.15 we obtain very similar

estimates with a bandwidth twice larger or twice smaller than the optimal bandwidth used in our baseline estimation. Third, we show that our results are robust to using a second order local polynomial instead of the default local linear regression of Calonico et al. (2014). Fourth, we show robustness to different choices of kernel. Finally, Appendix F shows that our results are robust to using the independent randomization inference procedure from Cattaneo et al. (2015).

Construction of the general index. In Appendix Table D.8, we show that our results for the general index are robust to alternative constructions of this index, namely: excluding each of the components in turn; only keeping elections for which we have data on all components; using a weighted index à la Pocock (1997), which gives less weight to components which are more correlated with each other; defining the general index as the simple average of all outcomes used in the components instead of the simple average of the components; excluding observations in large geographical regions; and excluding observations in each decade of the sample.

Restricting to major elections. Many countries, including the U.S., hold both presidential and parliamentary elections. In these cases, our main sample includes both types of elections. Appendix Table D.8 shows that our results are robust to restricting the sample to major elections in each country, namely presidential elections in presidential systems, and parliamentary elections in parliamentary systems.

Excluding elections coinciding with a regime change. Turnovers might improve outcomes by fostering an immediate change in the nature of the political regime in a direction that tends to promote performance. However, such episodes are unlikely to explain our results since turnovers do not lead to a discontinuous jump in the likelihood of a regime change in the year of the election (see Appendix Figure C.4). In addition, our results are robust to excluding these elections from the sample: the impact of turnovers remains nearly identical (0.27 SD) in this case (Appendix Table D.8).²²

Alternative outcomes and data sources. To choose our main outcomes, we prioritized data availability and reliability. Appendix Tables D.1 to D.4 show additional results for a wider set of outcomes. For example, we show that our results are unchanged when we use growth in GDP instead of GDP per capita growth, and that electoral turnovers decrease poverty and have no discernible effects on inequality.

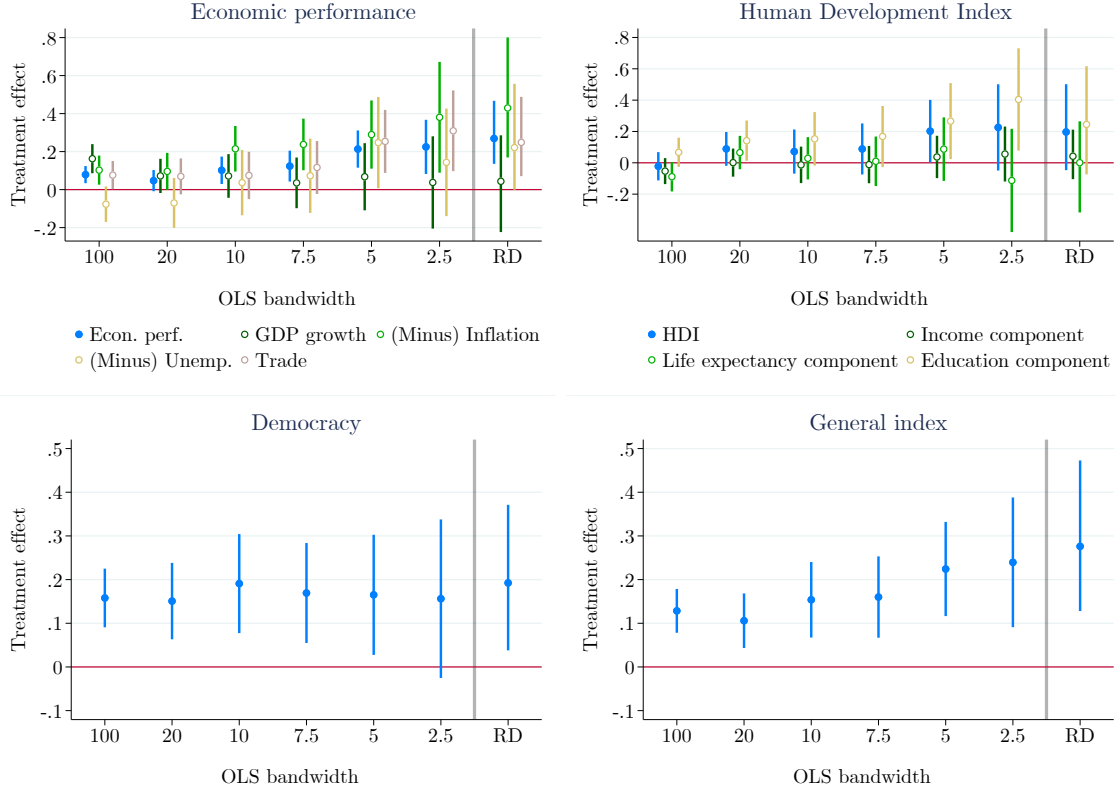
2.4.4 Comparing Effects in Close Elections and Other Elections

OLS estimation and endogeneity. Figure 2.7 reports OLS estimates of the effects of electoral turnovers. The OLS estimates can be interpreted as difference-in-differences estimates since we difference out each outcome with its value in the year preceding the election, as in the RD estimation. We report these estimates across windows of different

²²Appendix Figure C.5 further shows that electoral turnovers do not affect the likelihood of concomitant constitutional events, defined as the adoption of a new constitution or constitutional amendment.

sizes around the RD cutoff, ranging from 2.5 to 100 percentage points. In line with our RD results, the OLS estimates indicate that turnovers are associated with improved performance. Moreover, the magnitude of OLS estimates for very close elections is similar to that of RD estimates: turnovers in elections settled with a victory margin under 5 p.p. increase both economic performance and overall performance by approximately 0.20 SD. The effects on human development and democracy are also positive and statistically significant for these elections.

Figure 2.7: Comparing RD with OLS estimates



Notes: This figure compares OLS and RD estimates of the effects of electoral turnovers on country performance. For OLS estimates, we plot effects for the full sample, as well as for the subsamples of elections with a running variable in $[-20, 20]$, $[-10, 10]$, $[-7.5, 7.5]$, $[-5, 5]$, and $[-2.5, 2.5]$. All point estimates are displayed with 90% confidence bands.

OLS estimates remain positive, but their magnitude is smaller when we consider elections which are less close. In line with our earlier discussion (see Section 2.3.4), several forces could explain these differences. First, OLS estimates might be biased downward if economic decline before the election is associated with lower reelection probabilities for incumbents *and* a further degradation of performance after the election. Second, incumbents who decisively win reelection may be more positively selected than challengers who decisively get elected, relative to closely elected candidates of either type. This would bias the effect of turnovers downward for elections which are less close, in the OLS estimation.

Furthermore, unlike OLS, the RDD estimates a local effect of electoral turnovers: specifically, we measure the effect of turnovers *exactly* at the cutoff. As

mentioned in Section 2.3.2 and shown in Appendix Figure B.3, close elections are more likely to be held in more developed and democratic countries, and in contexts with slightly more favorable economic conditions.

To assess whether our results hold for elections which are less close, we implement two exercises. First, building on Arezki et al. (2022), we look at a subsample of “unlucky incumbents” who run for reelection in a context of unfavorable global oil prices. Second, we follow Angrist and Rokkanen (2015) to estimate RD treatment effects in a larger window around the cutoff.

Global oil prices. To explore the possibility that our results are driven by a selection of low-quality incumbents at the RD threshold, it is informative to look at national elections that happened to be close not as a result of poor incumbent performance, but due to factors outside the incumbent’s control.

Incumbents are often penalized by their electorate when elections take place in the aftermath of an oil shock (Arezki et al., 2022), possibly because voters over-attribute observable outcomes to leaders or fail to take into account these adverse circumstances (Glaeser and Ponzetto, 2017). Unfavorable global oil prices thus provide exogenous variation affecting the occurrence of close national elections. While some incumbents struggle to get reelected because of their own poor performance, other unlucky incumbents face stiffer competition due to adverse oil prices. In Appendix Table E.4, we estimate the effects of turnovers in elections conducted in a context of unfavorable oil prices. We first compute the annual growth in the worldwide price of crude oil g_t using the World Bank Commodity Price Data. Distinguishing between oil-importing and oil-exporting countries, we then associate an election taking place at time t_E with the following variable:

$$s_E = \text{OilNetImporter}_{c,t_E} \times \frac{g_{t_E-1} + g_{t_E-2}}{2},$$

where g_t corresponds to the annual growth in the worldwide price of crude oil from the World Bank’s World Integrated Trade Solution (WITS) dataset, and $\text{OilNetImporter}_{c,t}$ is a variable equal to 1 for net oil importers and to -1 for net oil exporters.

We then show how the estimated effects of electoral turnovers change when we restrict the sample to contexts that were increasingly unfavorable to the incumbent, i.e., when the s_E variable is above the sample median, in the top tercile, and in the top quartile (Appendix Table E.4, columns 2 to 4). As expected, the running variable is on average 2 percentage points higher when s_E is above the median than when s_E is below the median: unfavorable oil prices tend to improve the performance of challengers. Yet, the positive effects of turnovers hold in all subsamples, with effect sizes slightly smaller than in the full sample of elections and ranging between 0.13 and 0.19 SD. Column 5 further shows the effects following the 1973, 1979, and 2007-08 global economic crises. Effects on economic performance and on the general index are actually larger there than in the full sample, and they are significant at the 1% and 10% level, respectively. In sum, the incumbents who are closely reelected under adverse global economic circumstances also tend to perform worse than counterfactual challengers.

Estimating effects away from the cutoff. Appendix Table E.5 and Appendix Figures E.5–E.6 report results from the procedure of Angrist and Rokkanen (2015). This procedure relies on a testable conditional independence assumption (CIA): in a window around the cutoff, potential outcomes are assumed to be mean-independent of the running variable conditional on a set of controls. We focus on the $[-10\text{pp}, +10\text{pp}]$ window, which encompasses about 30% of elections in our sample. In Appendix Figure E.5, we test the CIA hypothesis and fail to reject its validity for our main outcomes. We then construct two CIA-based estimators: a linear reweighting estimator discussed by Kline (2011), and a version of the Hirano et al. (2003) propensity score estimator. Appendix Table E.5 reports these estimates. We find effects which are all positive and consistent with our main results, although smaller in magnitude. On average, elections won by the challenger by 10 percentage points at most increase the general index by 0.16–0.18 SD, and we fail to reject equality of the Calonico et al. (2014) and CIA-based estimates for all outcomes. These results indicate that the effects of electoral turnovers are not limited to close elections.

2.4.5 The Role of Leader Characteristics

The effects of turnovers on country performance may in principle result from differences in the characteristics of elected leaders. Recent work shows how certain types of leaders have affected national performance throughout history (Dube and Harish, 2020; Ottinger and Voigtländer, 2021; Funke et al., 2023). For this to be the case in our context, leader characteristics would need to systematically differ after an electoral turnover. For example, elected challengers should have systematically different ideologies, or their average quality should differ from that of reelected incumbents.

We first investigate whether electoral turnovers coincide with the victory of candidates with specific ideological characteristics. Here, we estimate equation (2.1) using as our dependent variable the left-right ideology, the populism score, and the illiberalism score of the winning party. Appendix Figure E.8 reports the corresponding results. Elected challengers tend to be more to the right, more populist, and less illiberal than reelected incumbents, but most of these effects are non-significant. We interpret these results with much caution: the sample size in these regressions is small, and these analyses rely on V-Dem’s classification of party ideologies. Nonetheless, the evidence does not point to systematically different ideologies between elected challengers and incumbents.²³

Next, we examine the role of leader quality. Since quality is unobserved, we build a proxy measure based on observed improvements in electoral performance. Intuitively, an increase in the winner’s vote share between one election and the next indicates that voters are satisfied with the leader’s performance during this timeframe. We compute the difference between the incumbent’s vote share in the next ($t+1$) election and the winner’s vote share in the current (time t) election, and we examine the extent to which turnovers explain variation in this outcome. Appendix Table E.8 suggests there is no jump in this proxy for leader quality

²³We additionally look at effects on the leader’s age. The point estimate (-2.0 years) is non-significant again. In this case, we restrict the sample to elections that lead to the designation of a member of the executive. Note that the positive effects of turnovers are unlikely to be driven by changes in leader gender. Our sample only includes 133 electoral races where exactly one female candidate is among the top two candidates, and our results are robust to dropping these elections.

(column 1). Then, we regress this outcome on country performance measured before the (time t) election as well as characteristics of the election (such as the number of candidates), and we use the residuals from this regression as proxies for unobserved leader quality. We also do not find evidence of a change in this measure: if anything, the point estimate is negative (column 2). In column 3, we additionally control for endogenous measures of leader performance between t and $t + 1$, namely the same indicators of country performance that we use in our main analysis. In this case, the dependent variable can be interpreted as a proxy for unobserved quality, net of observable pre-existing conditions and post-election performance. Again, we find no evidence that turnovers increase leader quality. These findings are consistent with the fact that overall, turnovers have no measurable effect on the probability of a turnover in the next election (Appendix Figure E.2, panel b). Overall, these results suggest that quality differentials are unlikely to play a major role in our results.²⁴ Nonetheless, we acknowledge that this channel may still be present as quality cannot be observed directly.

2.5 Mechanisms

We now highlight two main mechanisms driving the positive effect of turnovers on performance. We first show that these effects come, in part, from shifts in economic policy (Section 2.5.1). We then explore the role of accountability mechanisms (Section 2.5.2).

2.5.1 Policy Changes

A likely channel driving the positive effects of turnovers, in addition to the increase in international openness discussed in Section 2.4.2, is that challengers may implement different *policies* than incumbents would have in the counterfactual electoral outcome. Thus, we first ask whether turnovers foster more change in economic policy, proxied by the level of government intervention in the economy. We consider four measures of intervention: government expenditure, tax revenue, national debt (all measured as a share of GDP), and a standardized index combining these outcomes.²⁵

Appendix Figure E.7, panel a, shows that the effect of turnovers on the *level* of government intervention is small and non-significant across all four measures. However,

²⁴In Appendix Table E.9, we implement a different exercise built on the intuition that unobserved quality differences between elected challengers and elected incumbents can be bounded using information on the previous government experience of challengers. Challengers who have previously served as incumbents are plausibly of higher quality than those without recent government experience, and we should expect turnovers to be more beneficial in such instances. When considering only challengers who have not previously served in government in the previous ten years, turnovers increase economic performance by 0.24 SD and the index of overall country performance by 0.23 SD. Both coefficients are statistically significant at the 5% level. When considering only challengers who have previously been in power, turnovers increase economic performance by 0.29 SD and overall performance by 0.29 SD (both coefficients are significant at the 5% level). Thus, the two types of challengers deliver similar improvements in country performance.

²⁵Government intervention is typically one of the key policy levers that newly elected leaders employ to improve economic performance early in their tenure. In the U.S., Franklin D. Roosevelt passed most of the New Deal legislation within his first 100 days in office, while major economic stimulus bills were adopted in the early days of several recent presidencies.

challengers could still enact new policies that are better tailored to the country’s needs, whereas incumbents prefer the status quo. For instance, challengers might increase government intervention (relative to incumbents) during economic downturns and reduce it when the economy is overheating. Such mechanisms would only be captured by the effect of turnovers on *non-directional* outcomes.

Accordingly, we examine as an outcome the absolute value of the difference between the post-election average and the pre-election value of each policy measure. We find positive effects of turnovers on changes in government expenditure, national debt, and tax revenue. The latter effect (but not the two others) is significant at the 10% level (Figure E.7, panel b). The effect on the overall change in government intervention, of 0.21 SD, is also significant at the 10% level. In Appendix Table E.6, we show additional results on the absolute value of the difference of various types of economic policies, including a measure of central bank independence (Garriga, 2016), government expenditure composition (Ortiz-Ospina, 2016), taxation composition from the World Bank, and financial liberalization (Abiad et al., 2010).²⁶ Most estimates (25 out of 28) in this table are positive, suggesting that policy shifts take place across a range of dimensions. Overall, this suggests that non-directional differences in the policies implemented by incumbents and challengers might contribute to the positive effect of turnovers.

Appendix Table E.2 provides additional evidence. Elected challengers deliver achievements that are better suited to their country’s needs: the beneficial effects of turnovers on inflation and unemployment are concentrated in contexts where these indicators are deteriorated. When inflation in the pre-election year is high (above the sample median), turnovers decrease inflation by 0.92 SD, while the corresponding effect is -0.06 SD when pre-election inflation is low (columns 1 and 2). Similarly, when unemployment in the pre-election year is high (resp. low), turnovers are associated with a 0.71 SD (resp. 0.04 SD) drop in unemployment (columns 3 and 4). Turnovers are also more beneficial when leaders have more policy leverage. As shown in columns 5 and 6, the effect of turnovers on inflation is larger (-1.05 SD) when the independence of the central bank before the election is low than when it is high (-0.44 SD).

2.5.2 Political Accountability

Turnovers may also improve performance because newly elected challengers face stronger incentives to perform. If this were the case, we should observe that standard proxies for politician effort, such as corruption (e.g., Besley, 2007; Ferraz and Finan, 2011), also improve as a result of turnovers.

Appendix Figure E.9 shows that turnovers positively affect several measures of perceived corruption and accountability. We use an index of government accountability,²⁷ indices of

²⁶Appendix Table E.7 reports the corresponding directional effects. While some estimates are statistically significant, as one would expect, we do not find robust evidence that electoral turnovers systematically move policies in one direction or another.

²⁷The accountability index of V-Dem aggregates measures of vertical, diagonal, and horizontal accountability. Vertical accountability captures the extent to which citizens can hold the government accountable. Diagonal accountability covers the mechanisms that citizens, civil society, and the media can use to hold the government accountable. Finally, horizontal accountability captures the power of

executive corruption and public sector corruption from V-Dem, the World Bank’s index on the control of corruption, and a standardized index aggregating these measures. The effects we estimate are large (ranging from 0.16 SD for public sector corruption to 0.44 SD for the control of corruption) and they are generally significant at the 1% or 5% level. Appendix Table E.10 shows results for a broader set of variables. Among 20 outcomes signed such that higher values indicate greater accountability, all but two are positively affected by turnovers (18 estimates are larger than 0.10 SD, and seven are statistically significant). Finally, Appendix Table E.11 shows that the effects on corruption are larger for presidential elections, for elections held in regimes with fewer checks and balances, and for those where the executive leader holds more power. This suggests that turnovers are especially conducive to good governance in countries with weaker constraints on the executive, where reelected incumbents may otherwise use their power to extract rents.

As with measures of democracy explored in Section 2.4, one might fear that the coding of these indicators is endogenous to the occurrence of a turnover. In that case, corruption indices might drop sharply during the year of the turnover, or shortly thereafter. To explore this, we estimate dynamic effects of electoral turnovers on corruption, using specifications in the form of equation (2.2). Instead, we find that the effects of a turnover on corruption are initially small and increase over time (Appendix Figure E.10). Moreover, turnovers have a 0.12 SD (non-significant) effect on the reduction of actual corruption incidents measured in the Global Incidents of Corruption Index (GICI) from Furceri et al. (2019). Overall, this provides suggestive evidence that improvements in governance and corruption might contribute to the overall improvement in country performance, which follows similar dynamics after turnovers.²⁸

Term limits. Prior work suggests that term limits could explain why corruption falls after turnovers: in the absence of reelection incentives, incumbents who are reelected but term-limited might exert less effort and perform more poorly (Ferraz and Finan, 2011; Fourinaies and Hall, 2021). However, Table 2.2 already provided evidence at odds with this mechanism: electoral turnovers also improve economic performance (by 0.20 SD) and country performance (by 0.22 SD) under parliamentary systems, where leader term limits are almost always absent.

Nonetheless, term limits could play a role in the context of presidential elections. To explore this, we identify regimes with presidential term limits, using data from the Comparative Constitutions Project or CCP (Elkins et al., 2021). In these regimes, we determine whether the incumbent and challenger would face a term limit, should they win.²⁹

state institutions to oversee the government. The effects of electoral turnovers are strongest for horizontal accountability, followed by diagonal accountability and vertical accountability (Appendix Table E.10).

²⁸A large literature has documented the relationship between corruption and economic performance (Mauro, 1995) through theft of government resources (Olken, 2006; Ferraz et al., 2012), effects on firms (Svensson, 2003; Sequeira and Djankov, 2014), misallocation of capital (Khwaja and Mian, 2005), and demand for regulation (Di Tella and MacCulloch, 2007).

²⁹The CCP covers 59% of our presidential elections. Within this subsample, term limits are not specified in the constitution in 12% of elections, and are explicitly nonexistent in 4% of elections. Furthermore, in 52% of elections, term limits exist but are not differentially binding for the incumbent and the challenger. This occurs because: presidents can only serve one term (13%), the candidate representing the incumbency is not the incumbent themselves (35%), the challenger had already been in power in the past (2%), or the

In total, term limits exist and are differentially binding for the incumbent and the challenger in less than half of the presidential elections covered by the CCP. Appendix Table E.1, column 6 reports the effects of turnovers in a subsample restricted to parliamentary elections and presidential elections in which there is no differentially binding term limit. Electoral turnovers improve the general index of country performance by 0.22 SD in this subsample, which is similar to the point estimate in the full sample. Thus, presidential term limits are unlikely to be driving our results.³⁰

Interpretation. Even in the absence of *de jure* term limits, turnovers could still improve accountability through related mechanisms akin to career concerns (Holmström, 1999). In regimes that hold regular elections, elected leaders may want to build reputation by exerting more effort early in their tenure, and use their later terms in office to extract rents. In the late terms of a given leader or party, the incentives to not misbehave are diminished because voters have already formed precise beliefs about the incumbent’s type, and additional new information is unlikely to change these beliefs (as in Ashworth, 2005).

This mechanism is compatible with learning in corruption by incumbents, who might learn over time how to extract rents, and delayed corruption or “golden goose” effects (Niehaus and Sukhtankar, 2013). While leaders elected in close elections might have lower legitimacy and stronger incentives to perform, close elections could also signal to incumbents that their days as national leaders are numbered, a mechanism akin to a “de facto” term limit. This can explain why the effects of turnovers appear slightly larger close to the RD threshold: closely reelected incumbents may face lower incentives to perform and engage in more rent extraction than decisively reelected incumbents if a narrow margin of victory signals to reelected incumbents that future reelection is unlikely. Finally, incumbents might simply be experiencing government fatigue—an erosion of their motivation and power due to the effect of time.

Implementing reforms is more difficult for individuals and parties which have held power for a long time. The effects we find on bureaucratic quality, also reported in Appendix Table E.10, are consistent with this interpretation. We use the bureaucracy score from Khan et al. (2022), which captures the presence of meritocratic recruitment and of an impartial administration, to evaluate bureaucratic effectiveness in our sample. If anything, the effect of turnovers on this measure of bureaucratic quality is positive (0.12 SD, not significant), consistent with the notion that turnovers do not deteriorate bureaucratic quality and perhaps mitigate bureaucratic inertia under long-serving incumbents.

Overall, the evidence we present suggests that a combination of these accountability mechanisms contributes to explain shifts in performance and policy taking place in the aftermath of turnovers.

candidate representing the incumbency had already been in power in the past but the term limit is of three terms or more (2%).

³⁰The small number of elections with a differentially binding term limit unfortunately means that we do not have sufficient sample size and power to estimate the effect of electoral turnovers in these elections specifically.

2.6 Conclusion

Since the end of World War II, most countries have held regular presidential or parliamentary elections to determine the composition of their government. A key function associated with these elections is to allow citizens to ask for continuity or change in their country's leadership: dismissing incumbents in the ballot box can allow citizens to chart a new course for their country. To evaluate the merits of electoral democracy, understanding how the outcome of national elections affects country-level performance—including, but not limited to economic performance—is essential. To a large extent, the benefits of electoral transitions are commensurate with their ability to deliver improvements in citizens' welfare.

Overall, we find that voting for change matters: turnovers deliver improvements in country-level performance along many dimensions. The largest effects of turnovers are on measures of international openness, inflation, and democratic quality, and they tend to be larger in settings where governments face weaker checks and balances on their power. We also observe large effects on indices of corruption and on policy change. We hypothesize that the main force driving these effects is the role turnovers play in giving the reins of government to new leaders facing stronger incentives and reputation concerns.

These novel findings call for future research exploring the mechanisms linking turnovers to specific dimensions of economic policy and performance. For example, future work may investigate the role of countercyclical policy responses in explaining how turnovers enhance macroeconomic discipline. Our results also provide reasons to be cautiously optimistic about the potential of electoral democracies to deliver superior performance to their citizens. While recent research shows that populist leaders deteriorate economic performance (Funke et al., 2023), our paper shows that regular power transitions enabled by democracy are conducive to better performance. Together, this body of evidence suggests that healthy democracies shall continue to outperform autocracies and populist regimes over the *longue durée*. To what extent citizens perceive these performance differentials across regime types, however, remains an open question for future research.

Bibliography

- Abiad, Abdul, Enrica Detragiache, and Thierry Tressel**, “A New Database of Financial Reforms,” *IMF Staff Papers*, June 2010, 57 (2), 281–302.
- Acemoglu, Daron, Suresh Naidu, Pascual Restrepo, and James A. Robinson**, “Democracy Does Cause Growth,” *Journal of Political Economy*, 2019, 127 (1), 47–100.
- Akhtari, Mitra, Diana Moreira, and Laura Trucco**, “Political Turnover, Bureaucratic Turnover, and the Quality of Public Services,” *American Economic Review*, 2022, 112 (2), 442–93.
- Alesina, Alberto, Sule Özler, Nouriel Roubini, and Phillip Swagel**, “Political Instability and Economic Growth,” *Journal of Economic growth*, 1996, 1 (2), 189–211.
- Alt, James, Ethan Bueno de Mesquita, and Shanna Rose**, “Disentangling Accountability and Competence in Elections: Evidence from U.S. Term Limits,” *The Journal of Politics*, 2011, 73 (1), 171–186.
- Anagol, Santosh and Thomas Fujiwara**, “The Runner-up Effect,” *Journal of Political Economy*, 2016, 124 (4), 927–991.
- Angrist, Joshua D. and Miikka Rokkanen**, “Wanna Get Away? Regression Discontinuity Estimation of Exam School Effects Away From the Cutoff,” *Journal of the American Statistical Association*, 2015, 110 (512), 1331–1344.
- Archives, Records Administration National**, “The Electoral College,” 2020.
- Arezki, Rabah, Simeon Djankov, Ha Nguyen, and Ivan Yotzov**, “The Political Costs of Oil Price Shocks,” *CESifo Working Paper Series 9763*, 2022.
- Ashworth, Scott**, “Reputational Dynamics and Political Careers,” *The Journal of Law, Economics, and Organization*, 08 2005, 21 (2), 441–466.
- Ban, Pamela, Elena Llaudet, and James M. Snyder JR.**, “Challenger Quality and the Incumbency Advantage,” *Legislative Studies Quarterly*, 2016, 41 (1), 153–179.
- Beland, Louis-Philippe**, “Political Parties and Labor-Market Outcomes: Evidence from US States,” *American Economic Journal: Applied Economics*, 2015, 7 (4), 198–220.
- Besley, Timothy**, *Principled Agents? The Political Economy of Good Government*, Oxford University Press Scholarship Online, 2007.
- Boix, Carles, Michael Miller, and Sebastian Rosato**, “Boix-Miller-Rosato Dichotomous Coding of Democracy, 1800-2015,” 2018.
- Bolt, Jutta, Robert Inklaar, Herman de Jong, and Jan Luiten van Zanden**, “Rebasing ‘Maddison’: New Income Comparisons and the Shape of Long-Run Economic Development,” *GGDC Research Memorandum*, 2018, 174.
- Brancati, Dawn**, “Global Elections Database,” *Global Elections Database, New York.*, 2016, 15.
- Brender, Adi and Allan Drazen**, “How Do Budget Deficits and Economic Growth Affect Reelection Prospects? Evidence from a Large Panel of Countries,” *American Economic Review*, December 2008, 98 (5), 2203–20.

- Budge, Ian, Hans-Dieter Klingemann, Andrea Volkens, Judith Bara, Eric Tanenbaum et al., *Mapping Policy Preferences: Estimates for Parties, Electors, and Governments, 1945-1998*, Vol. 1, Oxford University Press on Demand, 2001.
- Calonico, Sebastian, Matias D. Cattaneo, and Rocio Titiunik, “Robust Nonparametric Confidence Intervals for Regression-Discontinuity Designs,” *Econometrica*, 2014, 82 (6), 2295–2326.
- Canay, Ivan A and Vishal Kamat, “Approximate Permutation Tests and Induced Order Statistics in the Regression Discontinuity Design,” *The Review of Economic Studies*, 10 2017, 85 (3), 1577–1608.
- Carlin, Ryan E, Jonathan Hartlyn, Timothy Hellwig, Gregory J Love, Cecilia Martínez-Gallardo, and Matthew M Singer, “Executive Approval Database 2.0,” Available for download at www.executiveapproval.org, 2019.
- Carlino, Gerald, Thorsten Drautzburg, Robert Inman, and Nicholas Zarra, “Partisanship and fiscal policy in economic unions: Evidence from us states,” *American Economic Review*, 2023, 113 (3), 701–737.
- Carr, Adam, *Psephos: Adam Carr’s Election Archive*, Adam Carr, 2003.
- Cattaneo, Matias D, Brigham R Frandsen, and Rocio Titiunik, “Randomization Inference in the Regression Discontinuity Design: an Application to Party Advantages in the US Senate,” *Journal of Causal Inference*, 2015, 3 (1), 1–24.
- , Michael Jansson, and Xinwei Ma, “Manipulation Testing Based on Density Discontinuity,” *The Stata Journal*, 2018, 18 (1), 234–261.
- , Rocio Titiunik, and Gonzalo Vazquez-Bare, “Inference in Regression Discontinuity Designs under Local Randomization,” *The Stata Journal*, 2016, 16 (2), 331–367.
- Cheibub, José Antonio, Jennifer Gandhi, and James Raymond Vreeland, “Democracy and Dictatorship Revisited,” *Public Choice*, 2010, pp. 67–101.
- Clogg, Clifford C, Eva Petkova, and Adamantios Haritou, “Statistical Methods for Comparing Regression Coefficients Between Models,” *American Journal of Sociology*, 1995, 100 (5), 1261–1293.
- Colomer, Josep M, “Comparative European Politics,” 2008.
- Colonnelli, Emanuele, Mounu Prem, and Edoardo Teso, “Patronage and selection in public sector organizations,” *American Economic Review*, 2020, 110 (10), 3071–3099.
- Coppedge, Michael, John Gerring, Carl Henrik Knutsen, Staffan I Lindberg, Jan Teorell, David Altman, Michael Bernhard, Agnes Cornell, M Steven Fish, Lisa Gastaldi et al., “V-Dem Codebook v11.1,” 2021.
- Cruz, Cesi, Philip Keefer, and Carlos Scartascini, “Database of Political Institutions 2017 (DPI2017),” *Inter-American Development Bank. Numbers for Development*, 2018.
- Döring, Holger and Philip Manow, “Parliament and Government Composition Database (ParlGov),” *An Infrastructure for Empirical Information on Parties, Elections and Governments in Modern Democracies. Version*, 2012, 12 (10).
- Dube, Oeindrila and SP Harish, “Queens,” *Journal of Political Economy*, 2020, 128 (7),

2579–2652.

- Eggers, Andrew C., Anthony Fowler, Jens Hainmueller, Andrew B. Hall, and James M. Snyder**, “On the Validity of the Regression Discontinuity Design for Estimating Electoral Effects: New Evidence from Over 40,000 Close Races,” *American Journal of Political Science*, 2015, 59 (1), 259–274.
- ElectionGuide, IFES**, “ElectionGuide: Democracy Assistance & Election News,” 2017.
- Elkins, Zachary, Tom Ginsburg, and James Melton**, “Characteristics of National Constitutions, version 3.0,” Technical Report 2021.
- Fair, Ray C**, “Presidential and Congressional Vote-Share Equations,” *American Journal of Political Science*, 2009, 53 (1), 55–72.
- Feenstra, Robert C, Robert Inklaar, and Marcel P Timmer**, “The Next Generation of the Penn World Table,” *American Economic Review*, 2015, 105 (10), 3150–82.
- Ferraz, Claudio and Frederico Finan**, “Electoral Accountability and Corruption: Evidence from the Audits of Local Governments,” *American Economic Review*, June 2011, 101 (4), 1274–1311.
- , – , and **Diana Moreira**, “Corrupting Learning: Evidence from Missing Federal Education Funds in Brazil,” NBER Working Papers 18150, National Bureau of Economic Research, Inc 2012.
- Fouirnaies, Alexander and Andrew B. Hall**, “How Do Electoral Incentives Affect Legislator Behavior? Evidence from U.S. State Legislatures,” *American Political Science Review*, 2021, pp. 1–15.
- Fujiwara, Thomas and Carlos Sanz**, “Rank Effects in Bargaining: Evidence from Government Formation,” *The Review of Economic Studies*, 2020, 87 (3), 1261–1295.
- Funke, Manuel, Moritz Schularick, and Christoph Trebesch**, “Populist Leaders and the Economy,” *American Economic Review*, 2023, 113 (2), 3249–88.
- Furceri, Davide, Chris Papageorgiou, and Hites Ahir**, “Global incidents of corruption index,” *Work. Pap., Int. Monet. Fund, Washington, DC*, 2019.
- Garriga, Ana Carolina**, “Central Bank Independence in the World: A New Data Set,” *International Interactions*, 2016, 42 (5), 849–868.
- Girardi, Daniele**, “Partisan Shocks and Financial Markets: Evidence from Close National Elections,” *American Economic Journal: Applied Economics*, October 2020, 12 (4), 224–52.
- Giuliano, Paola, Prachi Mishra, and Antonio Spilimbergo**, “Democracy and Reforms: Evidence from a New Dataset,” *American Economic Journal: Macroeconomics*, October 2013, 5 (4), 179–204.
- Glaeser, Edward L and Giacomo A.M. Ponzetto**, “Fundamental Errors in the Voting Booth,” Working Paper 23683, National Bureau of Economic Research August 2017.
- Gordon, Sanford C., Gregory A. Huber, and Dimitri Landa**, “Challenger Entry and Voter Learning,” *American Political Science Review*, 2007, 101 (2), 303–320.
- Granzier, Riako, Vincent Pons, and Clémence Tricaud**, “Coordination and

- Bandwagon Effects: How Past Rankings Shape the Behavior of Voters and Candidates,” *American Economic Journal: Applied Economics*, 2023, 15 (4), 177–217.
- Gratton, Gabriele, Luigi Guiso, Claudio Michelacci, and Massimo Morelli**, “From Weber to Kafka: Political Activism and the Emergence of an Inefficient Bureaucracy,” *The American Economic Review*, 2015.
- Guriev, Sergei and Daniel Treisman**, “Informational Autocrats,” *Journal of Economic Perspectives*, November 2019, 33 (4), 100–127.
- **and Elias Papaioannou**, “The Political Economy of Populism,” *Journal of Economic Literature*, September 2022, 60 (3), 753–832.
- Gygli, Savina, Florian Haelg, Niklas Potrafke, and Jan-Egbert Sturm**, “The KOF globalisation index–revisited,” *The Review of International Organizations*, 2019, 14, 543–574.
- Hegel, Georg Wilhelm Friedrich**, *Elements of the Philosophy of Right* 1820.
- Hirano, Keisuke, Guido W Imbens, and Geert Ridder**, “Efficient Estimation of Average Treatment Effects Using the Estimated Propensity Score,” *Econometrica*, 2003, 71 (4), 1161–1189.
- Holmström, Bengt**, “Managerial Incentive Problems: A Dynamic Perspective,” *The Review of Economic Studies*, 01 1999, 66 (1), 169–182.
- Horowitz, Shale, Karla Hoff, and Branko Milanovic**, “Government turnover: Concepts, measures and applications,” *European Journal of Political Research*, 2009, 48 (1), 107–129.
- Johnson, Simon, William Larson, Chris Papageorgiou, and Arvind Subramanian**, “Is Newer Better? Penn World Table Revisions and their Impact on Growth Estimates,” *Journal of Monetary Economics*, 2013, 60 (2), 255–274.
- Jones, Benjamin F and Benjamin A Olken**, “Do Leaders Matter? National Leadership and Growth Since World War II,” *The Quarterly Journal of Economics*, 2005, 120 (3), 835–864.
- Khan, Adnan, Guo Xu, Robin Burgess, and Timothy Besley**, “Bureaucracy and Development,” *Annual Review of Economics*, August 2022, 14 (1), 397–424.
- Khwaja, Asim Ijaz and Atif Mian**, “Do Lenders Favor Politically Connected Firms? Rent Provision in an Emerging Financial Market*,” *The Quarterly Journal of Economics*, 11 2005, 120 (4), 1371–1411.
- Kline, Patrick**, “Oaxaca-Blinder as a Reweighting Estimator,” *American Economic Review*, 2011, 101 (3), 532–37.
- Kling, Jeffrey R, Jeffrey B Liebman, and Lawrence F Katz**, “Experimental Analysis of Neighborhood Effects,” *Econometrica*, 2007, 75 (1), 83–119.
- Klingemann, Hans-Dieter, Andrea Volkens, Ian Budge, Judith Bara, and Michael D McDonald**, *Mapping Policy Preferences II: Estimates for Parties, Electors, and Governments in Eastern Europe, European Union, and OECD 1990-2003*, Vol. 2, Oxford University Press on Demand, 2006.

- Kollman, Ken, Allen Hicken, Daniele Caramani, David Backer, and David Lublin**, “Constituency-Level Elections Archive,” *Ann Arbor, mich.: Center for Political studies, University of michigan*. At <http://www.electiondataarchive.org>, accessed may, 2011.
- Levitsky, Steven and Daniel Ziblatt**, *How Democracies Die*, Penguin Random House, 2019.
- Lührmann, A, N Düpont, M Higashijima, YB Kavasogly, KL Marquardt, M Bernhard, H Döring, A Hicken, M Laebens, SI Lindberg et al.**, “Varieties of Party Identity and Organization (V-Party) Dataset V1,” 2020.
- Mauro, Paolo**, “Corruption and Growth,” *The Quarterly Journal of Economics*, 08 1995, 110 (3), 681–712.
- McCrary, Justin**, “Manipulation of the Running Variable in the Regression Discontinuity Design: A Density Test,” *Journal of Econometrics*, 2008, 142 (2), 698–714.
- Niehaus, Paul and Sandip Sukhtankar**, “Corruption Dynamics: The Golden Goose Effect,” *American Economic Journal: Economic Policy*, November 2013, 5 (4), 230–69.
- Nohlen, Dieter**, “Elections in the Americas. A Data Handbook: North America, Central America, and the Caribbean. 2 vols. Vol. 1,” *Elections Worldwide*, 2005.
- **and Philip Stöver**, *Elections in Europe*, Nomos Verlagsgesellschaft mbH & Co. KG, 2010.
- **et al.**, *Elections in the Americas: A Data Handbook: Volume 2 South America*, Vol. 2, Oxford University Press on Demand, 2005.
- **, Florian Grotz, and Christof Hartmann**, “Elections and Electoral Systems in Asia and the Pacific,” *Elections in Asia and the Pacific: A Data Handbook: Volume II: South East Asia, East Asia, and the South Pacific*, 2001, 2, 1.
- **, – , and –**, *Elections in Asia and the Pacific: A Data Handbook: Volume I: Middle East, Central Asia, and South Asia*, OUP Oxford, 2001.
- **, Michael Krennerich, Bernard Thibaut et al.**, *Elections in Africa: A Data Handbook*, Oxford University Press, 1999.
- Nunley, Albert C**, “African Elections Database,” 2007, 13.
- Nunn, Nathan, Nancy Qian, and Jaya Wen**, “Distrust and Political Turnover during Economic Crises,” Working Paper 24187, National Bureau of Economic Research January 2018.
- Olken, Benjamin**, “Corruption and The Costs of Redistribution: Micro Evidence from Indonesia,” *Journal of Public Economics*, 2006, 90 (4-5), 853–870.
- Olson, Mancur**, *The Rise and Decline of Nations*, Yale University Press, 1984.
- Ortiz-Ospina, Esteban**, “Government Spending,” *Our World in Data*, 2016. <https://ourworldindata.org/government-spending>.
- Ottinger, Sebastian and Nico Voigtländer**, “History’s Masters: The Effect of European Monarchs on State Performance,” Technical Report, National Bureau of Economic Research 2021.

- Papaioannou, Elias and Gregorios Siourounis**, “Democratisation and Growth,” *Economic Journal*, 02 2008, 118, 1520–1551.
- Pemstein, Daniel, Kyle L Marquardt, Eitan Tzelgov, Yi ting Wang, Joshua Krusell, and Farhad Miri**, “The V-Dem Measurement Model: Latent Variable Analysis for Cross-National and Cross-Temporal Expert-Coded Data,” *V-Dem Working Paper*, 2018, 21.
- Persson, Torsten and Guido Tabellini**, *Political Economics: Explaining Economic Policy*, MIT press, 2002.
- Pocock, Stuart J**, “Clinical Trials with Multiple Outcomes: A Statistical Perspective on their Design, Analysis, and Interpretation,” *Controlled clinical trials*, 1997, 18 (6), 530–545.
- Przeworski, Adam**, *Democracy and the Market: Political and Economic Reforms in Eastern Europe and Latin America*, Cambridge University Press, 1991.
- Rodrik, Dani and Romain Wacziarg**, “Do Democratic Transitions Produce Bad Economic Outcomes?,” *The American Economic Review*, 2005, 95 (2), 50–55.
- Rose, Richard and Neil Munro**, *Parties and Elections in New European Democracies*, Ecpr Press, 2009.
- Sequeira, Sandra and Simeon Djankov**, “Corruption and Firm Behavior: Evidence from African Ports,” *Journal of International Economics*, 2014, 94 (2), 277–294.
- Snowberg, Erik, Justin Wolfers, and Eric Zitzewitz**, “Partisan Impacts on the Economy: Evidence from Prediction Markets and Close Elections,” *The Quarterly Journal of Economics*, 05 2007, 122 (2), 807–829.
- Svensson, Jakob**, “Who Must Pay Bribes and How Much? Evidence from a Cross Section of Firms,” *The Quarterly Journal of Economics*, 2003, 118 (1), 207–230.
- Tella, Rafael Di and Robert MacCulloch**, “Why Doesn’t Capitalism Flow to Poor Countries?,” Working Paper 13164, National Bureau of Economic Research June 2007.
- Union, IPU Inter-Parliamentary**, “PARLINE Database on National Parliaments,” *Ukraine. Verkhovna Rada (Parliament)*, Geneva, 2017.
- University, OAS Georgetown**, “Political Database of the Americas,” 2002.
- Volkens, Andrea, Tobias Burst, Werner Krause, Pola Lehmann, Theres Matthie, Nicolas Merz, Sven Regel, Bernhard Weels, and Lisa Zehnter**, “The Manifesto Data Collection. Manifesto Project (MRG/CMP/MARPOR). Version 2020b,” 2020.

Chapter 3

The Global Incumbency Advantage

3.1 Introduction

In both democracies and autocracies, incumbents are often in a dominant position when competing for reelection and seeking to maintain control of political power (Gelman and King, 1990; Levitt, 1994). The size of this incumbency advantage has important implications for political accountability and the health of democracy (Ashworth, 2012). Strong incumbency effects may indicate that voters reward competent leaders and value stability, but they may also reflect direct benefits from holding office, such as access to institutional resources, superior fundraising capacity, and greater name recognition and visibility. In turn, such effects might influence political selection by deterring high-quality challengers from competing against experienced incumbents (Cox and Katz, 1996). They may weaken electoral incentives and reduce performance, or conversely, foster long-term policy planning.

Since the 1970s, a large literature has studied the incumbency advantage within countries, with an early focus on U.S. congressional and state elections (Ferejohn, 1977; Fiorina, 1977). This inspired more recent empirical work leveraging causal identification methods, in particular regression discontinuity designs, to estimate the incumbency advantage (Lee, 2008) and the effects of candidates' rankings in past elections (Anagol and Fujiwara, 2016; Granzier et al., 2023). Many other studies examine the incumbency advantage at the subnational level beyond the U.S. context. However, we do not know to what extent winning a national election affects a party's or a candidate's subsequent electoral performance and political trajectory. At the national level, the existence of large incumbency effects would have important implications for the likelihood of consequential political turnovers (Marx et al., 2025) and the functioning of democracy (Levitsky and Ziblatt, 2019).

This paper provides the first estimates of the incumbency advantage at the national level. Doing so poses new conceptual challenges because national incumbents not only enjoy the common sources of advantage described in the literature, but also have the ability to affect the timing of elections and the conditions in which elections are fought. Thus, while previous work often equates the incumbency advantage with the effect of an electoral win on subsequent electoral performance, we must consider the possibility that incumbents may

determine whether competitive elections take place at all, when they take place, and under which conditions (Fearon, 2011; Egorov and Sonin, 2014).

We introduce a novel characterization of the incumbency advantage as the causal effect of winning an election on the probability that a candidate or party remains in office beyond the scheduled end of their term. This allows us to consider a variety of settings, including countries with heterogeneous levels of democracy where leaders retain power through various means, including—but not restricted to—winning elections. Using data from all national elections conducted globally since 1945, we study how electoral victories affect a party’s or a candidate’s chances of remaining in office beyond the next scheduled election. Our analysis compares the power trajectories (i.e., the probability of being in power τ years after a given election or after the scheduled term end) of winners and losers of close elections conducted at time t .¹ We also estimate how incumbency affects the probability of winning the next election, conditional on the election taking place. In our setting, incumbency effects on subsequent electoral performance constitute one of the mechanisms driving a leader’s ability to retain political power, in addition to leader-driven changes in the fairness and the timing of elections.

We first review the rich empirical literature studying the incumbency advantage *within countries*. While previous work has studied incumbency effects in many contexts, existing estimates show considerable variation in magnitude and even sign (e.g., Ferreira and Gyourko, 2009; Eggers et al., 2015; Fiva and Smith, 2018; Klačnja and Titunik, 2017; Kartik and Weelden, 2019). Overall, we lack a unified understanding of the key insights from this literature. The first part of the paper presents a meta-analysis of this body of work, covering 94 estimates of the incumbency advantage obtained via regression discontinuity designs across 58 studies and 29 countries. Large incumbency advantages are observed in OECD countries, which are generally high-income and democratic. Emerging and developing countries, by contrast, tend to display incumbency disadvantages. We conduct several empirical exercises to identify the sources of this heterogeneity and highlight the role of three variables—GDP per capita, corruption, and democratic quality—as important predictors of the incumbency advantage at the subnational level.

We then turn to our core objective: providing a causal estimate of the *national* incumbency advantage across all presidential and parliamentary elections held since 1945, at the level of political parties and individual candidates. Using the nonparametric regression discontinuity approach from Calonico et al. (2014), we first conduct extensive placebo checks to verify the validity of our empirical design. There is no evidence of electoral manipulation overall and across various subsamples, except in the subsets of elections conducted in autocracies and elections classified (ex post) as not free and fair by the Varieties of Democracy project (Coppedge et al., 2021). Having established this pattern, we report results separately for the full sample of elections and for the subsample of free and fair elections.

Across all presidential and parliamentary elections, we document the existence of a positive but short-lived party-level incumbency advantage at the national level. Political parties that win national elections experience different power trajectories relative to runner-

¹Our analysis accounts for the possibility of legal transition periods: we report the effect of marginally winning an election on the likelihood of being in power at least six months (or longer) after the scheduled end of an incumbent’s term in office.

up parties: election winners are 13 percentage points (p.p.) more likely to be in power five years after a given election (which is significant at the 10% level) and 10 p.p. more likely to be in power six months beyond the scheduled end of their term (which corresponds to 27% of the runner-up party's probability of being in power at that point and is significant at the 10% level). However, seven years after a given election and three years after the scheduled term end, the winning party is no more likely to be in power, on average. Estimates of the incumbency advantage of individual candidates in presidential elections deliver similar insights. Winning presidential candidates benefit from a positive (10 p.p.) incumbency advantage on average, but this effect is also short-lived. All these results are robust to only considering national elections that are classified as free and fair and to various sample restrictions and alternative specification choices.

We generally do not observe substantial differences between incumbency effects in presidential elections and parliamentary elections. However, mirroring the insights from our meta-analysis of subnational results, these incumbency effects vary substantially across the world. Incumbent parties in OECD countries—in particular, those in North America and Western Europe—as well as those in African countries enjoy a more pronounced advantage. The same insights broadly hold when we consider individual candidates (rather than political parties) running in presidential elections in those regions. By contrast, incumbent parties face a disadvantage in Latin America and in Asia and Oceania.² The evidence from national elections diverges from within-country estimates for low-income countries. In those settings, the literature has documented large incumbency disadvantages in local elections, while we estimate a positive party-level incumbency advantage at the national level. Finally, elections classified as not free and fair yield very large incumbency advantages for the parties and presidential candidates that win them.

We explore two sets of mechanisms to explain these findings: incumbency effects on subsequent electoral performance, and manipulation of the fairness and the timing of follow-up elections. First, we examine how incumbents fare in subsequent elections, extending the standard characterization of the incumbency advantage to the national level. We find non-significant incumbency effects on the joint probability of competing in and winning the next national election. However, this null average result masks substantial heterogeneity across subsamples that mirrors the heterogeneity we find for the power trajectories of election winners and runners-up. Election winners in North America and Western Europe have a substantially larger probability of winning the next election,³ while incumbents in two regions (Latin America, and Asia and Oceania) face a lower probability of winning. In line with the findings from our meta-analysis, these effects tend to be larger in regions with higher levels of income, lower levels of corruption, and higher levels of democratic quality.

Second, we consider the possibility that changes in the fairness and the timing of subsequent elections affect incumbents' ability to manipulate the outcome of these elections and to retain power beyond their scheduled term. We first provide evidence that in several subsamples (e.g., elections in autocracies and those in Africa), winning a national election increases the probability of winning a follow-up election that is not free and fair. To explore

²We group countries in North America and Western Europe into one subsample, and countries in Asia and Oceania into another (see Section 3.4).

³Mayhew (2008) provides consistent descriptive evidence in the context of U.S. presidential elections.

the timing mechanism, we compute the difference between the dates of constitutionally mandated elections and actual election dates. National officeholders in several subsamples (e.g., autocracies and low-income regions) retain power by postponing elections or by canceling them outright. Across the least democratic countries in our sample, only 56% of national elections are followed by another election held on schedule, 26% by an election held early, and 17% by a late election or no election at all. By contrast, in the most democratic countries—where snap elections are common but elections are rarely postponed—the corresponding figures are 74%, 24%, and 2%.

In the final part of the paper, we provide a decomposition of the incumbency advantage at the national level by distinguishing between three mutually exclusive scenarios. Election winners may retain power beyond the scheduled end of their term because (i) they have won a follow-up election that was free and fair, (ii) they have won a follow-up election that was not free and fair, or (iii) no election has taken place. This decomposition delivers intuitive insights. For instance, the positive incumbency effects we observe among OECD countries and democratic countries are largely driven by the likelihood of winning follow-up elections that are free and fair. By contrast, elections held in autocratic regimes and, more generally, elections that are not free and fair yield a sizable incumbency advantage that is almost entirely driven by the absence of free and fair follow-up elections. Taken together, these results give rise to a non-linear relationship between our measure of the national-level incumbency advantage on the one hand, and levels of economic development and democratic quality on the other. While subnational estimates of the incumbency advantage correlate positively with GDP per capita, our estimates of the national incumbency advantage are lowest for middle-income countries with intermediate values of institutional quality, and largest for the most democratic and the least democratic countries in our sample.

This pattern arises due to different mechanisms in highly democratic and less democratic countries. In high-income and democratic countries, winning a national election substantially improves the winning side’s performance in the subsequent election. In those settings, incumbents must typically secure re-election via democratic means, and the effects we observe may reflect legitimate electoral support for high-performing parties and candidates who successfully tap into the electoral benefits of incumbency (as in, e.g., [King and Gelman, 1991](#); [Cox and Katz, 1996](#); [Levitt and Wolfram, 1997](#)). This might lead to a virtuous circle shaped by voter learning, positive selection, and long-term policy-making, as described in the models of, e.g., [Aruoba et al. \(2019\)](#) and [Kartik et al. \(2025\)](#), and the reviews of [Ashworth \(2012\)](#) and [Duggan and Martinelli \(2017\)](#). Alternatively, strong incumbency effects could be sustained by clientelistic exchanges between a loyal supporter base and strong incumbents who provide targeted transfers to the former (see [Weingast et al., 1981](#); [Hicken, 2011](#); [Golden and Min, 2013](#)), although we note that these forces are unlikely to be more pronounced in highly democratic settings. In less democratic countries, election winners are more likely to retain office beyond their scheduled term mainly due to their ability to manipulate the timing and the competitiveness of elections. This can generate large incumbency advantages, allowing electoral autocracies and other authoritarian regimes to solidify and persist ([Egorov and Sonin, 2024](#)).

Overall, our results shed new light on the nuanced relationship between incumbency effects and the quality of democracy across and within countries. They also illustrate the importance of quantifying the incumbency advantage at an aggregate level—and uncovering

its main determinants—to track the performance of democracies over time. We show that similarly large incumbency advantages can reflect radically different forces across different institutional settings.

The rest of the paper is organized as follows. Section 3.2 presents our meta-analysis of the literature studying the incumbency advantage within countries. Sections 3.3 and 3.4 outline the data and empirical framework we use to estimate the incumbency advantage at the national level. Section 3.5 reports our main results, Section 3.6 examines mechanisms, and Section 3.7 concludes.

3.2 Meta-analysis

We start our analysis of the global incumbency advantage with a review of the existing subnational evidence in economics and political science. We surveyed the literature to recover as many estimates of the incumbency advantage as possible and to identify robust conclusions from this body of work.

Our meta-analysis focuses on estimates of the incumbency advantage obtained via regression discontinuity designs (RDDs). This focus serves two purposes. First, while the literature provides estimates computed via other methods (e.g., [Ansolabehere et al., 2000](#)), RDDs have become the state-of-the-art approach to measure the incumbency advantage (see [Lee, 2008](#); [Lee and Lemieux, 2010](#)). Second, considering estimates from a single empirical approach allows us to pinpoint heterogeneity driven by contextual rather than methodological differences.

3.2.1 Analysis sample

In order to causally identify the incumbency advantage, close-elections RDDs examine how candidates or political parties who marginally won an election perform in the subsequent election, compared to marginal losers. Specifically, one can estimate

$$Y_{c,t+1} = \alpha + \beta T_{ct} + \gamma X_{ct} + \delta T_{ct} X_{ct} + \varepsilon_{c,t+1}, \quad (3.1)$$

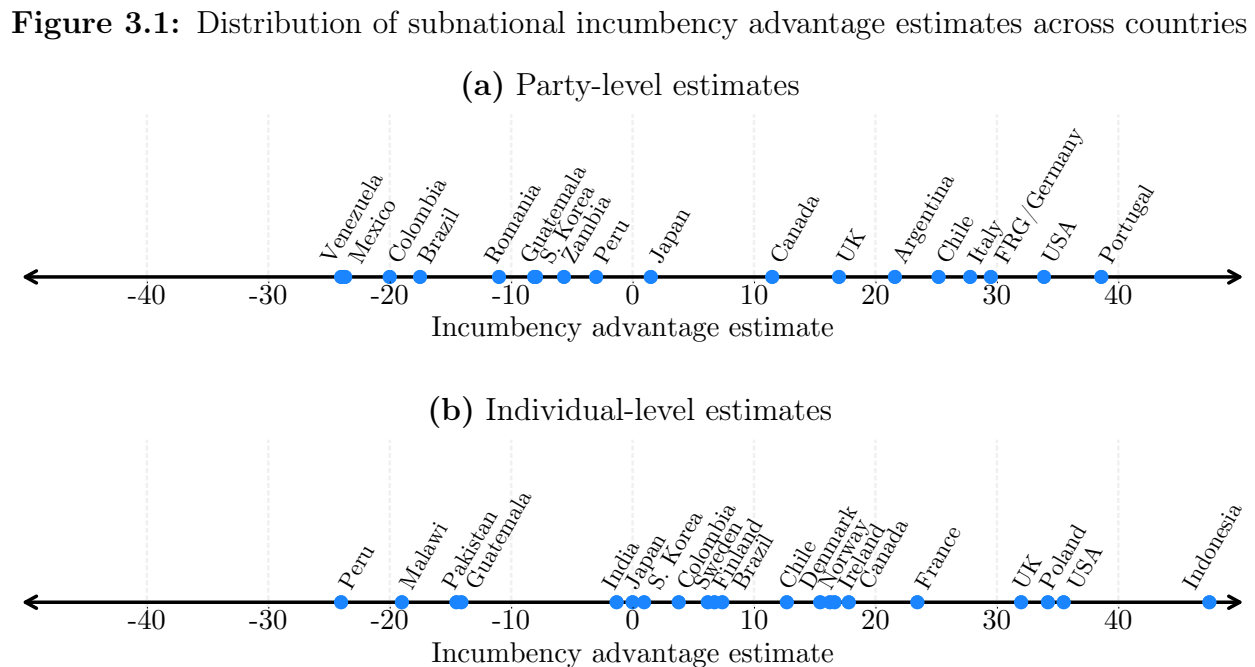
where X_{ct} represents the margin of victory of candidate c in the election held at time t , T_{ct} is a dummy indicating candidate c 's victory (it is equal to one when X_{ct} is positive, and zero otherwise), and $Y_{c,t+1}$ measures the electoral performance of candidate c in the subsequent election in the same constituency or electoral unit. The coefficient of interest β measures the electoral advantage that winners of the election conducted at time t enjoy in the election conducted at time $t + 1$.

In our meta-analysis, we focus on the effect of winning an election on the probability of winning the next one, instead of other outcomes (such as the probability of running in the next election or the candidate's vote share conditional on running) because these other outcomes are not consistently reported across studies. Appendix A.1 describes the methodology we followed to comprehensively survey the literature and Appendix Table A.1 lists all the studies included in the meta-analysis. In total, we recovered 94 estimates from 58 studies, covering a total of 29 countries. Most of these estimates (58) measure the

incumbency advantage for a local office (e.g., in mayoral, gubernatorial, regional, or state legislature elections). The remaining estimates (36) measure the incumbency advantage in constituencies for the national parliament (e.g., congressional districts for the U.S. House of Representatives or parliamentary constituencies in the United Kingdom). Studies of the incumbency advantage also differ in whether they measure effects of incumbency for individual candidates or their parties. We recover 46 estimates of party-level effects and 48 estimates of candidate-level effects.

3.2.2 Findings

Studies focusing on congressional or state elections in the U.S. have repeatedly documented a large incumbency advantage. For instance, [Lee \(2008\)](#) shows that when the Democratic Party marginally wins a seat in the U.S. House of Representatives, its probability of winning the same seat again in the subsequent election is 36 p.p. higher than if it had marginally lost. By contrast, estimates of the incumbency advantage outside the U.S. are typically much lower, and many countries feature an incumbency disadvantage. Figure 3.1 shows the wide range of available estimates across countries.

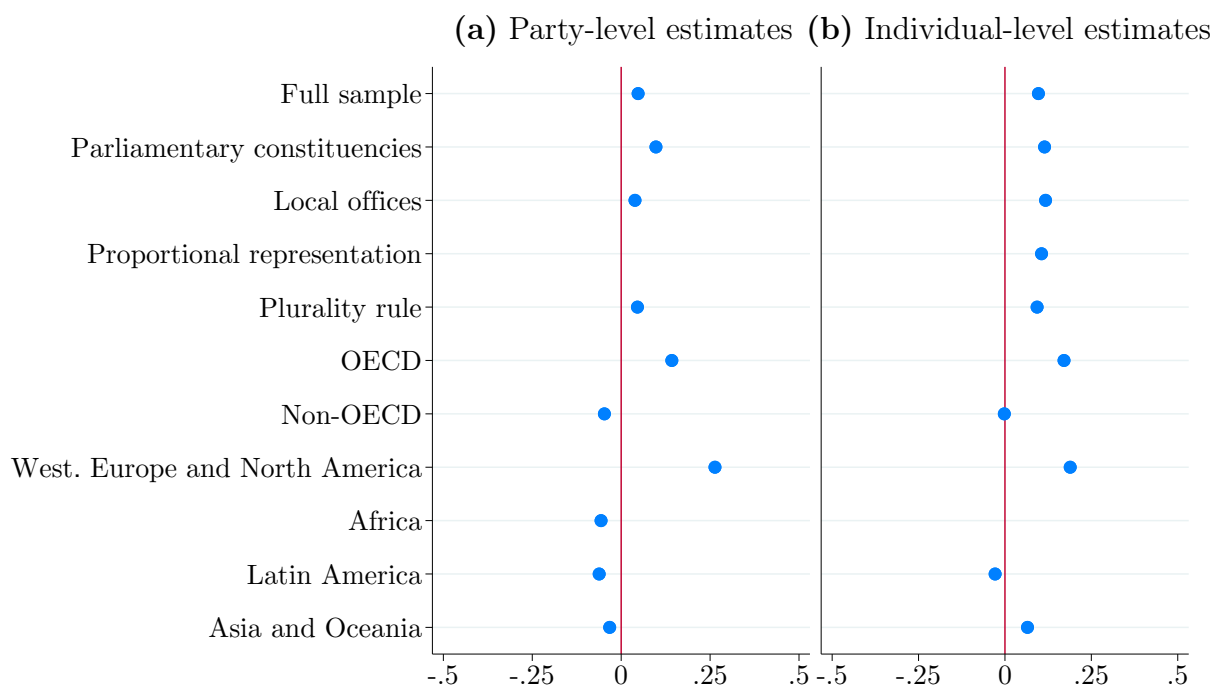


Notes: This figure shows the distribution of subnational estimates of the incumbency advantage across countries. Panel (a) (resp., b) shows party-level (resp., individual-level) estimates. When several studies measure the incumbency advantage in a country, we report the average of available estimates.

The largest incumbency advantages are typically measured in high-income countries. In Figure 3.2, we report the average incumbency advantage by region, electoral system, and election level. Panel (a) reports estimates at the party level. There is a sharp contrast between OECD countries, in which there is a large positive average incumbency advantage

(14 p.p.), and the rest of the world, in which there is an average incumbency disadvantage (-5 p.p.). The incumbency advantage is positive and very large on average in Western Europe and North America (26 p.p.) but negative in developing regions such as Africa (-6 p.p.) and Latin America (-6 p.p.). We also observe major differences between OECD and non-OECD countries at the candidate level—see Panel (b).

Figure 3.2: Average incumbency advantage at the subnational level



Notes: This figure reports the average incumbency advantage across studies of our meta-analysis, in the full sample and in various subsamples. To avoid over-weighting countries that are overrepresented in the literature, we first take averages of estimates within each country before taking averages within each subsample. We do not report results for subsamples with estimates from a single country.

Potential determinants of the incumbency advantage. OECD countries display large incumbency advantages, while countries outside the OECD often show an incumbency disadvantage. This suggests that economic development and institutional quality may be important explanatory factors. Indeed, previous work has identified several channels through which economic development might shape incumbency effects.

Based on the literature, we first consider three potential economic determinants of the incumbency advantage: GDP per capita, as measured in the Penn World Tables (Feenstra et al., 2015), unemployment, from the International Labour Organization, and income inequality, as measured by the Gini index, from the World Inequality Database. Across many studies, satisfactory economic performance has been linked to support for incumbent governments (see, e.g., Pacek, 1994; Duch, 1995, 2001; Anderson et al., 2003). Uppal (2009) shows that, in India, the incumbency disadvantage is higher in states with lower incomes, higher unemployment, and worse public goods quality. Bernhard and Karakoç

(2011) find that rising income inequality in post-communist countries is associated with higher party volatility and larger electoral losses for incumbents. Finally, Roberts (2008) shows that unemployment shapes voters' decision to sanction outgoing governments, in a sample of post-communist European countries where all workers were previously guaranteed employment.

Corruption is also frequently cited as a source of incumbency disadvantage (Klašnja, 2015). Indeed, when corruption is widespread, voters may assume that most incumbents are corrupt, creating an anti-incumbent bias. Klašnja and Titunik (2017) argue that this effect increases with the length of an incumbent's tenure. Since setting up effective rent-extraction networks takes time, voters can limit the total amount of corruption by limiting reelection probabilities (see Coviello and Gagliarducci, 2017, for evidence consistent with this mechanism). To explore the role of these factors in shaping the incumbency advantage, we consider as potential predictors the measures of corruption and accountability from the Varieties of Democracy (or V-Dem) dataset (Coppedge et al., 2021).

Finally, Ariga (2015) interprets the absence of an incumbency advantage in Japan as the consequence of the strong intraparty competition in the country's multi-member parliamentary districts. When several candidates share similar ideologies, voters can easily shift away from incumbents to ideologically close competitors, weakening any incumbency advantage. To account for the influence of ideological proximity between competitors, we consider political polarization (from V-Dem) as an additional potential determinant of the incumbency advantage.

Findings. We conduct several empirical exercises to explore how the incumbency advantage correlates with the seven covariates described above. In Appendix Figure A.1, we regress party-level estimates of the incumbency advantage on its potential determinants using three different approaches. In all regressions, we control for the voting rule (plurality rule or proportional representation) as well as the level at which the election is conducted (local offices or parliamentary constituencies). Furthermore, to avoid giving excessive weight to countries covered by numerous studies (such as the U.S.), we weigh incumbency advantage estimates so as to give equal weights to all countries.⁴

We first run bivariate OLS regressions in which we successively regress the party-level incumbency advantage on each of the aforementioned covariates. This exercise identifies four predictors that are statistically significant at the 5% level: the incumbency advantage is positively correlated with GDP per capita and accountability, and negatively correlated with corruption and political polarization.

Second, we estimate a multivariate OLS regression of the party-level incumbency advantage on the full set of explanatory variables described above. The estimates are less precise, reflecting the substantial multicollinearity among the covariates,⁵ and none of them is statistically significant, but their sign is consistent with the bivariate regressions.

Third, we run a Lasso regression using all covariates, choosing the penalty with a tenfold cross-validation to minimize the mean squared error. This analysis selects corruption and

⁴In Appendix Figures A.3 through A.6, we perform the same exercise with weights equal to the inverse standard error around each estimate of the incumbency advantage or without weights.

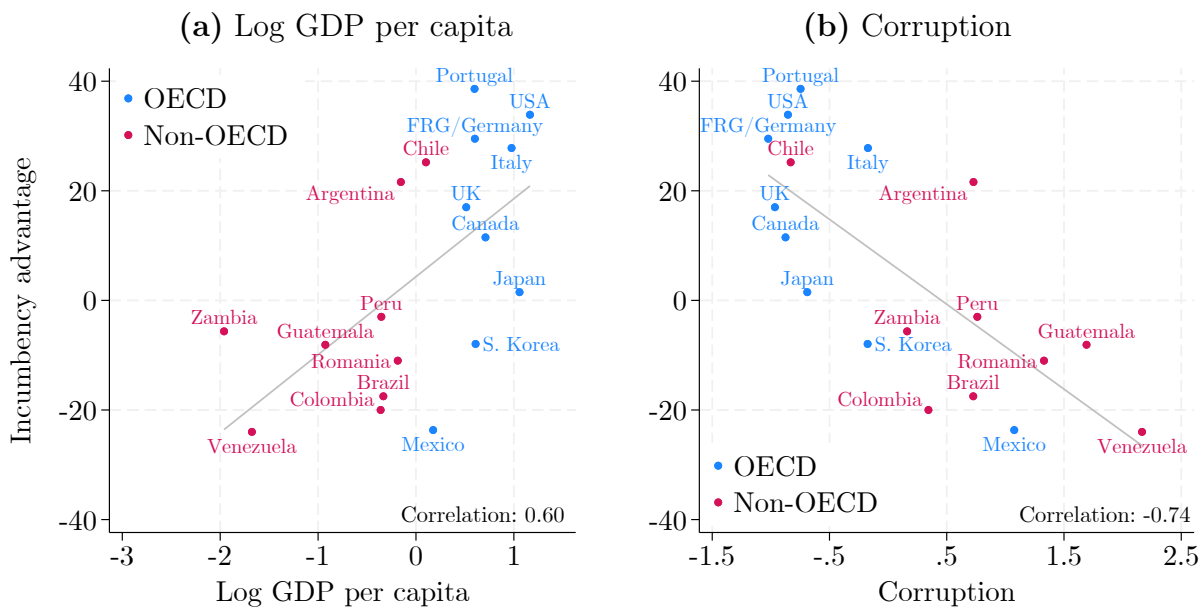
⁵For instance, the correlation between a country's log GDP per capita and its corruption level is 0.69.

accountability as the most relevant correlates of the party-level advantage. When running a multivariate OLS regression on these two covariates, only corruption is significantly correlated with the incumbency advantage.

In Appendix Figure A.2, we conduct the same set of exercises for candidate-level estimates of the incumbency advantage. This analysis yields similar results, with GDP per capita, unemployment, accountability, and corruption being the only statistically significant determinants (at the 10% level) in the bivariate analysis. In the Lasso estimation, (log) GDP per capita is selected as the strongest correlate of the incumbency advantage (p-value 0.034).

Overall, GDP per capita and corruption stand out as the strongest predictors of the incumbency advantage. In Figure 3.3, we plot the average party-level incumbency advantage in each country against these covariates. The incumbency advantage steadily increases with GDP per capita and decreases with corruption. The same patterns hold in Appendix Figure A.7, where we plot individual-level estimates of the incumbency advantage against GDP per capita and corruption. These results are in line with existing descriptive evidence from [Golden and Nazrullaeva \(2023\)](#), who document a strong positive correlation between legislators' reelection rates and GDP per capita.

Figure 3.3: Correlates of subnational incumbency advantage effects



Notes: This figure compares estimates of the subnational party-level incumbency advantage to (log) GDP per capita (as measured by the Penn World Tables) and corruption (as measured by V-Dem). Both variables are standardized, and we report Pearson correlation coefficients along with the line of best fit. Each coefficient in our meta-analysis is associated with the average GDP per capita and corruption level in the country and time period of study. We then report the average estimate by country.

We plot the corresponding figures for the other potential determinants of the party-level advantage in Appendix Figure A.8 and for those of the individual-level advantage in Appendix Figure A.9. The takeaways from these additional results are similar. For example,

both the party-level and the individual-level advantage correlate positively with political accountability.

Taking stock. In the remainder of the paper, we link the subnational estimates of the incumbency advantage surveyed in this section with our own estimates of the incumbency advantage at the national level. An important insight from our meta-analysis is that subnational incumbents benefit from large electoral advantages in settings with higher income and lower corruption that are often associated with high levels of democratic quality. Conversely, they tend to face a disadvantage in contexts with high corruption and low GDP per capita, which often characterize less democratic societies. In fact, we observe a strong 0.73 correlation between subnational-level estimates of the incumbency advantage and the V-Dem index of democratic quality across countries (see Appendix Figure A.10).

These results can be interpreted based on theoretical models in which politicians differ both in terms of their intrinsic type and the effort they exert. In democratic regimes, incumbents face the prospect of competitive reelection battles, and voters reward competent politicians who exert high effort, creating strong incentives for incumbents to perform well. In these settings, “positive” accountability forces can give rise to large incumbency advantages at the subnational level and high reelection rates in equilibrium (Ferejohn, 1986; Fearon, 1999; Besley, 2007). A vast theoretical literature describes the existence of such political equilibria and the conditions under which they emerge (e.g., Aruoba et al., 2019; Duggan and Martinelli, 2017, 2019; Kartik et al., 2025).

Incumbents may also benefit from structural advantages in terms of visibility (Franklin, 1993) or fundraising (Levitt, 1994; Pastine and Pastine, 2012; Fourniaies and Hall, 2014; Avis et al., 2022) and they may be able to use state resources to sustain clientelistic exchanges with voters (Hicken, 2011; Golden and Min, 2013). Large incumbency effects arising from such forces would have different normative implications, both in terms of what they reveal about the nature of the political equilibrium and the risk that they decrease officeholders’ incentives to exert effort. However, we note that clientelistic forces are unlikely to explain the positive correlation between the incumbency advantage and democratic quality. Indeed, these forces are, if anything, more likely to be at play in less democratic contexts, where incumbents can more easily tap into public funds to build a loyal voter base.

Conversely, incumbency disadvantages at the subnational level are more likely to emerge in settings with low levels of democracy, where disaffected voters may form beliefs that most politicians are dishonest, and incumbents are systematically punished for participating in a system perceived as corrupt. Then, elections cease to discipline politicians and incumbents face shorter time horizons in office, providing incentives for them to engage in corruption while they can, and inducing more dishonest candidates to enter politics. Incumbents and voters may therefore be trapped in a self-fulfilling equilibrium with poor performance, high corruption, and pessimistic expectations (Svolik, 2013; Klačnja, 2015; Klačnja and Titiumik, 2017). Furthermore, the selection effects induced by such forces might imply that the overall distribution of politician quality is worse in less democratic settings, which could directly and negatively affect the incumbency advantage.

The sign and magnitude of incumbency effects may differ at the national level. In particular, the positive correlation between the incumbency advantage and democratic

quality could break down at this level because country leaders in less democratic settings have many tools at their disposal to avoid competitive elections (unlike local mayors and parliamentary representatives, who typically cannot alter the electoral schedule or cancel elections outright). Thus, incumbency effects at the country level may result from the additional benefits conferred by the control of national offices, including the ability to manipulate the timing, competitiveness, and results of national elections to their advantage. In some cases, a large incumbency advantage could simply indicate that low-quality leaders have effectively silenced political competition. Overall, one cannot assess the welfare implications of national incumbency effects without also asking whether elections serve to reward positive selection and performance or get confiscated by certain individuals or parties.

3.3 Data

We now turn to estimating the incumbency advantage at the national level. For this analysis, we leverage data on all presidential and parliamentary elections held worldwide since 1945, combined with data on national leaders, parties, and political regimes. This section briefly describes the data leveraged in our analysis, and a detailed description is provided in Appendix B.

Elections and electoral results. We use the results of presidential and parliamentary elections held between 1945 and 2023 from the National Elections Database (Marx et al., 2025). Our data include 859 presidential elections and 2,162 parliamentary elections, for a total of 3,021 elections in 213 countries.⁶ In presidential elections, we observe each candidate’s party affiliation and vote share. In parliamentary elections, we observe each party’s seat share. We use these data to identify election winners for each type of election, as described in Section 3.4.

Leaders and political parties. We build a database of individual leaders of the executive branch for each country, including the dates at which they gained and lost executive power (see Appendix B.1). We then link these data to parties and candidates competing in elections to check whether election winners stay in power beyond their constitutionally planned term.

To track the performance of political parties over time, we build a database of parties using data from PartyFacts (Döring and Regel, 2019), V-Parties (Lührmann et al., 2020; Pemstein et al., 2018), and Wikidata. Importantly, we account for name changes and organizational restructurings over time.⁷ Failing to do so would generate mistakes, such as considering that a party has lost power when it retained power under a different name. We rely on Wikidata to identify these continuous linkages. Specifically, a party B is coded as the successor of another party A if Wikidata lists party A as party B’s sole

⁶The data also cover non-sovereign entities with significant autonomy such as Hong Kong, Niue, and Anguilla.

⁷For instance, in Kazakhstan, the “Otan” (Fatherland) party, founded in 1999, was relabeled “Nur Otan” in 2006, and then “Amanat” in 2022. In our analysis, we group these three entities as a single party.

predecessor and, reciprocally, B as A’s sole successor.⁸ This conservative rule aims to minimize misclassifications by ensuring that succession links reflect true continuity rather than factional splits.⁹ The resulting database includes 8,040 parties (see Appendix B.2).

Political regimes. Finally, we collect data on the institutional contexts in which national elections are held. In particular, we determine whether each election in our sample led to the designation of a member of the executive branch, and the type of regime in which it took place (see Appendix B.3). We also use V-Dem (Lührmann et al., 2020), the Comparative Constitutions Project (Elkins and Ginsburg, 2022), Baturo and Elgie (2019), and the text of national constitutions (recovered from various sources, including the vLex legal database), to identify the constitutionally defined term lengths of those appointed after each election, i.e., the length of a presidential term or the duration for which new MPs are elected in parliamentary elections (see Appendix B.4).

3.4 Empirical Framework

3.4.1 Power trajectories

When elections are held at regular constitutionally mandated intervals, the incumbency advantage can be measured as the effect of marginally winning an election on the probability of winning the next election. This is the object of interest of most studies surveyed in Section 3.2. However, this approach does not fully capture how winning a national election affects a winner’s chances of retaining power beyond their initial term in office, whether elections take place or not. For instance, national leaders can postpone elections or cancel them outright. They may also call early elections when conditions are favorable to them. In our dataset, this is not a marginal phenomenon: 39% of elections were not held within six months of the incumbent’s constitutionally planned term end (or were canceled).

Therefore, to measure the incumbency advantage at the national level, we estimate the effect of marginally winning versus losing an election on two outcomes: the likelihood of being in power τ years after the election conducted at time t , which we call the winner and the runner-up’s *power trajectory*; and the likelihood of being in power $\tilde{\tau}$ years after the end of the constitutionally planned term, an effect we refer to as the *post-term incumbency advantage*.

Power trajectories in presidential elections. We first estimate the effect of marginally winning an election on the likelihood of being in power over the ten years following the election. For each year after the election, we record whether the leader or party in power aligns with the election winner, the runner-up, or neither.

⁸This detection is performed using variables “follows” (P155), “replaces” (P1365), “followed by” (P156) and “replaced by” (P1366).

⁹For instance, the Italian Communist Party, which was active between 1921 and 1991, has two successors: the Communist Refoundation Party and the Democratic Party of the Left. We therefore treat these parties as three distinct entities rather than grouping them together.

For presidential elections, we compare the sitting president in each relative year τ to the winner and the runner-up of the election conducted at time t . In our main specification, we code the president as aligned with the election winner if they are the individual who won the election, if they belong to the party that won the election, or if they have been unambiguously designated as the representative or successor of the election winner. We use the same rules to determine whether the president is aligned with the runner-up.¹⁰ Given this definition, the estimates we obtain can be interpreted as a measure of the party-level incumbency advantage at the national level. To account for post-election transition periods, outcomes for relative year τ are measured τ years and 180 days after the date of the election conducted at time t .¹¹ In an alternative specification, we estimate the individual-level incumbency advantage in presidential elections, coding the president as aligned with the election winner only if they are the same individual.

To illustrate our approach, consider the 2008 U.S. presidential election. This election took place on November 4th, 2008. The winner, Barack Obama from the Democratic Party, was inaugurated on January 20th, 2009, replacing George W. Bush from the Republican Party. Barack Obama remained in power until January 20th, 2017, when Donald Trump from the Republican Party became the U.S. president. For this election, outcomes for relative year $\tau = 0$ are measured on May 3rd, 2009; for relative year $\tau = -1$ on May 3rd, 2008; and for relative year $\tau = 1$ on May 3rd, 2010. For all relative years in which Barack Obama was in office (namely for $\tau \in [0, 7]$), the sitting president was the election winner, so we code the president as aligned with the winner and not aligned with the runner-up. In relative years $\tau \in [8, 10]$, the sitting president, Donald Trump, had not competed in the 2008 election, but he ran for the Republican Party, which was the party of John McCain, the 2008 election runner-up. Therefore, we code the leader in power in relative years $\tau \in [8, 10]$ as aligned with the runner-up. We proceed similarly for $\tau = -1$ and $\tau = -2$ since Barack Obama's predecessor, George W. Bush, also belonged to the Republican Party. Table 3.1, panel (a) reports the power trajectories for both candidates in this election.

Power trajectories in parliamentary elections. We define the winner of a parliamentary election held at time t as the party or coalition that secured a plurality of seats in that election.¹² This definition allows us to identify an election winner in systems where parliamentary elections do not lead to the designation of a member of the executive branch, such as the United States.¹³ Furthermore, we group together the results of parties

¹⁰In a few rare cases, we code the sitting president as aligned with both the winner and the runner-up. For example, in the October 1981 Iranian presidential election, both candidates of interest were affiliated with the Islamic Republican Party (IRP), so the reelected president Ali Khamenei is coded as aligned with both of them. This configuration arises in 0.4% of presidential elections and 1.4% of parliamentary elections in our sample.

¹¹We motivate this decision in Appendix C.2. We measured the duration of transition periods for all presidential elections followed by a leader transition. For these elections, we observe that 99.7% of power transitions had taken place within 180 days of the election.

¹²In the rare case of a tie, we rank parties by vote shares.

¹³In principle, one could be interested in estimating an individual-level incumbency advantage in parliamentary elections that lead to the designation of a member of the executive, mirroring the individual-level specification we use for presidential elections, but data limitations prevent us from doing so. Indeed, individual candidates for executive leadership cannot be systematically identified before the

Table 3.1: Power trajectories for the 2008 elections in the United States

Relative year	-2	-1	0	1	2	3	4	5	6	7	8	9	10
(a) Presidential elections													
President aligned with winner (Obama, Democrat)	0	0	1	1	1	1	1	1	1	1	0	0	0
President aligned with runner-up (McCain, Republican)	1	1	0	0	0	0	0	0	0	0	1	1	1
(b) House of Representatives elections													
Party in power aligned with winner (Democratic Party)	1	1	1	1	0	0	0	0	0	0	0	0	1
Party in power aligned with runner-up (Republican Party)	0	0	0	0	1	1	1	1	1	1	1	1	0

Notes: This table reports the value of the outcome indicating the alignment of the president or party in power with the winner and runner-up of the 2008 U.S. presidential election, in panel (a), and of the 2008 U.S. House elections, in panel (b) for relative year $\tau \in [-2, 10]$.

that were part of coalitions formed before the election. Appendix B.5 describes how we distinguish such *ex ante* coalitions from *ex post* governing coalitions formed once the election results are known.

For each relative year, we consider that the party in power in parliament is the party that controlled a plurality of seats. Given the lack of systematic data on changes in the composition of parliaments between elections, we assume that parliamentary composition is always determined by the most recent parliamentary election.¹⁴ We then determine, for each relative year τ , whether the party in power is the election winner (or its clearly defined successor), the runner-up (or its successor), or neither. For instance, Table 3.1, panel (b) reports the power trajectories for the winner and runner-up parties in the 2008 elections to the U.S. House of Representatives. The Democratic Party won the 2008 elections, retaining a majority won in 2006. It was then defeated by the Republican Party, which won the 2010, 2012, 2014, and 2016 elections. The Democratic Party won a majority of seats again in 2018.

Post-term incumbency advantage. Tracking the power trajectories of election winners and runners-up also allows us to estimate the effect of winning an election on the likelihood of being in power after the constitutionally planned term has elapsed. When we measure this *post-term incumbency advantage*, we consider the number of years $\tilde{\tau}$ relative to the end of the constitutionally planned term, for $\tilde{\tau} \in [-3, 5]$, rather than the number of years relative to the election date. As before, we allow for power transition periods by measuring outcomes with a lag of 180 days. For instance, the U.S. Constitution mandates four-year presidential terms. To assess whether the winner of the 2008 U.S. presidential election was still in power after the end of the constitutionally planned term (at $\tilde{\tau} = 0$), we check who was the president on May 3rd, 2013, four years plus 180 days after the 2008 election took place. Outcomes for $\tilde{\tau} = -1$ and $\tilde{\tau} = 1$ are measured on May 3rd, 2012 and May 3rd, 2014, respectively.

Estimation. To estimate the incumbency advantage in our sample of national elections, our regressions include two observations for each election conducted in country i at time t :

election, as parties often refrain from naming a prospective head of government until after the results are known and coalition negotiations can begin.

¹⁴In case of a successful coup, we consider that no party is in power in parliament until new parliamentary elections are held.

the winner and the runner-up candidate or party, indexed by c .

We estimate the following equation for each year $\tau \in [-2, 10]$ relative to the election:

$$Y_{ic\tau} = \alpha + \beta_\tau T_{ict} + \gamma_\tau X_{ict} + \delta_\tau T_{ict} X_{ict} + \varepsilon_{ic\tau}. \quad (3.2)$$

In this equation, $Y_{ic\tau}$ is a dummy variable equal to one if party or candidate c is in power in relative year τ , and zero otherwise.¹⁵ The running variable X_{ict} is c 's margin of victory in the election held at time t . It is measured in terms of vote shares for presidential elections and seat shares for parliamentary elections.¹⁶ This running variable takes positive values for the winner ($T_{ict} = 1$) and negative values for the runner-up ($T_{ict} = 0$), and it is by construction symmetric around the threshold. The coefficient β_τ captures the effect of marginally winning an election on the likelihood of being in power τ years after the election. We further estimate the post-term incumbency advantage by using as outcome in equation (3.2) $Y_{ic\tilde{\tau}}$, a dummy variable equal to one if party or candidate c is in power $\tilde{\tau}$ years after the end of the term for which the winner was elected. For estimation, we use the nonparametric method of Calónico et al. (2020) and cluster standard errors at the election level.

To check that there is no imbalance in incumbency status at the RD threshold, we run placebo tests using outcomes measured one and two years before the election held at time t ($\tau = -1$ and $\tau = -2$). In the absence of electoral manipulation around the threshold, marginal winners should be no more likely than marginal runners-up to have been in power before the election.

Our sample for estimating equation (3.2) comprises 2,709 elections, including 753 presidential elections and 1,956 parliamentary elections. Appendix C.1 describes the rules we followed to determine which elections to include in the sample. 26% of these elections were held in OECD countries, 51% in democracies, and 65% were free and fair (according to V-Dem). In our heterogeneity analysis, we consider five regions: Africa (21% of the elections in the sample), Asia and Oceania (22%), Eastern Europe (11%), Latin America (27%), and Western Europe and North America (20%).

3.4.2 Next-election incumbency advantage

A key reason why election winners may remain in power beyond their constitutionally planned term is that winning an election may increase the probability of winning the subsequent one, conditional on that election taking place. We estimate this effect using the same RD design as the studies surveyed in Section 3.2:

$$Y_{ic,t+1} = \alpha + \beta T_{ict} + \gamma X_{ict} + \delta T_{ict} X_{ict} + \varepsilon_{ic,t+1}, \quad (3.3)$$

where $Y_{ic,t+1}$ is a dummy equal to one if party or candidate c who competed in the election at time t won the next election at time $t + 1$, and zero otherwise. β measures the *next-election incumbency advantage*, which may be driven by effects on both the probability of

¹⁵We provide additional information on the construction of our outcome variables in Appendix B.6. We verified the accuracy of our coding of outcome variables through several validity checks, described in Appendix B.7.

¹⁶When elections have several rounds, the margin of victory is measured using the results of the last round.

running in the next election and the probability of winning the next election conditional on running. We estimate equation (3.3) using a sample of 2,691 national elections, including 714 presidential elections and 1,977 parliamentary elections, for which we can define the outcome, the treatment, and the running variable.¹⁷ We impose the restriction that the time $t + 1$ election took place no more than ten years after the time t election. To illustrate how we define the outcome variable, consider the following examples:

- **1997 South Korean presidential election.** This election was won by Kim Dae-jung from the National Congress for New Politics (NCNP). During his term, the NCNP merged with the New People Party to form the Millennium Democratic Party (MDP). Kim Dae-jung retired at the end of his term, and the MDP was represented in the 2002 presidential election by Roh Moo-hyun. Moo-hyun won this election, defeating the candidate of the Grand National Party (GNP), Lee Hoi-chang. Thus, we consider that the winner of the 1997 election won the subsequent election.¹⁸
- **2017 German federal election.** The coalition between the Christian Democratic Union of Germany (CDU) and the Christian Social Union of Bavaria (CSU) won this election. However, having failed to secure an outright majority, the CDU/CSU engaged in a coalition with the runner-up party, the Social Democratic Party of Germany (SPD). Following the rules defined in Appendix B.5, we consider the CDU/CSU as a single entity because the CDU and CSU pledged to join forces before the election. By contrast, the SPD is considered as a distinct party because their union with the CDU/CSU was negotiated after the results. As the SPD won the next election in 2021, we consider that the runner-up of the 2017 election won the subsequent election.

To determine whether our estimates of β are driven by effects on the probability of running in the next election or by effects on electoral performance conditional on running, we also estimate equation (3.3) using a dummy indicating whether party or candidate c runs in the next election as the dependent variable. In addition, we estimate effects on the unconditional vote shares in subsequent presidential elections and the unconditional seat shares in subsequent parliamentary elections, coding the dependent variable as zero if a candidate or party does not run in the subsequent election.

¹⁷The sample size for this estimation is similar to the sample size in Section 3.4.1, but the composition of the two samples differs. On the one hand, the sample in Section 3.4.1 includes some elections that are not followed by another election, e.g., in all instances when a follow-up election was canceled by the incumbent. On the other hand, an election may be contained in the sample used in Section 3.4.1 but not the sample used in Section 3.4.2. This is the case whenever we cannot define the variable "party/candidate is in power 6 months after the end of the constitutionally-mandated term," which occurs for two reasons: when the term length is unknown (31 elections), and when the post-term outcome is measured after December 31st, 2023, but a follow-up election was held in the meantime (52 elections).

¹⁸In rare cases, a candidate may leave the party with which they ran and run with another in the next election. If the candidate's former party fields a new contender, we treat that new candidate as not aligned with the previous frontrunner. For instance, Lee Hoi-chang left the GNP after losing the 2002 South Korean election and ran as an independent in 2007 against the GNP candidate. Though the latter won the election, we consider that the runner-up of the 2002 election failed to win the subsequent election because Lee Hoi-chang lost the 2007 election.

3.5 Results

3.5.1 Testing the identification assumption

Our empirical strategy relies on the assumption that the only discrete change occurring at the threshold is the incumbency status of parties and candidates. This identification assumption would be violated, for instance, if incumbents could manipulate election results near the threshold to stay in power.

A common method to test for manipulation is the density test designed by McCrary (2008), which measures discontinuities in the distribution of the running variable at the cutoff. This test is not informative in our setting. Indeed, as we use exactly two observations for each election—one for the winner, on the right of the threshold, and one for the runner-up, on the left—with running variables equally distant from the threshold, the density of the running variable is symmetric by construction, with no discontinuous jump at the threshold (see Appendix Figure D.1).

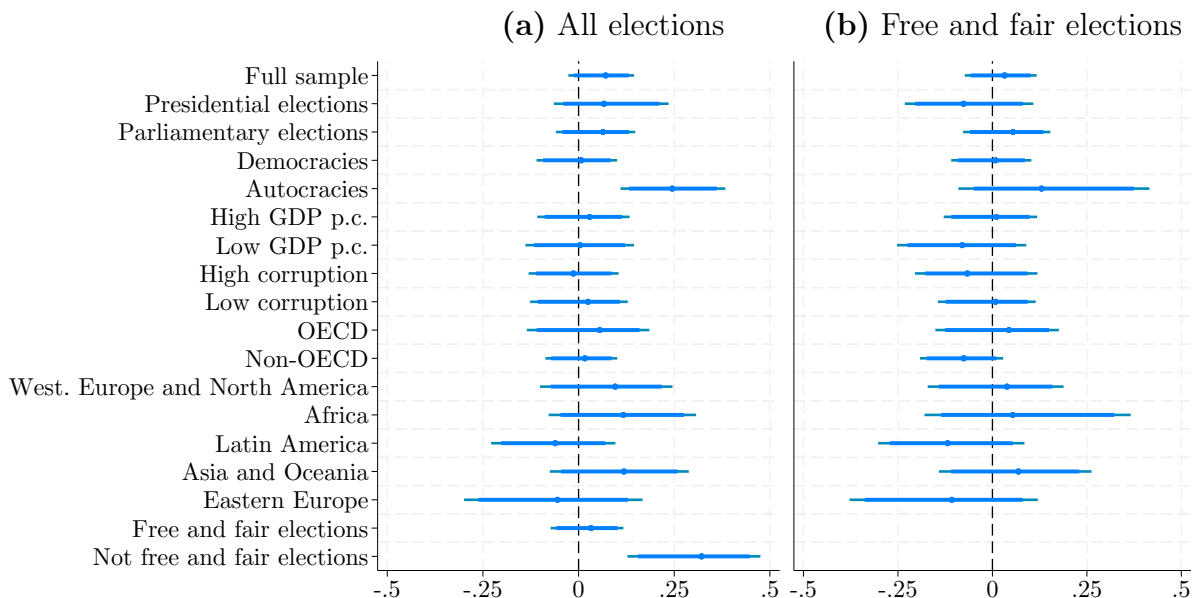
Studies leveraging a close-elections RDD to estimate incumbency effects at the subnational level generally consider elections held in democratic countries. They justify the identification assumption by pointing to the influence of exogenous factors such as weather conditions on election results, which generates quasi-randomness in the treatment assignment at the RD threshold (Eggers et al., 2015). While the same logic applies to national elections, the threat of manipulation looms larger in this context. Indeed, our sample includes elections in autocracies, where leaders may rig elections outright. To test whether incumbents systematically manipulate election results around the threshold, we implement the placebo test described in Section 3.4.1. Specifically, we examine whether marginally winning an election affects the probability of being in power six months *before* the election, corresponding to $\tau = -1$ in equation (3.2). Positive and significant effects would suggest that incumbents are able to manipulate election results to their benefit.

Panel (a) of Figure 3.4 presents placebo estimates for the full sample and various subsamples. We find no statistically significant effect in the full sample (p-value: 0.177), in presidential elections (p-value: 0.265), or in parliamentary elections (p-value: 0.402). There is no evidence of electoral manipulation by incumbents in most subsamples. However, we find positive and significant effects for elections classified as not free and fair by V-Dem and for those held in autocratic regimes, providing evidence of possible electoral manipulation in those settings. Appendix Figure D.2 shows similar results when we look instead at the candidates' likelihood of being in power 18 months before the election, corresponding to $\tau = -2$ in equation (3.2).¹⁹

In panel (b) of Figure 3.4, we display placebo estimates obtained when restricting the sample to elections classified as free and fair. We find no evidence of manipulation in that sample (p-value: 0.650) or in any of the corresponding subsamples. However, the classification of elections as free and fair may be endogenous to their results, with elections

¹⁹Appendix Figures D.3 and D.4 report the corresponding placebo tests for individual candidates in presidential elections, using candidates' likelihood of being in power before the election as outcome, for $\tau = -1$ (6 months before the election) and $\tau = -2$ (18 months before the election), respectively. We fail to detect a significant effect of electoral victories on this outcome except for elections held in autocracies (p-value: 0.003) and elections considered not free and fair (p-value: 0.006).

Figure 3.4: Placebo tests: Effect of election victory on being in power before the election



Notes: This figure shows RD estimates as well as 90% and 95% robust confidence intervals of the effect of winning an election on the likelihood of being in power six months before the election date, in our global sample of elections and in various subsamples. We include all elections, in panel (a), and restrict the sample to elections coded as free and fair by V-Dem, in panel (b). Our empirical strategy is described in Section 3.4.1.

marginally lost by the incumbent being more likely to be rated as free and fair. To avoid using a sample selection criterion that is endogenous to the electoral outcome, we therefore use the full sample of elections in our baseline estimation, and show in Appendix E that our results are robust to restricting the sample to free and fair elections. We also check that our results are robust to using the sample of elections *following* free and fair elections. This inclusion criterion should be less sensitive to the issue of endogenous coding while mostly selecting elections that are free and fair, given the strong autocorrelation (76%) in election fairness over time.

3.5.2 Estimates of the national incumbency advantage

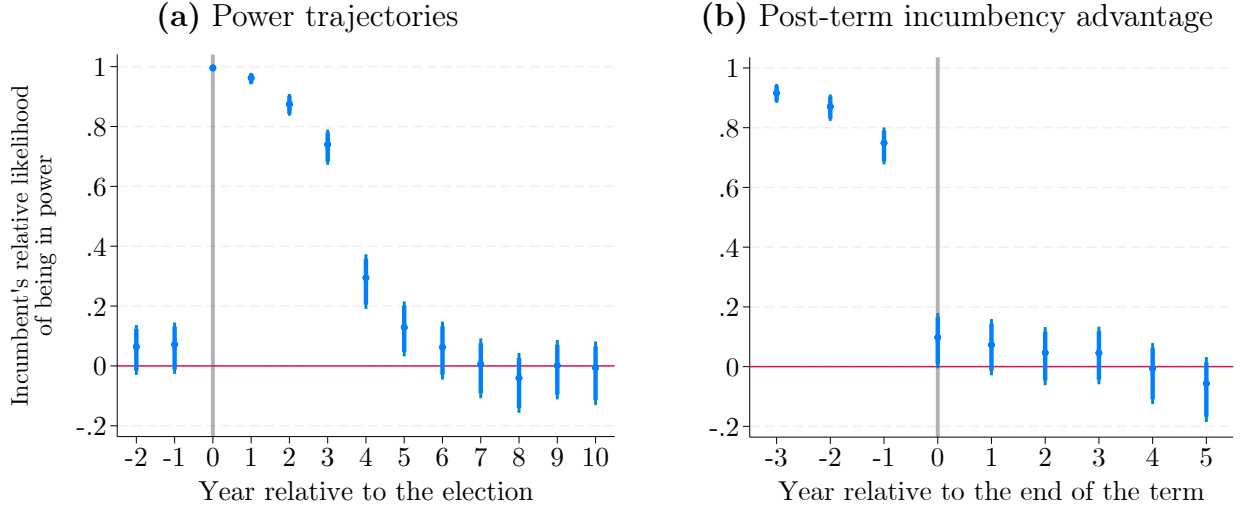
We now estimate the party-level incumbency advantage in our full sample including both presidential and parliamentary elections, using the outcomes defined in Section 3.4.1.

Figure 3.5(a) plots estimates of the national party-level incumbency advantage based on the comparison of the power trajectories of winners and runners-up for years $\tau \in [-2, 10]$ relative to the election, using equation (3.2). On average, winning an election increases a party's chances of being in power until approximately six years after the election, with the magnitude of the RD estimate dropping substantially between $\tau = 3$ and $\tau = 6$ years after the election. Naturally, this advantage is partly shaped by term lengths. The average term length in the full sample of elections is 4.5 years, with the majority of elections granting four-year (41%) or five-year (42%) terms.

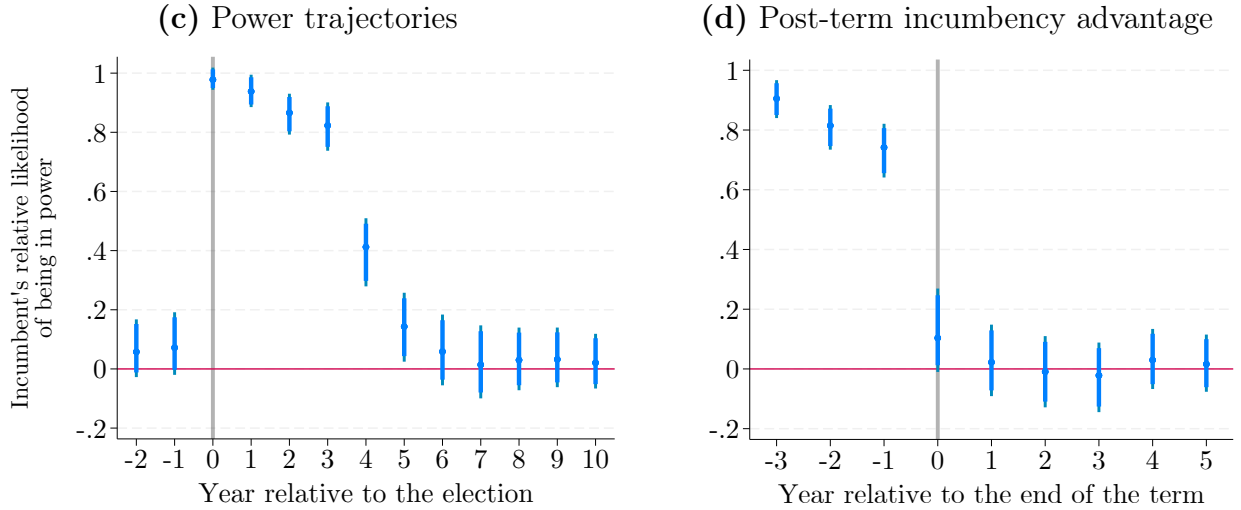
In Figure 3.5(b), we show the effect of a party's victory on the likelihood of being in

Figure 3.5: Incumbency advantage at the national level

Party-level estimates, all elections



Individual-level estimates, presidential elections



Notes: This figure reports RD point estimates as well as 90% and 95% robust confidence intervals for β_τ as defined in equation (3.2). Panels (a) and (b) report party-level estimates of the incumbency advantage across all elections, while panels (c) and (d) report individual-level estimates of the incumbency advantage in presidential elections. In panels (a) and (c), $\tau \in [-2, 10]$ represents years relative to the election, with a lag of 180 days. In panels (b) and (d), $\tilde{\tau} \in [-3, 5]$ is relative to the scheduled end of term.

power for each year $\tilde{\tau} \in [-3, 5]$ relative to the scheduled end of term. The positive coefficient for year $\tilde{\tau} = 0$ indicates that there is a global post-term incumbency advantage at the party level: winning an election increases the likelihood of being in power six months after the theoretical end of the term by 10 p.p., corresponding to a 27% increase relative to the average probability for close runner-up parties. This effect is significant at the 10% level

(p-value: 0.065). Appendix Figure D.5(a) shows the corresponding regression discontinuity plot. The gray dots are sample means of the likelihood of being in power 6 months after the scheduled end of term across 15 quantile-spaced bins on each side of the threshold, with a positive jump at the threshold. We also observe positive effects on the likelihood of being in power 18 months after the end of term and in the two subsequent years, as reported in Figure 3.5(b) but these effects are smaller and non-significant.

In Figures 3.5(c) and (d), we report estimates of the individual-level incumbency advantage in presidential elections.²⁰ The estimate for $\tilde{\tau} = 0$ is positive and significant at the 10% level (+10 p.p., p-value: 0.069) but the estimates for $\tilde{\tau} = 1$ and subsequent periods are again closer to zero and non-significant. Overall, the analysis of party-level and individual-level estimates yields evidence of a positive but short-lived average incumbency advantage at the national level. Appendix Tables D.1, D.2, D.3, and D.4 report all the estimates shown in Figure 3.5 for the full sample and various subsamples.

3.5.3 Heterogeneity and robustness

Our meta-analysis in Section 3.2 highlighted the wide variation in subnational estimates of the incumbency advantage across regions and levels of GDP per capita, corruption, and democratic quality. We now explore heterogeneity in the national incumbency advantage along similar dimensions.

Figure 3.6(a) presents the results from this heterogeneity analysis for party-level estimates across all elections. In this figure, each dot represents a separate RD estimate for $\tilde{\tau} = 0$, the effect of narrowly winning an election on a party's probability of being in office six months beyond the expected end of term for a given subsample. Appendix Figure D.6 reports the corresponding estimates for $\tilde{\tau} = 1$ and $\tilde{\tau} = 2$ (i.e., 18 and 30 months after the scheduled end of term, respectively). Together, Figure 3.6(a) and Appendix Figure D.6 paint a consistent picture: the heterogeneous effects we observe across different subsamples persist over an extended timeframe, and are not contingent on the specific choice of $\tilde{\tau}$ to measure the post-term incumbency advantage.

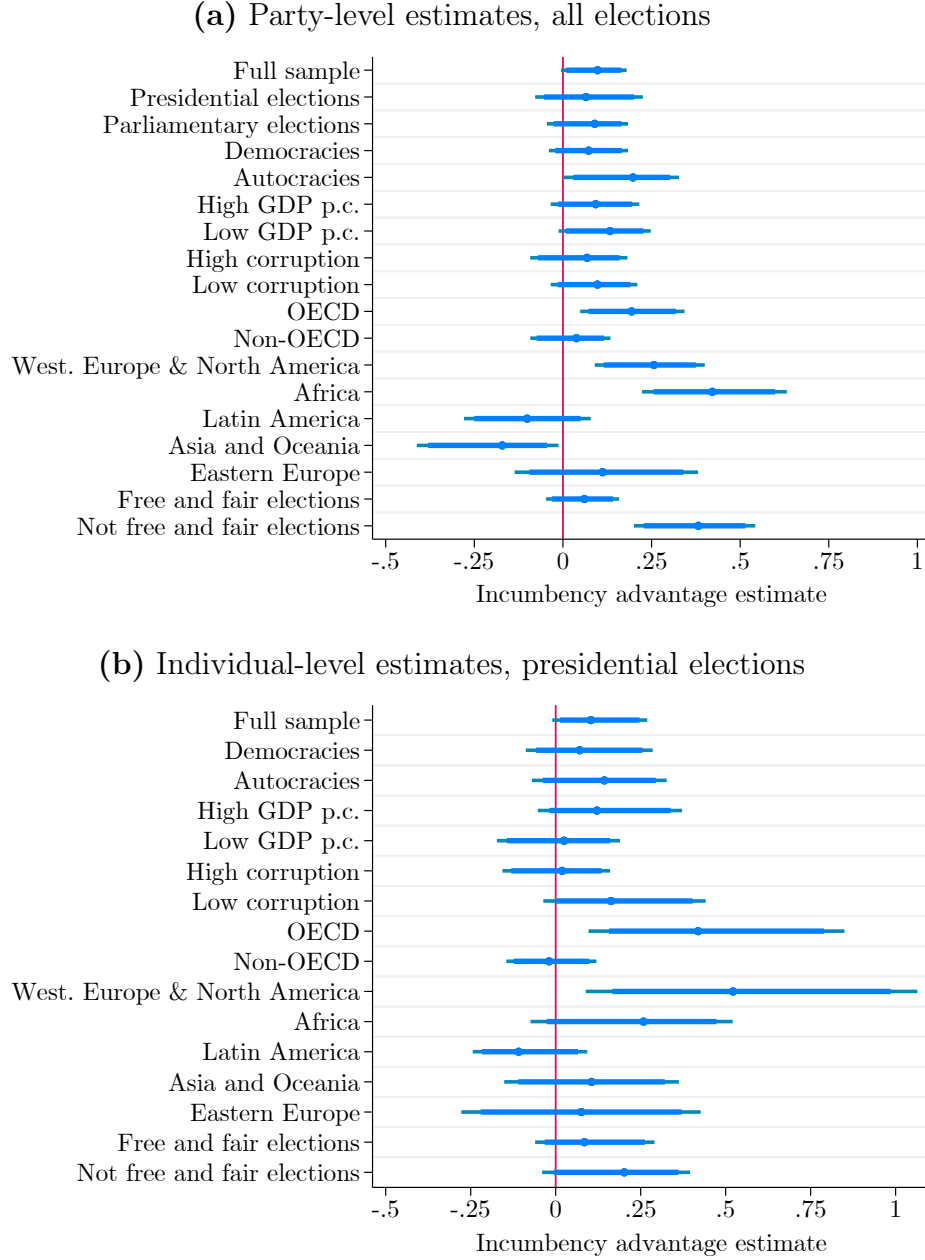
Mirroring our meta-analysis findings on the incumbency advantage at the subnational level, national elections in OECD countries are associated with a much larger incumbency advantage than those in non-OECD countries. In OECD countries, the party-level incumbency advantage is substantial (19 p.p.). This estimate is significant at the 1% level and corresponds to 55% of the likelihood of close runners-up of being in power six months after the scheduled end of term. Appendix Figures D.6 and D.7 show that this effect is persistent: incumbent parties in OECD countries enjoy a higher probability of being in power up to three-and-a-half years after the scheduled end of their term. By contrast, incumbency effects outside OECD countries are much smaller and non-significant.

Estimates of the party-level incumbency advantage in different regions reveal even more striking differences.²¹ Elections in North America and Western Europe are associated with

²⁰As explained in Section 3.4.1, data limitations prevent us from estimating the individual-level incumbency advantage in parliamentary elections that lead to the designation of a member of the executive (see footnote 13).

²¹Appendix Figure D.5 shows the regression discontinuity plots separately for elections held in each region, in panels b to f.

Figure 3.6: Heterogeneity in the post-term incumbency advantage



Notes: This figure plots RD estimates as well as 90% and 95% robust confidence intervals of the effect of winning an election on the likelihood of being in power six months after the scheduled end of term, in our global sample of elections and in various subsamples. Panel (a) reports party-level estimates across all elections while panel (b) reports individual-level estimates in presidential elections. Our empirical strategy is described in Section 3.4.1.

a large positive incumbency advantage (+26 p.p.). We also find a positive, albeit non-significant, incumbency advantage in Eastern European countries (+11 p.p.). Other regions are characterized instead by sizable incumbency disadvantages. In Asia and Oceania, winning an election reduces the likelihood of being in power six months after the scheduled end of

term by 17 p.p. on average, implying that close winners are 37% less likely to be in power after their term was scheduled to end than close runners-up. This disadvantage persists 18 months and 30 months after the election (Appendix Figure D.6). Latin American elections are also associated with an incumbency disadvantage (-10 p.p.), although this estimate is not statistically significant.

These differences across regions echo those found in Section 3.2. However, in contrast to the incumbency disadvantage found in Africa at the subnational level, this region is the one in which we find the largest national-level incumbency advantage. There, marginally winning a national election increases a party’s likelihood of staying in power beyond the scheduled end of term by a staggering 42 p.p., which is significant at the 1% level and amounts to almost tripling the probability of being in power relative to a close runner-up (+191%). We also find positive effects in autocracies (+20 p.p., significant at the 5% level), elections deemed not free and fair (+38 p.p., significant at the 1% level), and countries with low GDP per capita (+13 p.p., significant at the 10% level) or high corruption levels (+7 p.p., which is non-significant).²² Thus, while we documented monotonic relationships at the subnational level between the incumbency advantage on the one hand and economic development, corruption, and democratic quality on the other hand, our estimates of the national incumbency advantage paint a much more nuanced picture.

Appendix Figure E.2 shows similar effects when restricting the sample to elections that are free and fair or those that follow free and fair elections. Our results are also robust to restricting the sample to UN member states, excluding countries with fewer than 500,000 inhabitants (see Appendix Figure D.8), and including election-year fixed effects in the estimation (Appendix Figure D.9). Appendix Figure D.10 illustrates how little our main estimate for the full sample of elections varies when we alternatively include no fixed effects, election year fixed effects, region fixed effects, and region $\times$ election year fixed effects in the estimation. Finally, Appendix Figure D.11 compares our main results to those obtained using a RD bandwidth equal to half of the MSE-optimal bandwidth from Calonico et al. (2020); the latter estimates across different subsamples are more noisy but largely similar.

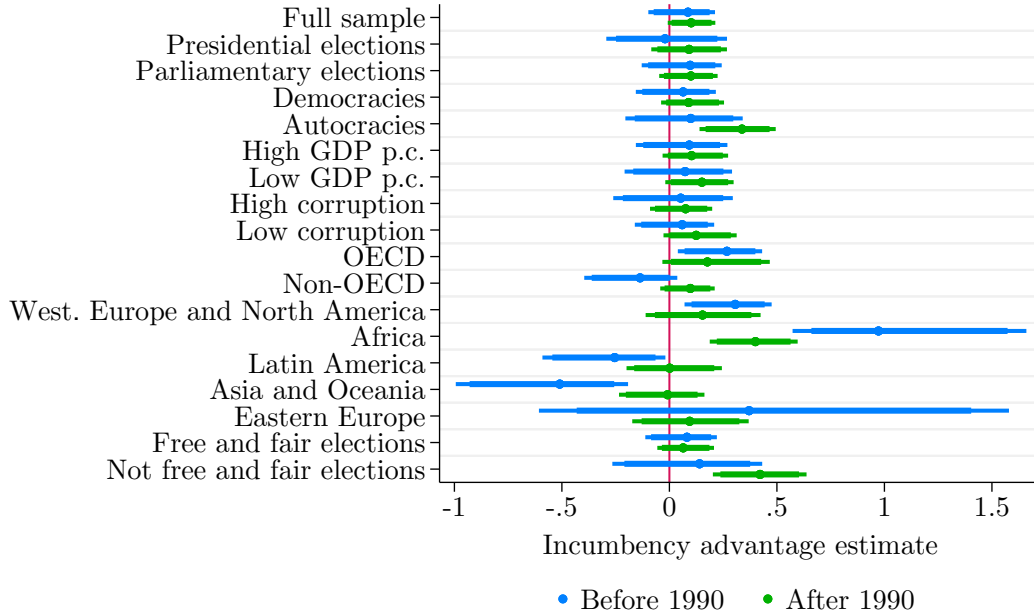
Figure 3.7 reports effects separately for elections held before and after 1990. While the main insights from Figure 3.6 hold throughout both time periods, the incumbency advantage has generally fallen in magnitude since 1990 in several subsamples. In the regions where the advantage is positive on average (Africa, Western Europe and North America), this advantage has declined slightly over time.²³ In regions where we observe a disadvantage on average (Asia and Oceania, Latin America), this disadvantage has converged towards zero since 1990. At odds with this general trend, the incumbency advantage in autocracies and that in non free and fair elections has become larger after 1990.

In Appendix Figures D.12(a) and D.12(b), we report party-level estimates of the incumbency advantage separately for presidential and parliamentary elections. The large effects observed in OECD countries are primarily driven by presidential elections. For parliamentary elections, the differences between OECD and non-OECD countries are modest

²²Note that there is substantial overlap between these subsamples. For instance, African elections constitute 39% of those held in autocracies, 41% of those classified as not free and fair, 33% of elections in high-corruption countries, and 37% of those in low-income countries.

²³This is consistent with evidence from Jacobson (2015) and Guriev et al. (2025) documenting the recent decline of the incumbency advantage in U.S. congressional elections.

Figure 3.7: Incumbency advantage before and after 1990



Notes: This figure plots RD estimates as well as 90% and 95% robust confidence intervals of the post-term incumbency advantage for different time periods. For each estimate, we show the incumbency advantage for the full sample of elections and the incumbency advantage for the subset of elections deemed free and fair by V-Dem.

and the effect is not statistically significant in either group. The incumbency advantage in Africa and that in North America and Western Europe are present and statistically significant in both presidential and parliamentary elections.

Finally, Figure 3.6(b) presents heterogeneity results for individual-level estimates of the incumbency advantage in presidential elections. Overall, incumbent presidents have a statistically significant advantage in presidential elections held in OECD countries (+42 p.p.) and in Western Europe and North America (+52 p.p.). This effect is also large in Africa (+26 p.p.) and in elections considered not free and fair, but both of these estimates fall short of statistical significance.

3.6 Mechanisms

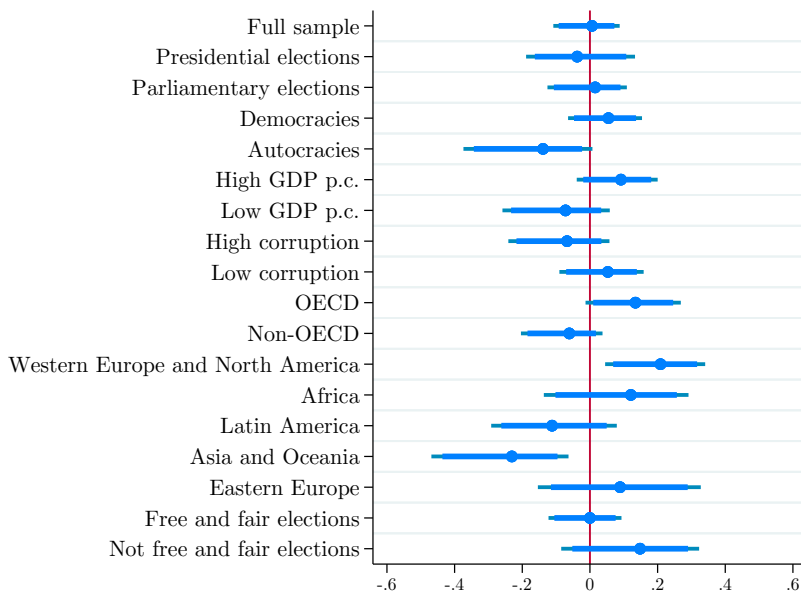
Several channels may underlie the incumbency effects we observe. First, winning a national election might influence the likelihood of competing in the subsequent one. Second, incumbency may improve electoral performance conditional on running. This includes the possibility that incumbents might tilt the electoral playing field in their favor, preventing elections from being free and fair to retain power. Third, power holders could manipulate the electoral cycle by calling early elections when conditions are favorable to them or by postponing or canceling elections altogether. In this section, we build on our RD framework to provide evidence on these potential mechanisms.

3.6.1 Next-election incumbency advantage

To assess the first two channels, we estimate the effect of marginally winning an election on the probability of running in and winning the subsequent one.

Impact on performance in the next election. We first estimate the effect of winning a national election on the joint probability that parties compete in and win the next election, using the approach described in Section 3.4.2. Our estimates are reported in Figure 3.8. The global incumbency advantage uncovered for the full sample of elections in Figure 3.5(b) does not stem from this channel. Indeed, on average, winning an election only increases a party’s probability of victory in the subsequent election by 1 p.p., which is not statistically significant. We also detect no effect on winning the next election in the subsamples of presidential or parliamentary elections.²⁴

Figure 3.8: Next-election incumbency advantage, party-level estimates



Notes: This figure plots RD estimates as well as 90% and 95% robust confidence intervals of the effect of winning an election on a party’s joint likelihood of running in and winning the next election, in our global sample of elections and in various subsamples. Our empirical strategy is described in Section 3.4.2.

However, we find substantial incumbency effects in several subsamples. In North America and Western Europe, close winners are 21 p.p. more likely to win the next election relative to close runners-up. This estimate is significant at the 5% level and corresponds to a 58% increase in the probability of winning. Similar effects are visible in other subsamples characterized by strong democracies, namely OECD countries (13 p.p.), countries with a high GDP per capita (9 p.p.), and those with low corruption levels (5 p.p.), although the

²⁴Appendix Table D.5 reports the coefficients displayed in Figure 3.8, as well as the corresponding robust standard errors and p-values. Our results are robust to including election-year fixed effects (see Appendix Figure D.13).

last two estimates are not statistically significant.²⁵ Instead, in Asia and Oceania as well as in Latin America, incumbents suffer from substantial electoral disadvantages (of -23 p.p., significant at the 1% level, and -11 p.p., non-significant, respectively). Appendix Figure D.14 reports estimates of the next-election incumbency advantage for candidates in presidential elections. Once again, we find the strongest effects in OECD countries and in North America and Western Europe, where candidates who marginally win an election are much more likely to win the subsequent presidential election than marginal losers.

While the next-election incumbency effects in Figure 3.8 often mirror the post-term incumbency effects reported in Figure 3.6, some subsamples exhibit stark differences, with negative effects on the probability of winning the next election but positive effects on the likelihood of being in power beyond the constitutionally planned term. This is particularly true in less democratic settings. In particular, autocracies exhibit a strong next-election incumbency disadvantage (-14 p.p.) but a large post-term incumbency advantage (+20 p.p.).

Impact on competing in the next election. The incumbency effects reported in Figure 3.8 compound two effects: an effect of incumbency on a party's likelihood of running in the next election, and an effect on electoral performance in the next election, conditional on running.

The effect on running in the next election may result from scare-off effects, whereby potential challengers are deterred from running against strong incumbents (Cox and Katz, 1996; Levitt and Wolfram, 1997; Hall and Snyder, 2015; Ban et al., 2016). Furthermore, winning parties may have a higher probability of survival, further contributing to a discontinuous increase in the probability of running in the next election (Dinas et al., 2015).²⁶ Conversely, in contexts where incumbents expect to be defeated, this might create a discontinuous drop at the threshold in the probability of running again.

Various channels could also affect the electoral performance of incumbents conditional on running again in the next election. For example, officeholders' electoral prospects may be enhanced if they allocate public funds strategically to increase electoral support, direct funds to local political organizations and campaigns, and exert influence over media outlets. Incumbents also typically benefit from greater name recognition and political experience accumulated in office than runners-up. Yet, holding office also subjects them to retrospective accountability: weak economic performance during their tenure could erode or even overturn these advantages.

To distinguish between these two components, we first estimate the effect of winning an election on the likelihood of competing in the next one in Appendix Figures D.15(a). We then separately report unconditional effects on seat shares in parliamentary elections and vote shares in presidential elections (with seat and vote shares coded as zero if the party does not run at time $t + 1$) in Appendix Figures D.15(b) and D.15(c), respectively.

²⁵These subsamples substantially overlap. For example, Western European and North American elections represent 66% of those in the OECD, 36% of those in countries with high levels of GDP per capita, and 39% of those in low-corruption countries.

²⁶Losing parties may also be more likely to change their names, which could make them harder to track and lead us to erroneously code them as not competing in the next election. To mitigate this risk, we carefully linked parties across elections and identified party name changes using Wikidata (see Appendix B.2 for details).

The estimates in Appendix Figure D.15(a) are generally small and non-significant. This is due to the fact that in national elections, almost all marginal winning parties (94%) and marginal runner-up parties (95%) compete in the next election. Only elections in Latin America stand out: there, incumbency leads to a marginally significant (p-value: 0.094) 7 p.p. drop in the likelihood of running in the next election. This effect contributes to the incumbency disadvantage observed in this region, suggesting that strategic withdrawals may prevent incumbent parties from contesting future elections. Overall, the small effect sizes in Appendix Figure D.15(a) suggest that the next-election incumbency advantage is primarily driven by effects on subsequent electoral performance conditional on running.

3.6.2 Fairness and timing of elections

In this section, we explore whether incumbents manipulate the fairness and the timing of follow-up elections. Theoretically, such mechanisms could affect both the *next-election advantage*, if incumbents time the next election strategically, change the conditions in which electoral campaigns are fought, or directly manipulate the results; as well as the *post-term advantage*, if incumbents postpone or cancel follow-up elections outright. Evidence on the relative importance of these mechanisms could, in turn, explain the wedge between post-term and next-election effects that we observe in some regions.

Manipulation of the electoral playing field. Incumbents can tilt elections in their favor by changing the conditions in which electoral campaigns are fought or by directly manipulating the results. To explore this mechanism, we re-estimate equation (3.3) using as an outcome $Y_{ic,t+1}$, a dummy variable equal to 1 if party or candidate c in the election held at time t won the subsequent election (held at time $t + 1$) and that subsequent election was not free and fair. Our estimates, displayed in Appendix Figure D.16, provide suggestive evidence that the manipulation of the electoral playing field contributes to the next-election incumbency advantage in less democratic contexts. Indeed, we find large positive effects of incumbency on the probability of winning a subsequent non free and fair election in autocracies (8 p.p., which is not significant) and for elections that were already not free and fair at time t (18 p.p., which is significant at the 5% level).

Manipulation of the electoral schedule. We then explore the extent to which incumbents manipulate the timing of elections. Such manipulation could contribute to the *next-election advantage*, if the incumbents strategically time elections to take place in favorable periods; and it may directly increase the *post-term advantage* simply because postponing or canceling the next election implies that the incumbent will remain in power beyond the scheduled end of their term.

In Appendix Figure D.17, we examine whether elections take place earlier or later than their constitutionally planned date, across settings with varying levels of democratic quality. We distinguish between subsequent elections held on schedule (within six months of the constitutionally planned end of term), subsequent elections held early (more than six months before the end of term), and subsequent elections held late (more than six months after the end of term) or not at all.

This analysis shows that the electoral calendar is most often manipulated in less democratic countries, and that manipulation involves both advancing and postponing elections. In countries in the bottom quintile of democratic quality, only 56% of elections were followed by another election held on schedule, compared to 74% in the third quintile and 74% in the top quintile of democratic quality. In those countries, 17% were held late or not at all, as opposed to only 7% of elections in the third quintile and 2% in the top quintile of democratic quality. By postponing elections, incumbents in the least democratic settings artificially extend their tenure and thereby secure an advantage. In democratic countries, elections are seldom postponed but snap elections are common: 24% of elections in the top quintile of democratic quality are held early, which is comparable to the fraction of elections held early in the least democratic countries (26%). Countries with intermediate levels of democratic quality are those that most closely follow their constitutionally planned electoral calendar: indeed, many of these countries are located in regions (Asia and Oceania, Latin America) where we estimate mostly null or negative electoral advantages. As a result, early elections might harm incumbents, while institutional constraints may be sufficiently robust to prevent the postponement or outright cancellation of elections.

3.6.3 Decomposing the incumbency advantage

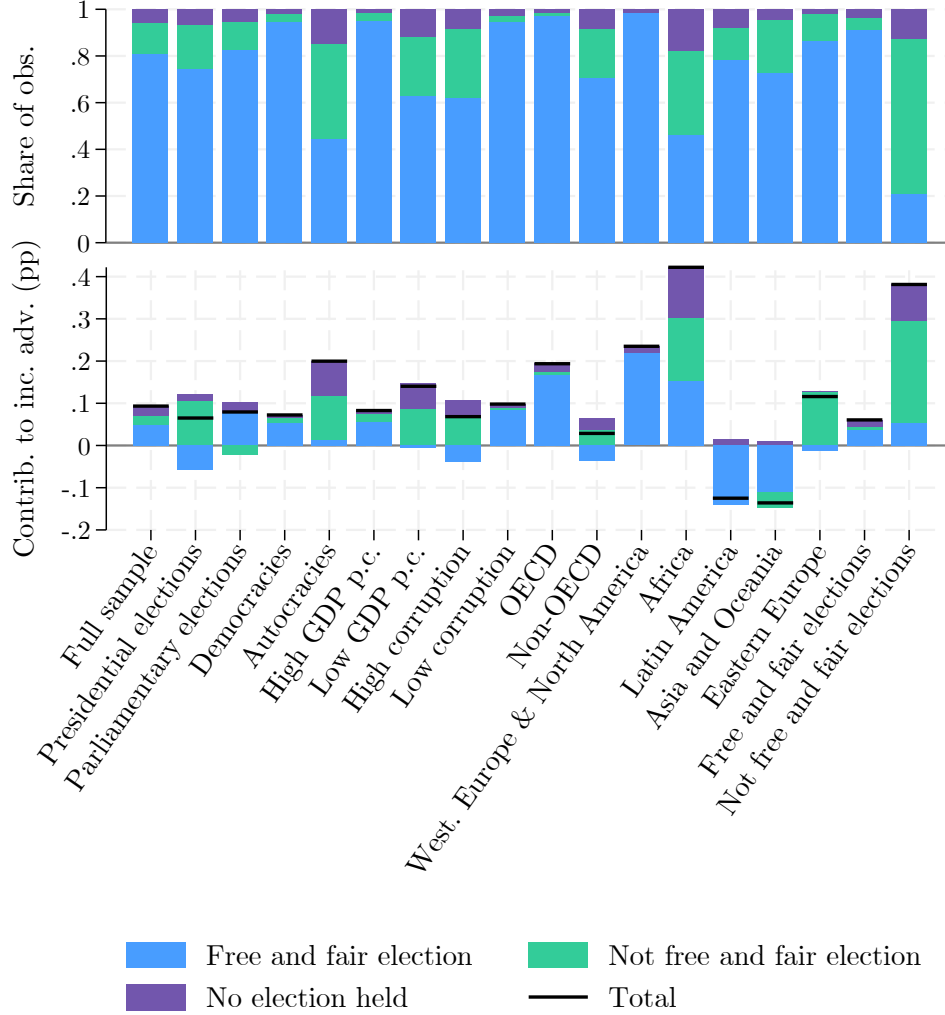
The evidence thus far shows that the incumbency effects in Figure 3.6 stem from multiple forces: an increased likelihood of winning the next election in a fair manner; in an unfair manner; or postponing or canceling the election altogether. To quantify the relative importance of these mechanisms, we conduct a simple decomposition exercise.

When estimating equation (3.2), thus far we have defined our outcome $Y_{ic\tilde{\tau}}$ as a dummy variable equal to one if candidate or party c (the winner or runner-up in election t) was in power in country i in year $\tilde{\tau}$, measured relative to the scheduled end of term. This outcome can be decomposed as the sum of three dummies, all measured in year $\tilde{\tau}$: (i) $Y_{ic\tilde{\tau}}^{\text{no election}}$, equal to one if c is in power and no election has been held since t ; (ii) $Y_{ic\tilde{\tau}}^{\text{free and fair}}$, equal to one if c is in power and at least one election has been held since t , the last of which was free and fair; and (iii) $Y_{ic\tilde{\tau}}^{\text{not free and fair}}$, equal to one if c is in power and at least one election has been held since t , the last of which was not free and fair. We can estimate equation (3.2) using $Y_{ic\tilde{\tau}}^{\text{no election}}$, $Y_{ic\tilde{\tau}}^{\text{free and fair}}$, and $Y_{ic\tilde{\tau}}^{\text{not free and fair}}$ as outcomes, allowing us to recover estimates of the effect of marginally winning an election on these outcomes: $\hat{\beta}_{\tilde{\tau}}^{\text{no election}}$, $\hat{\beta}_{\tilde{\tau}}^{\text{free and fair}}$, and $\hat{\beta}_{\tilde{\tau}}^{\text{not free and fair}}$. The overall post-term incumbency effect, $\hat{\beta}_{\tilde{\tau}}$, is the sum of these three effects, providing us with a decomposition of the overall incumbency effect into three terms: one reflecting the postponement or cancellation of elections, one capturing the incumbency advantage in free and fair elections, and one capturing the advantage in elections that are not free and fair.

We show the results of this decomposition in Figure 3.9, where outcomes are measured six months after the end of the constitutionally planned term. We also report the share of elections in the RD bandwidth for which, six months after the scheduled end of term, (i) no subsequent election had been held; (ii) at least one subsequent election had been held, the last of which was free and fair; and (iii) at least one subsequent election had been held, the last of which was not free and fair.

In the subsamples of OECD countries, countries in North America and Western Europe as well as Latin America, and countries with high GDP per capita and low corruption,

Figure 3.9: Decomposition of the incumbency advantage



Notes: In the top panel, we report, for our global sample of elections and for various subsamples, the share of elections in the RD bandwidth for which, six months after the constitutionally planned end of the term, (i) no subsequent election had been held; (ii) at least one subsequent election had been held, the last of which was free and fair; and (iii) at least one subsequent election had been held, the last of which was not free and fair. In the bottom panel, we decompose post-term incumbency effects using the procedure described in Section 3.6.3. The overall post-term incumbency effects—measured six months after the constitutionally planned end of the term and represented with horizontal black lines—are expressed as the sum of three components: one reflecting the postponement or cancellation of elections (“No election held”), one capturing the incumbency advantage in free and fair elections (“Free and fair election”), and one capturing the advantage in elections that are not free and fair (“Not free and fair election”). To categorize elections as free and fair or not, we rely on V-Dem. We exclude from the sample observations for which we could not compute an outcome because an election was not categorized by V-Dem. In all regressions, we use the optimal bandwidth selected by the procedure of [Calonico et al. \(2020\)](#) for the computation of the overall incumbency effect.

which mostly include strong democracies, free and fair elections are almost always held by the end of constitutionally planned terms (or shortly thereafter), and incumbency effects in these contexts reflect the advantage that incumbents benefit from in free and fair elections. However, in subsamples with lower levels of democratic quality (e.g., autocratic countries,

non-OECD countries, African countries, and countries with low GDP per capita and high corruption), it is common for close elections to be followed by non free and fair elections or for subsequent elections to be postponed, especially in non-democratic contexts. These substantially inflate the incumbency advantage, as estimates of $\hat{\beta}_{\tau}^{\text{no election}}$ and $\hat{\beta}_{\tau}^{\text{not free and fair}}$ are almost always positive and of a similar order of magnitude as $\hat{\beta}_{\tau}^{\text{free and fair}}$.

Appendix Figures D.18 and D.19 show the corresponding results for outcomes measured 18 months and 30 months after the end of the constitutionally scheduled term, with similar patterns to those of Figure 3.9. Incumbency effects in democratic settings (e.g., the subsamples of democracies, high-income countries, and OECD countries) are driven by a large advantage in free and fair elections, while those in less democratic settings (e.g., the subsamples of autocracies and elections not free and fair at time t) reflect a boost in the performance of incumbents in follow-up elections that are not free and fair.

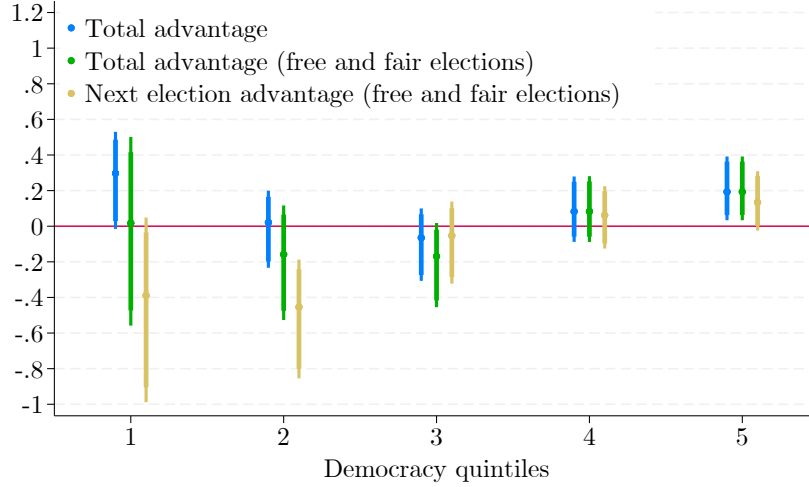
The evidence in Figure 3.9 could be partly driven by elections classified as not free and fair at time t . To address this concern, in Appendix Figures E.4 and E.5, we report the same decomposition exercise considering only free and fair elections at time t , or elections at time t that follow a free and fair election, respectively. While electoral manipulation at time $t + 1$ plays a slightly diminished role in these subsamples, these two figures corroborate the main insights from Figure 3.9: even when we focus on the winners of free and fair elections, the incumbency advantage in some subsamples (e.g., the subsamples of low-income countries or countries in Africa) continues to be partly driven by electoral manipulation, in the form of incumbents either ensuring that follow-up elections are not free and fair, or postponing or canceling follow-up elections altogether.

3.6.4 Democratic quality and the national-level incumbency advantage

Overall, the decomposition in Figure 3.9 helps reconcile our measures of the incumbency advantage in national elections with the results of our meta-analysis. At the subnational level, incumbents enjoy a much stronger advantage in contexts characterized by higher levels of economic and political development. By contrast, this pattern does not hold at the national level, where we find similar incumbency effects across contexts with different levels of GDP per capita and corruption. This reflects the ability of national leaders to postpone elections or to hold elections that are not free and fair. Indeed, the contribution of free and fair elections to the national incumbency advantage, $\beta_{\tau}^{\text{free and fair}}$, mirrors the pattern observed in the meta-analysis, with sizable advantages in contexts with high incomes, low corruption levels, and strong democracies, and muted or negative effects in poorer, more corrupt, and less democratic environments. However, in less democratic contexts, large advantages from non-free-and-fair elections and from postponing elections raise the overall incumbency advantage to levels comparable to those observed in strong democratic settings.

Figure 3.10 further illustrates the relationship between democratic quality and the incumbency advantage. There, we show three measures of the incumbency advantage for different quintiles of the democracy index: (i) the post-term incumbency advantage, measured using equation (3.2); (ii) the post-term incumbency advantage when restricting the sample of time t elections to those deemed free and fair by V-Dem; and (iii) the

Figure 3.10: Incumbency advantage measures by level of democracy



Notes: This figure reports RD estimates as well as 90% and 95% robust confidence intervals of the incumbency advantage for different quintiles of a democracy index (extracted from V-Dem). For each quintile, we show (1) the post-term incumbency advantage for the full sample of elections; (2) the post-term incumbency advantage for the subset of time t elections coded as free and fair by V-Dem; and (3) the next-election incumbency advantage for the subset of time t elections coded as free and fair by V-Dem. Democracy levels are measured as an average of V-Dem’s five corresponding variables: deliberative, egalitarian, liberal, participatory, and electoral democracy.

next-election incumbency advantage, measured using equation (3.3) and again restricting the sample to free and fair elections. In the most democratic contexts, the three measures almost coincide, reflecting an incumbency advantage arising from free and fair elections. As democratic quality deteriorates, the next-election incumbency advantage drops, reaching strong disadvantages in the least democratic contexts. However, in these contexts, manipulation of the timing and competitiveness of follow-up elections allows incumbents to retain power beyond the scheduled end of their term, despite the lack of positive incumbency effects on subsequent performance in free and fair elections, when they take place. Marginal election winners are least likely to stay in power beyond their term in countries with intermediate levels of democracy. In these contexts, voters regularly sanction incumbents in elections, and democratic institutions are strong enough that leaders cannot easily cling to power through undemocratic means. Overall, these forces yield a U-shaped relationship between democratic quality and the post-term incumbency advantage.

3.7 Conclusion

This paper provides the first causal estimates of the incumbency advantage at the country level across a global sample of national elections. We document a positive but short-lived average incumbency effect at this level: election winners are more likely to remain in office beyond the scheduled end of their term, yet this advantage dissipates within a few years. Importantly, the magnitude and direction of incumbency effects vary substantially across contexts. OECD countries and African countries are characterized by large positive incumbency effects that last for several years after the scheduled end of term, while

incumbency disadvantages emerge in other regions. These patterns are closely associated with varying levels of economic development, corruption, and democratic quality.

Our results imply that different factors contribute to the electoral advantage enjoyed by incumbents. In consolidated democracies, the national-level incumbency advantage primarily operates through improved electoral performance in free and fair elections, mirroring the effects at the subnational level established in our meta-analysis. In settings with lower democratic quality, incumbents suffer an electoral disadvantage when free and fair elections take place, echoing again the results from the meta-analysis, but the manipulation of the timing and the fairness of elections enables them to remain in power beyond the end of their term, translating into an overall incumbency advantage. These heterogeneous patterns underscore the need to consider both electoral and institutional dynamics when evaluating how incumbency effects affect political competition worldwide.

By extending the analysis of incumbency effects to the national level, we offer novel evidence on the determinants of political turnover and bring new insights to ongoing debates about the resilience and the future of democracy globally. In contexts where the incumbency advantage is sustained by institutional manipulation, the prospects for genuine political competition remain fragile. Conversely, in established democracies, where the incumbency advantage stems primarily from improved electoral prospects, incumbency effects may contribute to strengthening political selection and accountability, as political careers remain attractive to potential entrants and incumbents can expect that good performance will help them secure re-election democratically.

Finally, our results offer a different perspective on the evolution of democracy across countries since 1945. Estimates of the national-level incumbency advantage can potentially be interpreted as a measure of democratic quality and regime stability that does not rely on expert ratings or subjective assessments. Whether we consider the full sample of presidential and parliamentary elections, or only elections classified as free and fair, we find that the national-level incumbency advantage has generally fallen over the past decades, including the period 2005–2023. The next-election incumbency advantage has also remained stable during the period. Thus, despite the mounting evidence of democratic backsliding in the twenty-first century (illustrated in our data by the recent rise in the share of non free and fair elections), this trend has not yet materialized into more entrenched incumbents on a global scale. Future research should explore how secular fluctuations in democratic quality ultimately affect political turnover and regime durability, as well as implications for economic growth and voter welfare.

Bibliography

- Ade, Florian, Ronny Freier, and Christian Odendahl**, “Incumbency Effects in Government and Opposition: Evidence From Germany,” *European Journal of Political Economy*, December 2014, 36, 117–134.
- Afzal, Madiha**, “Voter Rationality and Politician Incentives: Exploiting Luck in Indian and Pakistani Elections,” 2007.
- Aguirre, Rodrigo and Matias Brum**, “Disentangling the Incumbent’s Advantage: New Estimates of Partisan and Candidate Separate Effects,” 2023.
- Anagol, Santosh and Thomas Fujiwara**, “The Runner-Up Effect,” *Journal of Political Economy*, 2016, 124 (4), 927–991.
- Anderson, Leslie, Michael Lewis-Beck, and Mary Stegmaier**, “Post-Socialist Democratization: A Comparative Political Economy Model of the Vote for Hungary and Nicaragua,” *Electoral Studies*, 2003, 22 (3), 469–484.
- Ansolabehere, Stephen, James M. Snyder, and Charles Stewart**, “Old Voters, New Voters, and the Personal Vote: Using Redistricting to Measure the Incumbency Advantage,” *American Journal of Political Science*, 2000, 44 (1), 17–34.
- Archambault, Jerome and Stanley L. Winer**, “Political Competitiveness, Regression Discontinuity and the Incumbency Effect,” *SSRN Electronic Journal*, 2023.
- Ariga, Kenichi**, “Incumbency Disadvantage Under Electoral Rules With Intraparty Competition: Evidence From Japan,” *The Journal of Politics*, July 2015, 77 (3), 874–887.
- , **Yusaku Horiuchi, Roland Mansilla, and Michio Umeda**, “No Sorting, No Advantage: Regression Discontinuity Estimates of Incumbency Advantage in Japan,” *Electoral Studies*, September 2016, 43, 21–31.
- Aruoba, S. Boragan, Allan Drazen, and Razvan Vlaicu**, “A Structural Model of Electoral Accountability,” *International Economic Review*, 2019, 60 (2), 517–545.
- Ashworth, Scott**, “Electoral Accountability: Recent Theoretical and Empirical Work,” *Annual Review of Political Science*, 2012, 15 (Volume 15, 2012), 183–201.
- Atsusaka, Yuki, Silviya Valeva, and Agustin Vallejo**, “The Margin of Victory in Ranked-Choice Voting,” July 2024.
- Avis, Eric, Claudio Ferraz, Frederico Finan, and Carlos Varjão**, “Money and Politics: The Effects of Campaign Spending Limits on Political Entry and Competition,” *American Economic Journal: Applied Economics*, October 2022, 14 (4), 167–99.
- Ban, Pamela, Elena Llaudet, and James M. Snyder Jr**, “Challenger Quality and the Incumbency Advantage,” *Legislative Studies Quarterly*, 2016, 41 (1), 153–179.
- Bartnicki, Sławomir, Maciej Alimowski, and Maciej A. Górecki**, “The Anomalous Electoral Advantage: Evidence From Over 17,000 Mayoral Candidacies in Poland,” *European Journal of Political Economy*, March 2022, 72, 102109.
- Bastos, Paulo and Cristian Sanchez**, “The Effects of Educated Leaders on Policy and Politics: Quasi-Experimental Evidence From Brazil,” *Latin American Economic Review*, 2024.

- Battocchio, Matias**, “La Ventaja Del Oficialismo en Las Elecciones Municipales De La Provincia De Buenos Aires,” *Económica*, December 2018, pp. 133–171.
- Baturo, Alexander and Robert Elgie**, *The Politics of Presidential Term Limits*, Oxford University Press, 2019.
- Benedetto, Marco Alberto De**, “Incumbency Advantage at Municipal Elections in Italy: A Quasi-Experimental Approach,” 2014.
- , “Personal or Partisan Incumbency Advantage? Evidence From an Electoral Reform at the Local Level in Italy,” *The B.E. Journal of Economic Analysis & Policy*, December 2019, 20 (1).
- Benedictis-Kessner, Justin De**, “Off-Cycle and Out of Office: Election Timing and the Incumbency Advantage,” *The Journal of Politics*, January 2018, 80 (1), 119–132.
- Bernhard, Michael and Ekrem Karakoç**, “Moving West or Going South?: Economic Transformation and Institutionalization in Postcommunist Party Systems,” *Comparative Politics*, 2011, 44 (1), 1–20.
- Besley, Timothy**, *Principled Agents? The Political Economy of Good Government*, Oxford University Press Scholarship Online, 2007.
- Bonifacio, Carlos Di, Guido Merzoni, and Federico Trombetta**, “Incumbency Effect in Competitive Autocracies: Evidence From Venezuela,” 2025.
- Broockman, David E.**, “Do Congressional Candidates Have Reverse Coattails? Evidence From a Regression Discontinuity Design,” *Political Analysis*, 2009, 17 (4), 418–434.
- Calonico, Sebastian, Matias D Cattaneo, and Max H Farrell**, “Optimal Bandwidth Choice for Robust Bias-Corrected Inference in Regression Discontinuity Designs,” *The Econometrics Journal*, 2020, 23 (2), 192–210.
- , **Matias D. Cattaneo, and Rocio Titiunik**, “Robust Nonparametric Confidence Intervals for Regression-Discontinuity Designs,” *Econometrica*, 2014, 82 (6), 2295–2326.
- Carrera, Ivan Morales**, “Efecto Incumbente en Elecciones Municipales: Un Análisis De Regresión Discontinua Para Guatemala,” *Revista De Análisis Económico*, October 2014, 29 (2), 113–150.
- Colomer, Josep**, *The Handbook of Electoral System Choice*, Springer, 2016.
- Coppedge, Michael, John Gerring, Carl Henrik Knutsen, Staffan I Lindberg, Jan Teorell, David Altman, Michael Bernhard, Agnes Cornell, M Steven Fish, Lisa Gastaldi et al.**, “V-Dem Codebook v11.1,” 2021.
- Coviello, Decio and Stefano Gagliarducci**, “Tenure in Office and Public Procurement,” *American Economic Journal: Economic Policy*, 2017, 9 (3), 59–105.
- Cox, Gary and Jonathan Katz**, “Why Did the Incumbency Advantage in US House Elections Grow?,” *American Journal of Political Science*, 1996, 40 (2), 478–497.
- da Fonseca, Mariana Lopes**, “Identifying the Source of Incumbency Advantage Through a Constitutional Reform,” *American Journal of Political Science*, 2017, 61 (3), 657–670.
- Dahlgaard, Jens Olav**, “You Just Made It: Individual Incumbency Advantage Under Proportional Representation,” *Electoral Studies*, 2016, 44, 319–328.

- Dano, Kevin, Francesco Ferlenga, Vincenzo Galasso, Caroline Le Pennec, and Vincent Pons**, “Coordination and Incumbency Advantage in Multi-Party Systems—Evidence From French Elections,” *Journal of the European Economic Association*, January 2025.
- Dinas, Elias, Pedro Riera, and Nasos Roussias**, “Staying in the First League: Parliamentary Representation and the Electoral Success of Small Parties,” *Political Science Research and Methods*, 2015, 3 (2), 187–204.
- Döring, Holger and Sven Regel**, “Party Facts: A Database of Political Parties Worldwide,” *Party Politics*, 2019, 25 (2), 97–109.
- Duch, Raymond M.**, “Economic Chaos and the Fragility of Democratic Transition in Former Communist Regimes,” *The Journal of Politics*, 1995, 57 (1), 121–158.
- , “A Developmental Model of Heterogeneous Economic Voting in New Democracies,” *American Political Science Review*, 2001, 95 (4), 895–910.
- Duggan, John and César Martinelli**, “The Political Economy of Dynamic Elections: Accountability, Commitment, and Responsiveness,” *Journal of Economic Literature*, September 2017, 55 (3), 916–84.
- and —, “Electoral Accountability and Responsive Democracy,” *The Economic Journal*, 12 2019, 130 (627), 675–715.
- Eggers, Andrew C. and Arthur Spirling**, “Incumbency Effects and the Strength of Party Preferences: Evidence From Multiparty Elections in the United Kingdom,” *The Journal of Politics*, July 2017, 79 (3), 903–920.
- Eggers, Andrew C, Anthony Fowler, Jens Hainmueller, Andrew B Hall, and James M Snyder Jr**, “On the Validity of the Regression Discontinuity Design for Estimating Electoral Effects: New Evidence From Over 40,000 Close Races,” *American Journal of Political Science*, 2015, 59 (1), 259–274.
- Egorov, Georgy and Konstantin Sonin**, “Incumbency Advantage in Non-Democracies,” Working Paper 20519, National Bureau of Economic Research September 2014.
- and —, “The Political Economics of Non-democracy,” *Journal of Economic Literature*, June 2024, 62 (2), 594–636.
- Elkins, Zachary and Tom Ginsburg**, “Characteristics of National Constitutions, Version 4.0,” Technical Report 2022.
- Erikson, Robert S. and Rocío Titiunik**, “Using Regression Discontinuity to Uncover the Personal Incumbency Advantage,” *Quarterly Journal of Political Science*, May 2015, 10 (1), 101–119.
- Fearon, James D.**, “Electoral Accountability and the Control of Politicians: Selecting Good Types versus Sanctioning Poor Performance,” in Adam Przeworski, Susan C Stokes, and Bernard Manin, eds., *Democracy, Accountability, and Representation*, Cambridge University Press, 1999, pp. 55–97.
- Fearon, James D.**, “Self-Enforcing Democracy,” *The Quarterly Journal of Economics*, 11 2011, 126 (4), 1661–1708.

- Feenstra, Robert C, Robert Inklaar, and Marcel P Timmer**, “The Next Generation of the Penn World Table,” *American Economic Review*, 2015, *105* (10), 3150–82.
- Feierherd, Germán and Adrián Lucardi**, “When the Partisan Becomes Personal: Mayoral Incumbency Effects in Buenos Aires, 1983–2019,” *Journal of Elections, Public Opinion and Parties*, October 2023, *33* (4), 684–704.
- Ferejohn, John**, “Incumbent Performance and Electoral Control,” *Public Choice*, 1986, *50* (1/3), 5–25.
- Ferejohn, John A.**, “On the Decline of Competition in Congressional Elections,” *American Political Science Review*, 1977, *71* (1), 166–176.
- Ferreira, Fernando and Joseph Gyourko**, “Do Political Parties Matter? Evidence From U.S. Cities,” *The Quarterly Journal of Economics*, February 2009, *124* (1), 399–422.
- Filho, George Avelino, Ciro Biderman, and Scott Desposato**, “Sources of the Incumbency (Dis)Advantage,” *Brazilian Political Science Review*, 2022, *16* (1).
- Fiorina, Morris P.**, “The Case of the Vanishing Marginals: The Bureaucracy Did It,” *The American Political Science Review*, 1977, *71* (1), 177–181.
- Fiva, Jon H. and Daniel M. Smith**, “Political Dynasties and the Incumbency Advantage in Party-Centered Environments,” *American Political Science Review*, August 2018, *112* (3), 706–712.
- **and Helene Lie Røhr**, “Climbing the Ranks: Incumbency Effects in Party-List Systems,” *European Economic Review*, January 2018, *101*, 142–156.
- Fouirnaies, Alexander and Andrew B. Hall**, “The Financial Incumbency Advantage: Causes and Consequences,” *The Journal of Politics*, 2014, *76* (3), 711–724.
- Fowler, Anthony**, “Disentangling the Personal and Partisan Incumbency Advantages: Evidence From Close Elections and Term Limits,” *Quarterly Journal of Political Science*, December 2014, *9* (4), 501–531.
- Franklin, Charles H.**, “Senate Incumbent Visibility over the Election Cycle,” *Legislative Studies Quarterly*, 1993, *18* (2), 271–290.
- Freier, Ronny**, “The Mayor’s Advantage: Causal Evidence on Incumbency Effects in German Mayoral Elections,” *European Journal of Political Economy*, December 2015, *40*, 16–30.
- Gelman, Andrew and Gary King**, “Estimating Incumbency Advantage Without Bias,” *American Journal of Political Science*, November 1990, *34* (4), 1142–1164.
- Golden, Miriam and Brian Min**, “Distributive Politics Around the World,” *Annual Review of Political Science*, 2013, *16* (Volume 16, 2013), 73–99.
- **and Eugenia Nazrullaeva**, *The Puzzle of Clientelism: Political Discretion and Elections Around the World*, Cambridge University Press, 2023.
- **and Lucio Picci**, “Incumbency Effects Under Proportional Representation: Leaders and Backbenchers in the Postwar Italian Chamber of Deputies,” *Legislative Studies Quarterly*, November 2015, *40* (4), 509–538.
- Granzier, Riako, Vincent Pons, and Clemence Tricaud**, “Coordination and

- Bandwagon Effects: How Past Rankings Shape the Behavior of Voters and Candidates,” *American Economic Journal: Applied Economics*, October 2023, *15* (4), 177–217.
- Guriev, Sergei, Nikita Melnikov, Joana Escorcio Silva, and Ekaterina Zhuravskaya**, “Mobile Broadband and the Decline of Incumbency Advantage,” 2025.
- Hall, Andrew B. and James M. Snyder**, “How Much of the Incumbency Advantage Is Due to Scare-Off?,” *Political Science Research and Methods*, September 2015, *3* (3), 493–514.
- Hicken, Allen**, “Clientelism,” *Annual Review of Political Science*, 2011, *14* (Volume 14, 2011), 289–310.
- Hyytinen, Ari, Jaakko Meriläinen, Tuukka Saarimaa, Otto Toivanen, and Janne Tukiainen**, “When Does Regression Discontinuity Design Work? Evidence From Random Election Outcomes,” *Quantitative Economics*, 2018, *9* (2), 1019–1051.
- Jacobson, Gary C.**, “It’s Nothing Personal: The Decline of the Incumbency Advantage in US House Elections,” *The Journal of Politics*, July 2015, *77* (3), 861–873.
- Jankowski, Michael and Stefan Müller**, “The Incumbency Advantage in Second-Order PR Elections: Evidence From the Irish Context, 1942–2019,” *Electoral Studies*, June 2021, *71*, 102331.
- Kang, Woo Chang, Won ho Park, and B.K. Song**, “The Effect of Incumbency in National and Local Elections: Evidence From South Korea,” *Electoral Studies*, December 2018, *56*, 47–60.
- Karnik, Ajit, Mala Lalvani, and Manali Phatak**, “Political Incumbency Effects in India: A Regional Analysis,” *Studies in Economics and Econometrics*, January 2023, *47* (1), 43–60.
- Kartik, Navin and Richard Van Weelden**, “Reputation Effects and Incumbency (Dis)Advantage,” *Quarterly Journal of Political Science*, 2019, *14* (2), 131–157.
- , **Elliot Lipnowski, and Harry Pei**, “Replacement and Reputation,” 2025.
- Kendall, Chad and Marie Rekkas**, “Incumbency Advantages in the Canadian Parliament,” *Canadian Journal of Economics/Revue Canadienne d’Économique*, November 2012, *45* (4), 1560–1585.
- King, Gary and Andrew Gelman**, “Systemic Consequences of Incumbency Advantage in the U.S. House,” *American Journal of Political Science*, 1991, *35* (1), 110–138.
- Klašnja, Marko**, “Corruption and the Incumbency Disadvantage: Theory and Evidence,” *The Journal of Politics*, October 2015, *77* (4), 928–942.
- and **Rocío Titiunik**, “The Incumbency Curse: Weak Parties, Term Limits, and Unfulfilled Accountability,” *American Political Science Review*, February 2017, *111* (1), 129–148.
- Kotakorpi, Kaisa, Panu Poutvaara, and Marko Tervii**, “Returns to Office in National and Local Politics: A Bootstrap Method and Evidence From Finland,” *The Journal of Law, Economics, and Organization*, August 2017, *33* (3), 413–442.
- Lee, Alexander**, “Incumbency, Parties, and Legislatures: Theory and Evidence From

- India,” *Comparative Politics*, January 2020, 52 (2), 311–331.
- Lee, David S.**, “The Electoral Advantage to Incumbency and Voters’ Valuation of Politicians’ Experience: A Regression Discontinuity Analysis of Elections to the U.S. House,” Working Paper 8441, National Bureau of Economic Research August 2001.
- , “Randomized Experiments From Non-Random Selection in U.S. House Elections,” *Journal of Econometrics*, February 2008, 142 (2), 675–697.
- **and Thomas Lemieux**, “Regression Discontinuity Designs in Economics,” *Journal of Economic Literature*, 2010, 48 (2), 281–355.
- Levitsky, Steven and Daniel Ziblatt**, *How Democracies Die*, Penguin Random House, 2019.
- Levitt, Steven and Catherine Wolfram**, “Decomposing the Sources of Incumbency Advantage in the U.S. House,” *Legislative Studies Quarterly*, 1997, 22 (1), 45–60.
- Levitt, Steven D.**, “Using Repeat Challengers to Estimate the Effect of Campaign Spending on Election Outcomes in the U.S. House,” *Journal of Political Economy*, 1994, 102 (4), 777–798.
- Lewis, Blane D., Hieu T.M. Nguyen, and Adrianus Hendrawan**, “Political Accountability and Public Service Delivery in Decentralized Indonesia: Incumbency Advantage and the Performance of Second Term Mayors,” *European Journal of Political Economy*, September 2020, 64, 101910.
- Linden, Leigh L**, “Are Incumbents Really Advantaged? The Preference for Non-Incumbents in Indian National Elections,” 2004.
- Lucardi, Adrián and Guillermo Rosas**, “Is the Incumbent Curse the Incumbent’s Fault? Strategic Behavior and Negative Incumbency Effects in Young Democracies,” *Electoral Studies*, December 2016, 44, 66–75.
- Lucas, Jack**, “The Size and Sources of Municipal Incumbency Advantage in Canada,” *Urban Affairs Review*, March 2021, 57 (2), 373–401.
- Lührmann, A, N Düpont, M Higashijima, YB Kavasogly, KL Marquardt, M Bernhard, H Döring, A Hicken, M Laebens, SI Lindberg et al.**, “Varieties of Party Identity and Organization (V-Party) Dataset V1,” 2020.
- Lundqvist, Heléne**, “Is It Worth It? On the Returns to Holding Political Office,” 2013.
- Macdonald, Bobbie**, “Incumbency Disadvantages in African Politics? Regression Discontinuity Evidence From Zambian Elections,” *SSRN Electronic Journal*, 2013.
- Magalhaes, Leandro De, Thomaz M. F. Gemignani, and Salomo Hirvonen**, “Political Careers in Brazil: The Effect of Winning Vs. Being the Runner-Up,” 2025.
- Magalhães, Leandro De and Isabel Spigel-Sinclair**, “Could Regression Discontinuity Estimates of Incumbency Effects Help Monitor Parliamentary Elections? Evidence From Malawi,” 2021.
- Magalhães, Leandro De and Salomo Hirvonen**, “A Second Chance Elsewhere. Estimating the Effect of Winning (Vs. Being the Runner-Up) on Future Electoral Prospects,” *Electoral Studies*, June 2023, 83, 102612.

- , **Dominik Hangartner, Salomo Hirvonen, Jaakko Meriläinen, Nelson A. Ruiz, and Janne Tukiainen**, “When Can We Trust Regression Discontinuity Design Estimates From Close Elections? Evidence From Experimental Benchmarks,” *Political Analysis*, July 2025, *33* (3), 258–265.
- Marx, Benjamin, Vincent Pons, and Vincent Rollet**, “Electoral Turnovers,” *The Review of Economic Studies*, 2025, *92* (5), 3306–3339.
- , – , and – , “National Elections Database (Version 2.0),” 2025.
- Mayhew, David R.**, “Incumbency Advantage in U.S. Presidential Elections: The Historical Record,” *Political Science Quarterly*, 02 2008, *123* (2), 201–228.
- McCrary, Justin**, “Manipulation of the Running Variable in the Regression Discontinuity Design: A Density Test,” *Journal of Econometrics*, 2008, *142* (2), 698–714.
- Miguel, Edward and Farhan Zaidi**, “Do Politicians Reward Their Supporters? Regression Discontinuity Evidence From Ghana,” 2003.
- Munoz, Erico Andres**, “Incumbency Advantage, Money, and Campaigns: A Note on Some Suggestive Evidence From Chile,” *Economics Bulletin*, 2021, *41* (3), 1203–1211.
- Nohlen, Dieter**, “Elections in the Americas. A Data Handbook: North America, Central America, and the Caribbean. 2 Vols. Vol. 1,” *Elections Worldwide*, 2005.
- and **Philip Stöver**, *Elections in Europe*, Nomos Verlagsgesellschaft mbH & Co. KG, 2010.
- **et al.**, *Elections in the Americas: A Data Handbook: South America*, Vol. 2, Oxford University Press on Demand, 2005.
- , **Florian Grotz, and Christof Hartmann**, “Elections and Electoral Systems in Asia and the Pacific,” *Elections in Asia and the Pacific: A Data Handbook: Volume II: South East Asia, East Asia, and the South Pacific*, 2001, *2*, 1.
- , – , and – , *Elections in Asia and the Pacific: A Data Handbook: Volume I: Middle East, Central Asia, and South Asia*, OUP Oxford, 2001.
- , **Michael Krennerich, Bernard Thibaut et al.**, *Elections in Africa: A Data Handbook*, Oxford University Press, 1999.
- Novaes, Lucas M. and Luis Schiumerini**, “Commodity Shocks and Incumbency Effects,” *British Journal of Political Science*, October 2022, *52* (4), 1689–1708.
- Nzabonimpa, Mélyne**, “Gender Differences in Politician Persistence and Incumbency Advantage,” *European Journal of Political Economy*, September 2023, *79*, 102404.
- Núñez, Lucas**, “Do Clientelistic Machines Affect Electoral Outcomes? Mayoral Incumbency as a Proxy for Machine Prowess,” *Electoral Studies*, October 2018, *55*, 109–119.
- Ochieng’Opalo, Kennedy**, “Institutions and Legislative Incumbency Advantage: A Regression Discontinuity Analysis of Legislative Elections in Africa,” *SSRN Electronic Journal*, 2014.
- Olson, Michael P. and Andrew R. Stone**, “The Incumbency Advantage in Judicial Elections: Evidence From Partisan Trial Court Elections in Six U.S. States,” *Political Behavior*, December 2023, *45* (4), 1333–1354.

- Pacek, Alexander C.**, “Macroeconomic Conditions and Electoral Politics in East Central Europe,” *American Journal of Political Science*, 1994, 38 (3), 723–744.
- Pastine, Ivan and Tuvana Pastine**, “Incumbency advantage and political campaign spending limits,” *Journal of Public Economics*, 2012, 96 (1), 20–32.
- Pemstein, Daniel, Kyle L Marquardt, Eitan Tzelgov, Yi ting Wang, Joshua Krusell, and Farhad Miri**, “The V-Dem Measurement Model: Latent Variable Analysis for Cross-National and Cross-Temporal Expert-Coded Data,” *V-Dem Working Paper*, 2018, 21.
- Powell, Jonathan M and Clayton L Thyne**, “Global Instances of Coups From 1950 to 2010: A New Dataset,” *Journal of Peace Research*, 2011, 48 (2), 249–259.
- Przeworski, Adam, S Newman, SK Park, Didac Queralt, Gonzalo Rivero, and Kong Joo Shin**, “Political Institutions and Political Events (PIPE) Data Set,” *Department of Politics, New York University*, 2013.
- Redmond, Paul and John Regan**, “Incumbency Advantage in a Proportional Electoral System: A Regression Discontinuity Analysis of Irish Elections,” *European Journal of Political Economy*, June 2015, 38, 244–256.
- Roberts, Andrew**, “Hyperaccountability: Economic Voting in Central and Eastern Europe,” *Electoral Studies*, 2008, 27 (3), 533–546.
- Roh, Jungho**, “The Incumbency Disadvantage in South Korean National Assembly Elections: Evidence From a Regression Discontinuity Approach,” *Electoral Studies*, April 2017, 46, 112–122.
- Salas, Christian**, “Incumbency Advantage in Multi-Member Districts: Evidence From Congressional Elections in Chile,” *Electoral Studies*, June 2016, 42, 213–221.
- Schiumerini, Luis and María Page**, “El Efecto “Cancha Inclinada”: Ventajas Del Oficialismo en La Política De Las Provincias Argentinas,” 2012.
- Sevi, Semra**, “Is Incumbency Advantage Gendered?,” *Legislative Studies Quarterly*, February 2023, 48 (1), 145–163.
- , “The Incumbency Advantage in Canadian Elections,” *Canadian Journal of Political Science*, June 2025, pp. 1–11.
- Svolik, Milan W.**, “Learning to Love Democracy: Electoral Accountability and the Success of Democracy,” *American Journal of Political Science*, 2013, 57 (3), 685–702.
- Titunik, Rocio**, “Incumbency Advantage in Brazil: Evidence From Municipal Mayor Elections,” 2009.
- Trounstone, Jessica**, “Evidence of a Local Incumbency Advantage,” *Legislative Studies Quarterly*, May 2011, 36 (2), 255–280.
- Uppal, Yogesh**, “The Disadvantaged Incumbents: Estimating Incumbency Effects in Indian State Legislatures,” *Public Choice*, January 2009, 138 (1-2), 9–27.
- , “Estimating Incumbency Effects in U.S. State Legislatures: A Quasi-Experimental Study,” *Economics & Politics*, July 2010, 22 (2), 180–199.
- Warshaw, Christopher**, “Local Elections and Representation in the United States,”

Annual Review of Political Science, May 2019, 22 (1), 461–479.

Weaver, Julie Anne, “Electoral Dis-Connection: The Limits of Reelection in Contexts of Weak Accountability,” *The Journal of Politics*, October 2021, 83 (4), 1462–1477.

Weiler, Johannes, “Once Mayor, Always Mayor? Incumbency Advantage of Austrian Mayors,” 2021.

Weingast, Barry R., Kenneth A. Shepsle, and Christopher Johnsen, “The Political Economy of Benefits and Costs: A Neoclassical Approach to Distributive Politics,” *Journal of Political Economy*, 1981, 89 (4), 642–664.

Zoorob, Michael, “There’s (Rarely) a New Sheriff in Town: The Incumbency Advantage for Local Law Enforcement,” *Electoral Studies*, December 2022, 80, 102550.